ARQ 201

...ance from p to V.
...meas-
...distance fr...
...shows the ...
...tations of a ...
...of the f...
...or ...ach ...nts. ...o be ...two

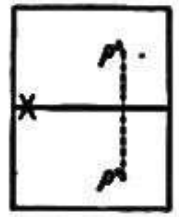


Fig. 3.

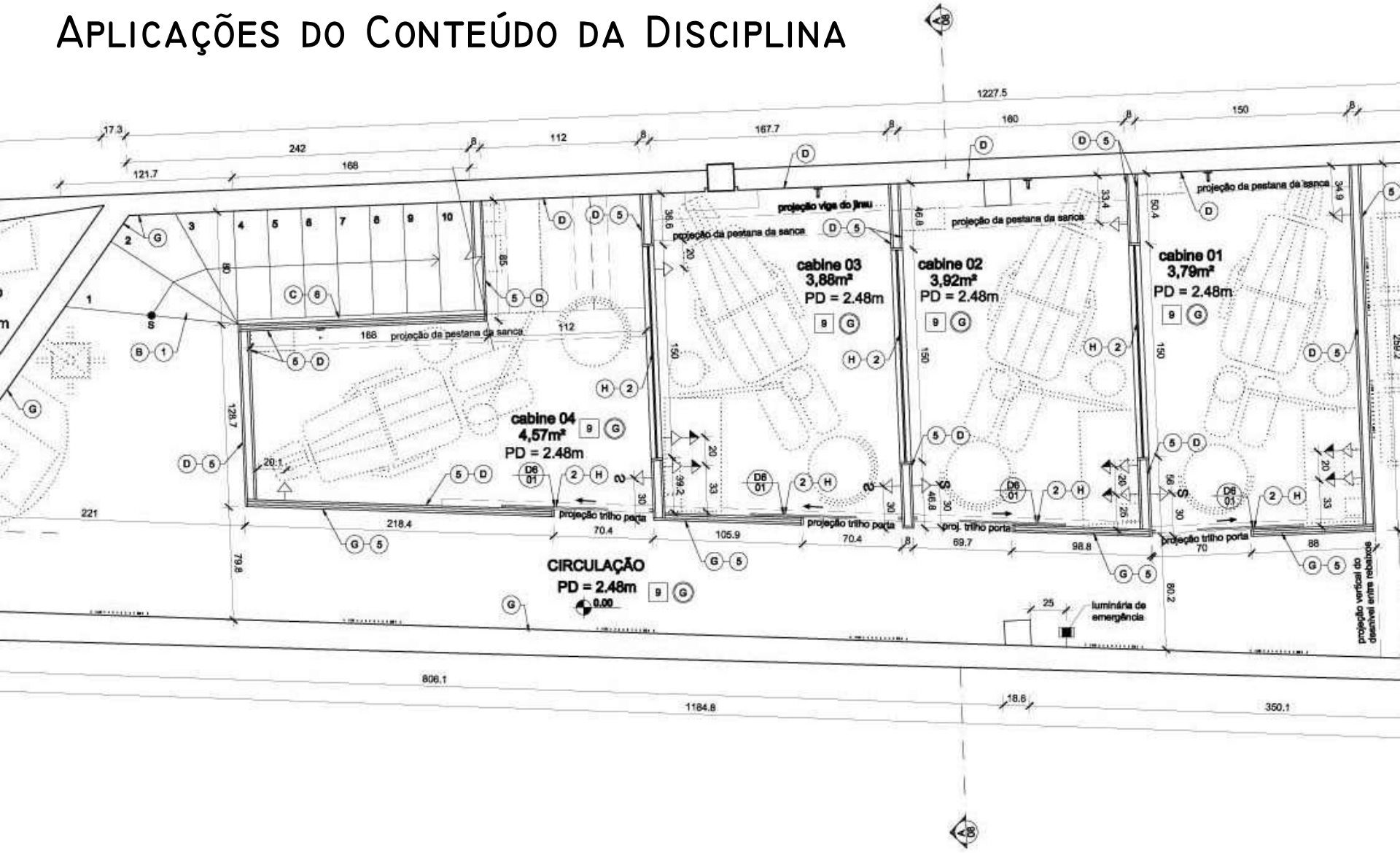
...the same perpen- ...be taken as the ...nt in space.

NOTATION.—Points in sp... will be designated by the small letters, as a, b, c. The V projections by the same letters with the exponent v, as a^v, b^v, c^v. The H projections with the same letters with the exponent h, as a^h, b^h, c^h. Successive positions of the same points will be denoted by subscripts, as a_1^h, a_2^h, a_3^h.

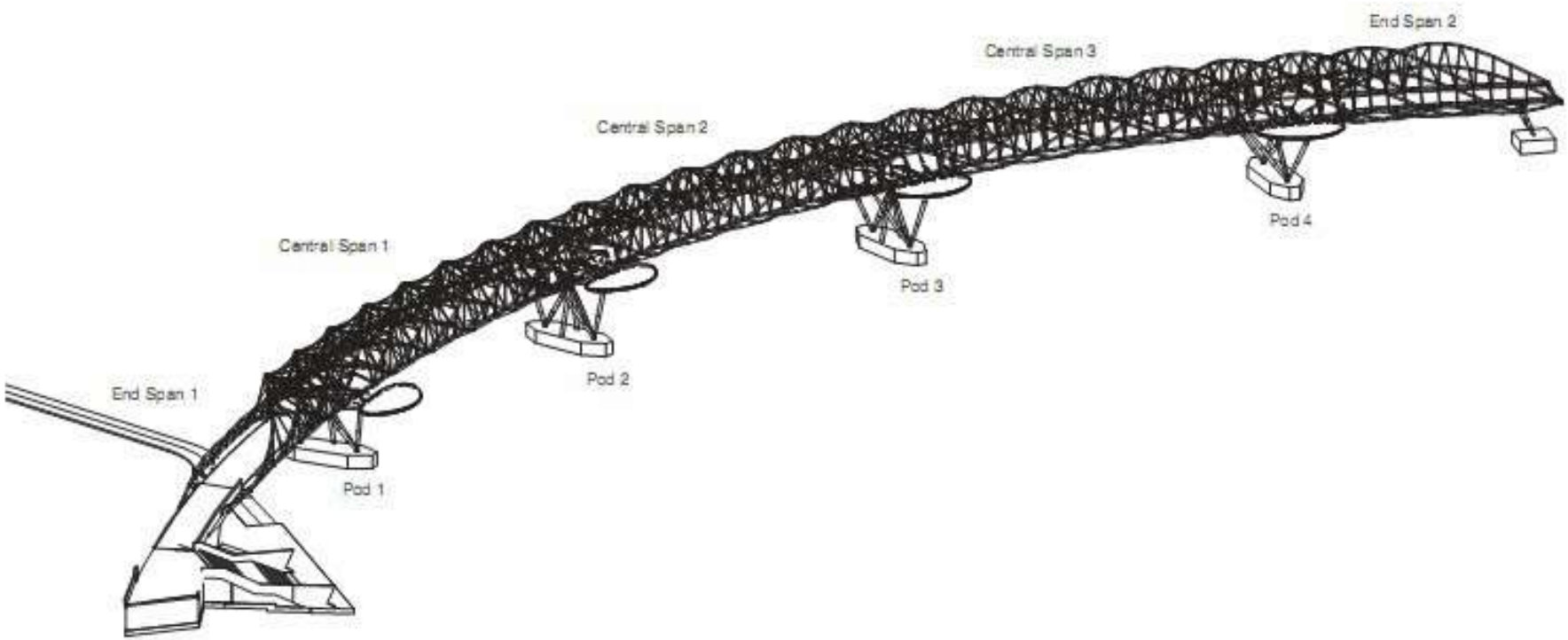
PROBLEM 1.—*Having the direction and distance of a point in space, from H and V, to draw its projections.*

Draw any per... X a... m X (above if the point be ... H ...elow ...e point be below H) the ...e o... poin... ...n H; this will be th... ojectio... the ... On the same per... ...lar set off... X ...e

Aplicações do Conteúdo da Disciplina



I. Leitura / interpretação de projetos



Helix Structural Frame

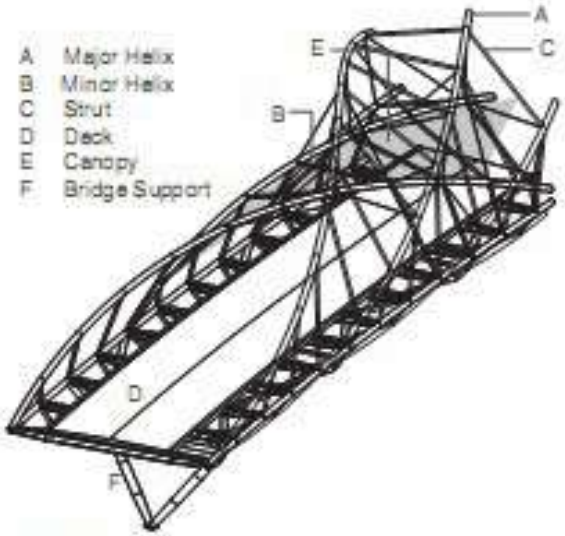
2. Desenho de edificações. (por que não é projeto?)

Architects Cox Architecture_ Architects 61

Raffles Avenue and Bayfront Bridge – Singapura - 2010

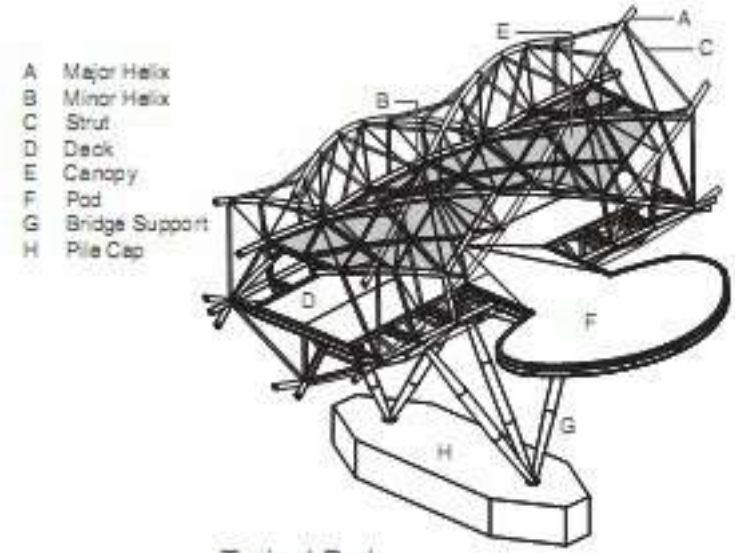
SGD$ 82 900 000,00



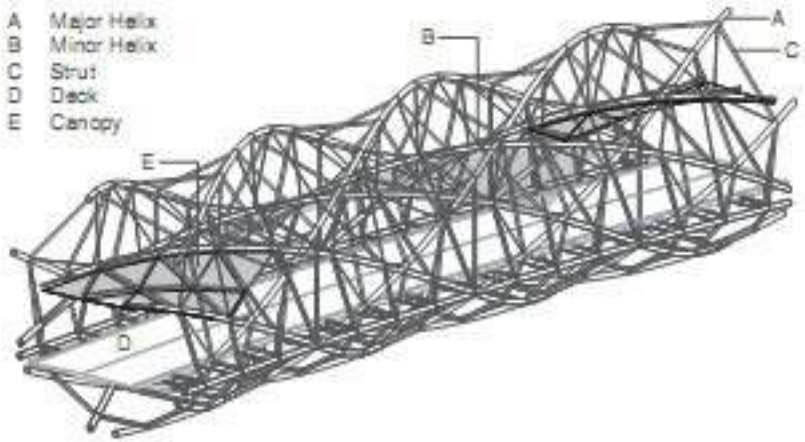


End Span

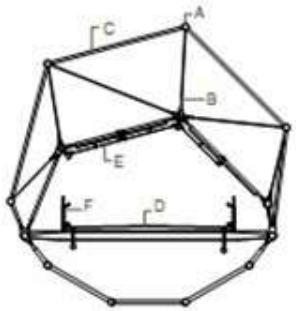




Typical Pod

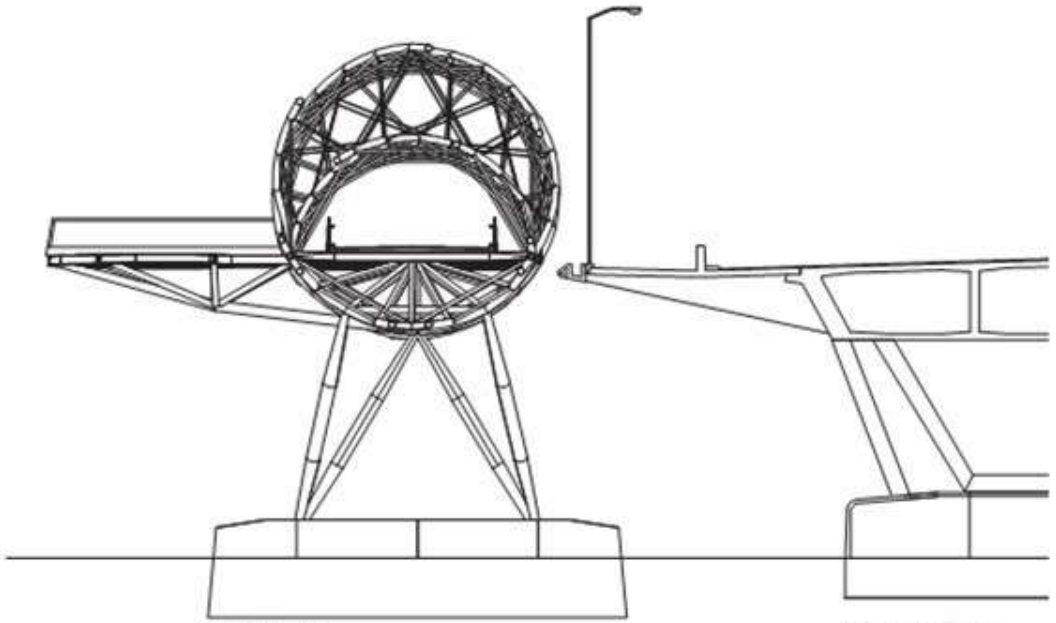


Central Span



A Major Helix
B Minor Helix
C Strut
D Deck
E Canopy
F Glass/Steel Balustrade

Bridge Section



Helix Bridge

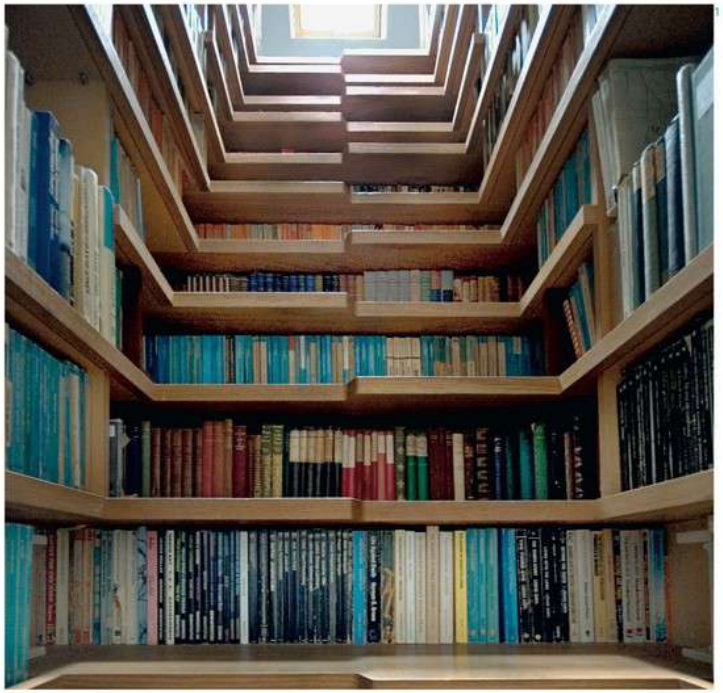
Bayfront Bridge

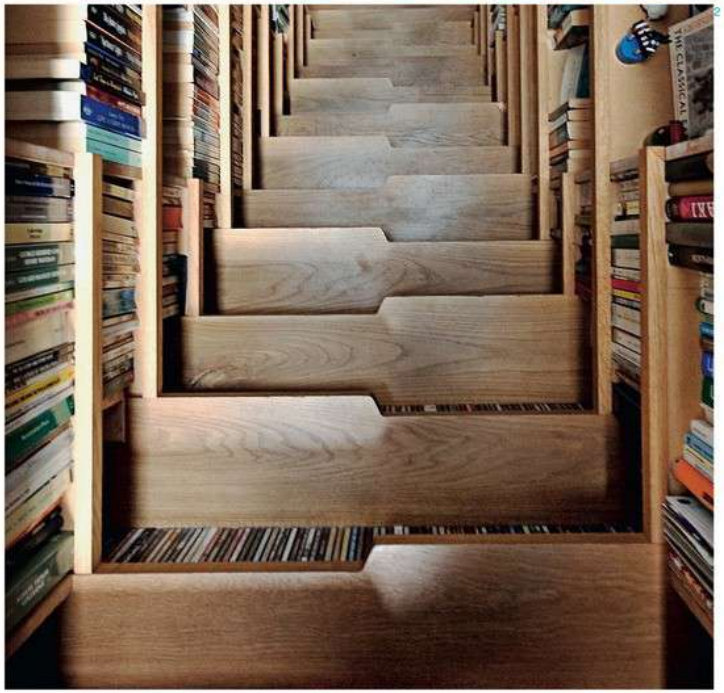
Section

1:250 0 2 4 6 8

COX


PAN PACIFIC

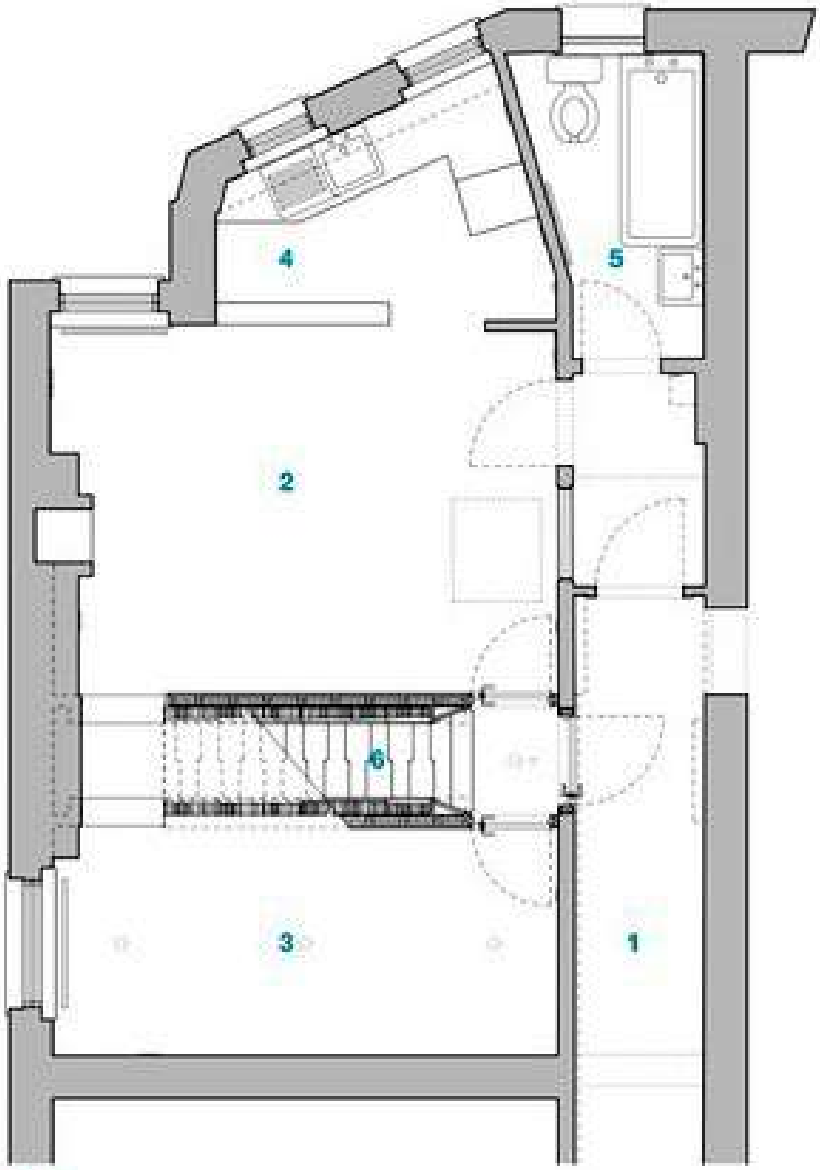




Levitate Architects

Library Staircase

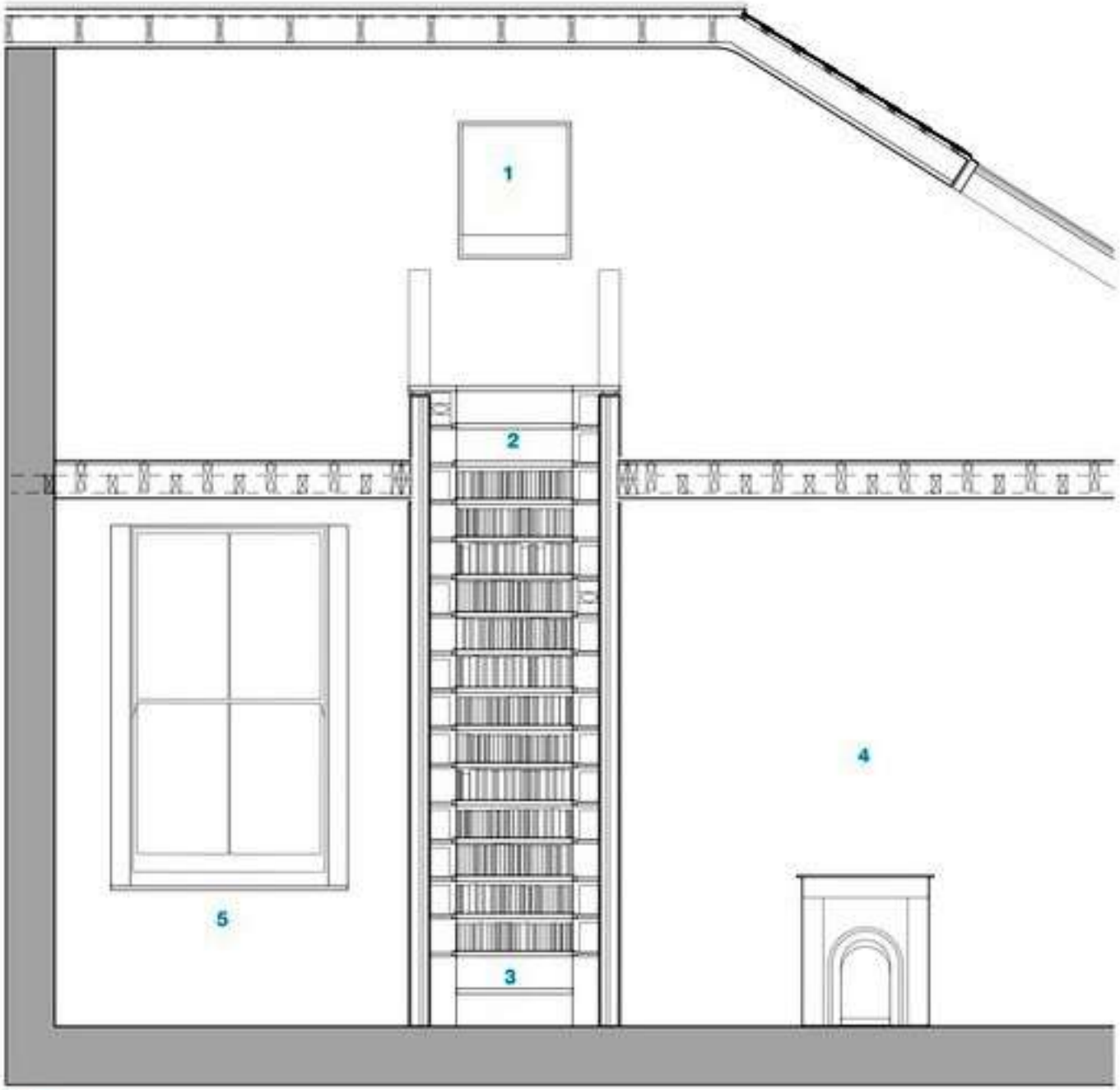
Londres



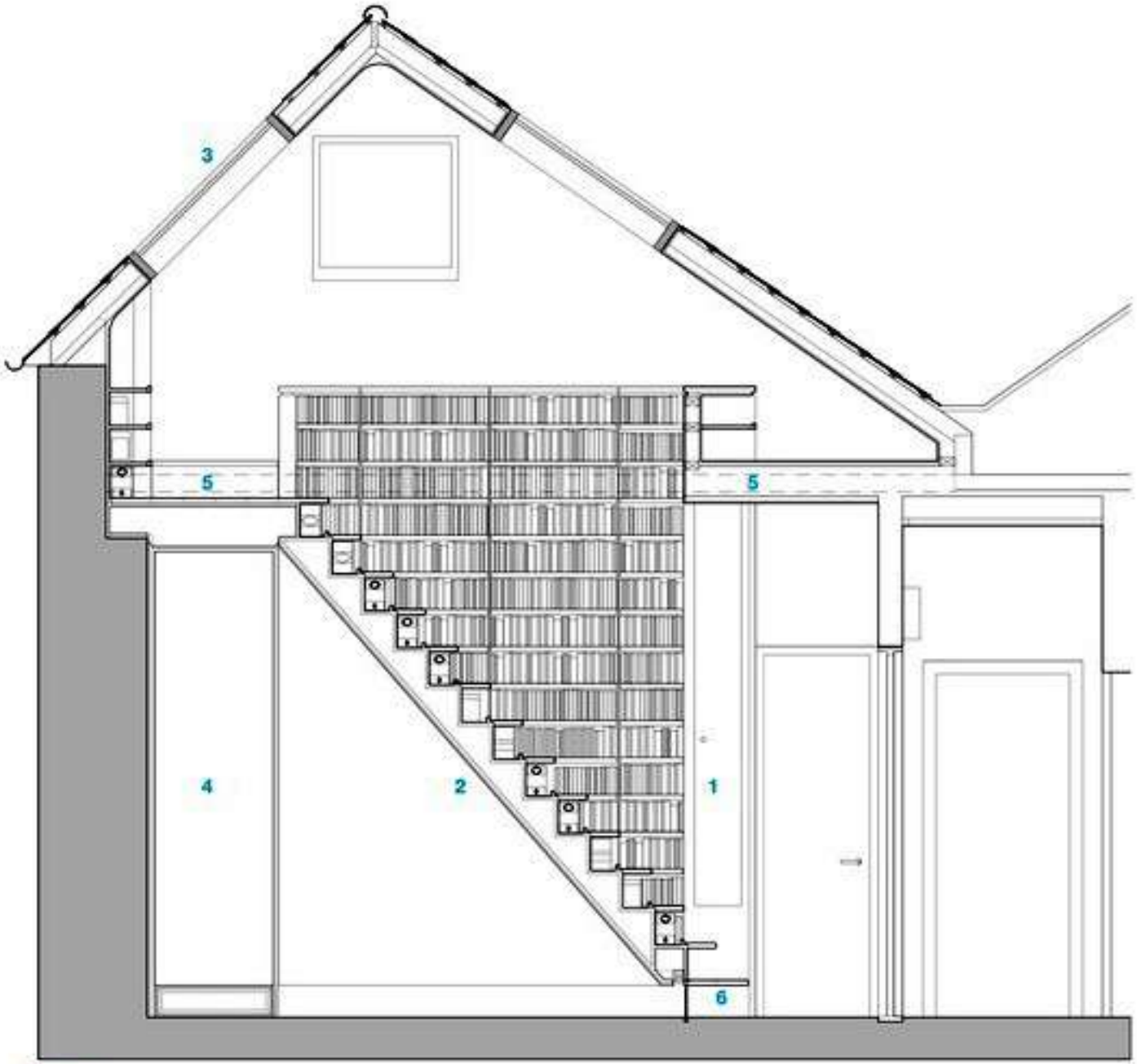
04.01

0 5m

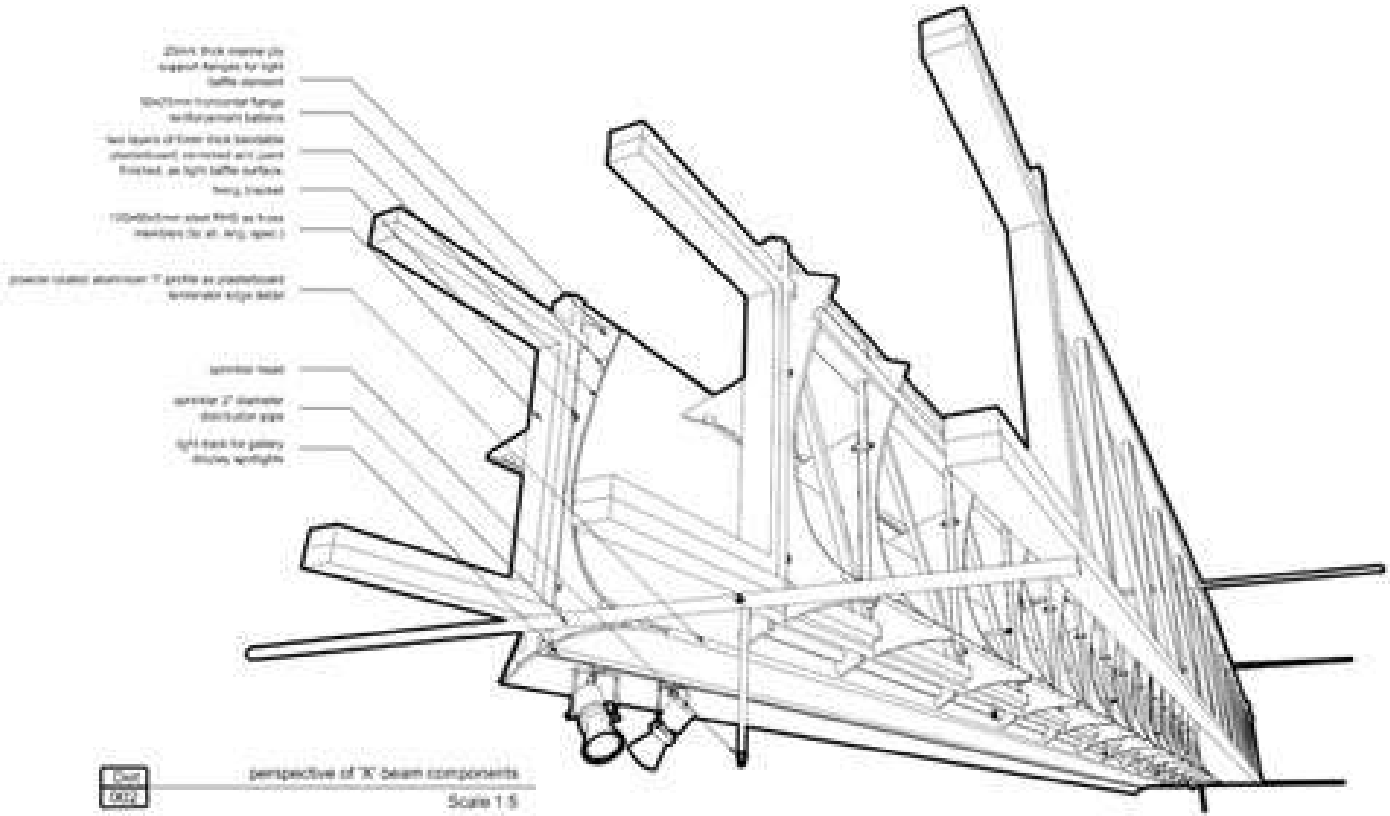
0 15ft



04.03

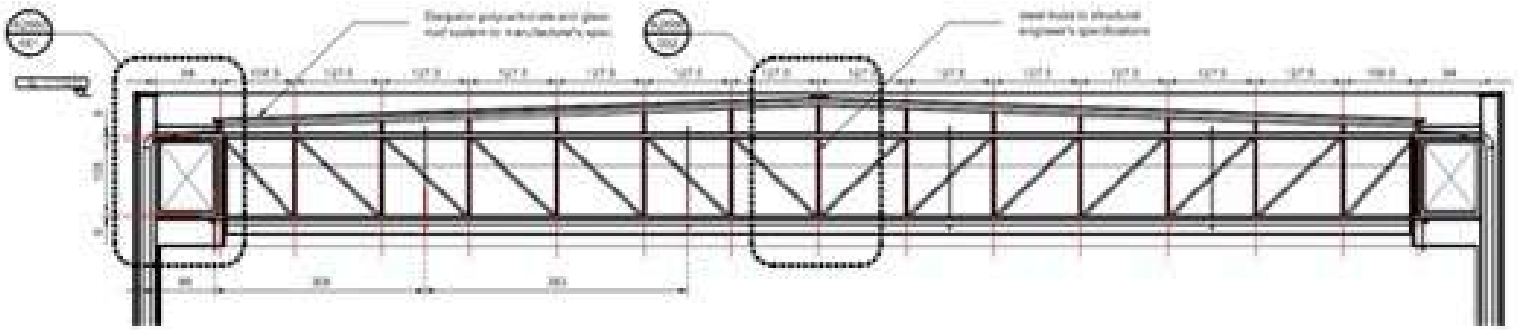


04.02



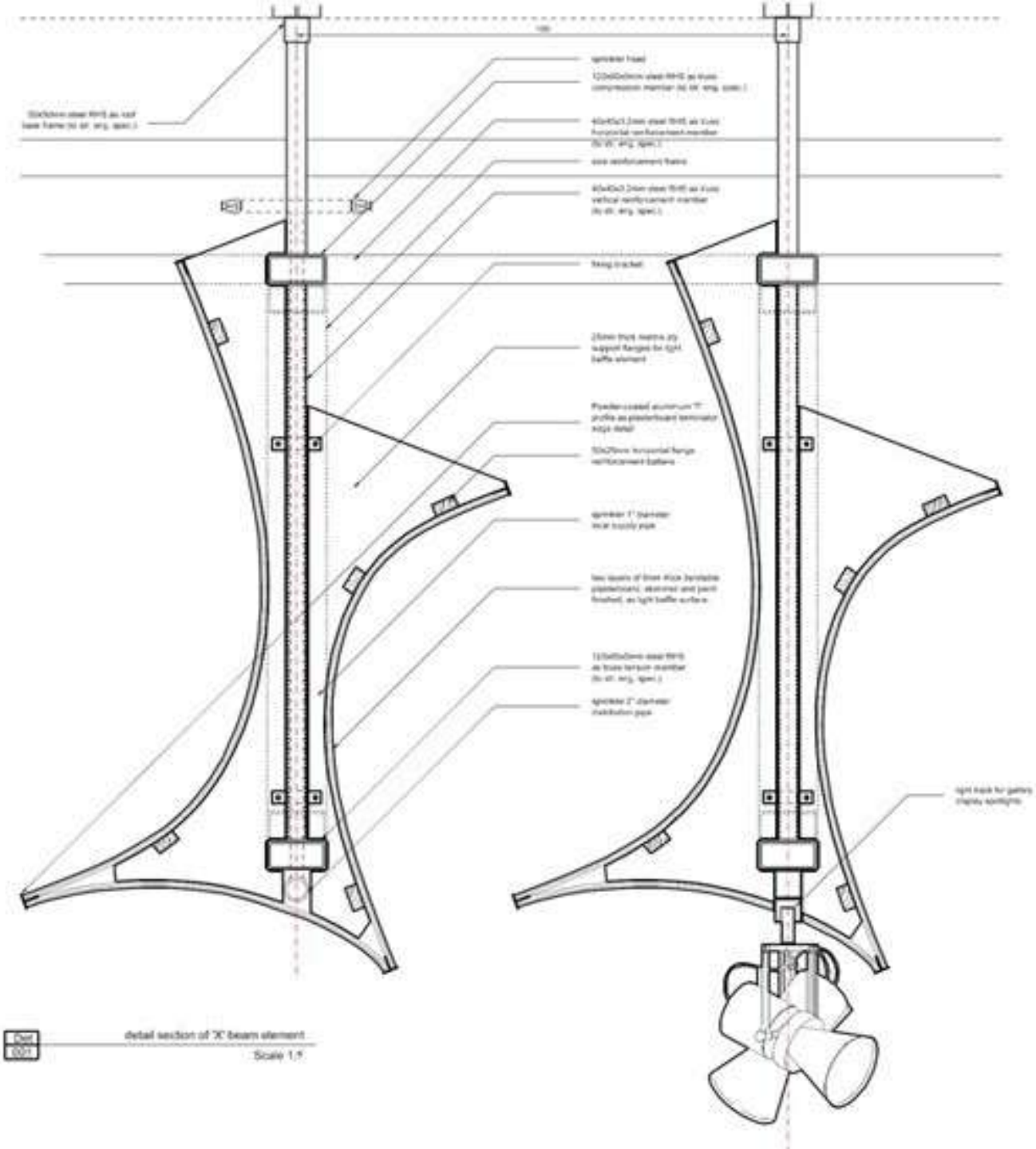
perspective of 'A' beam components

Scale 1:5



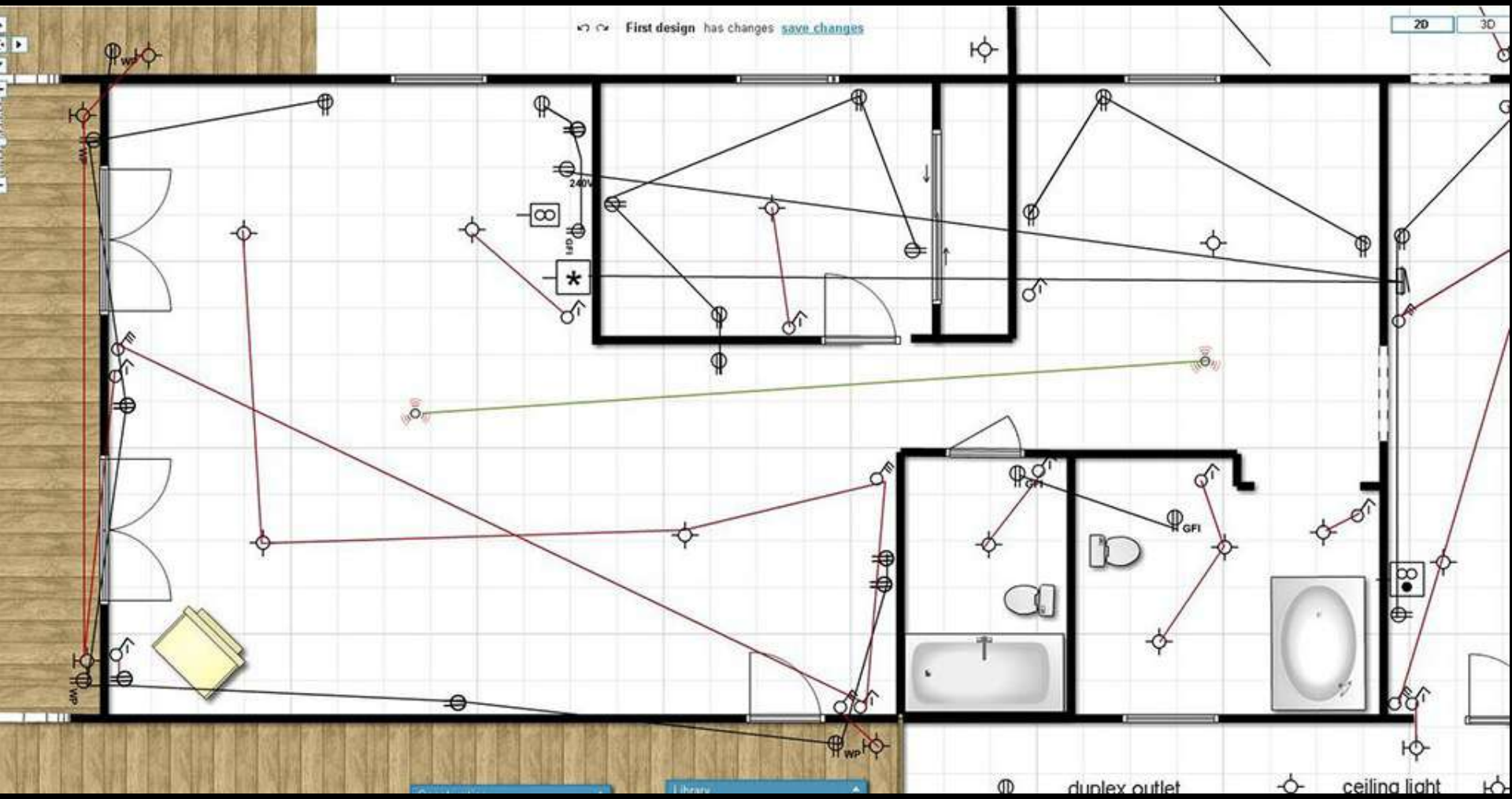
short section of roof type #1 (Gallery 1)

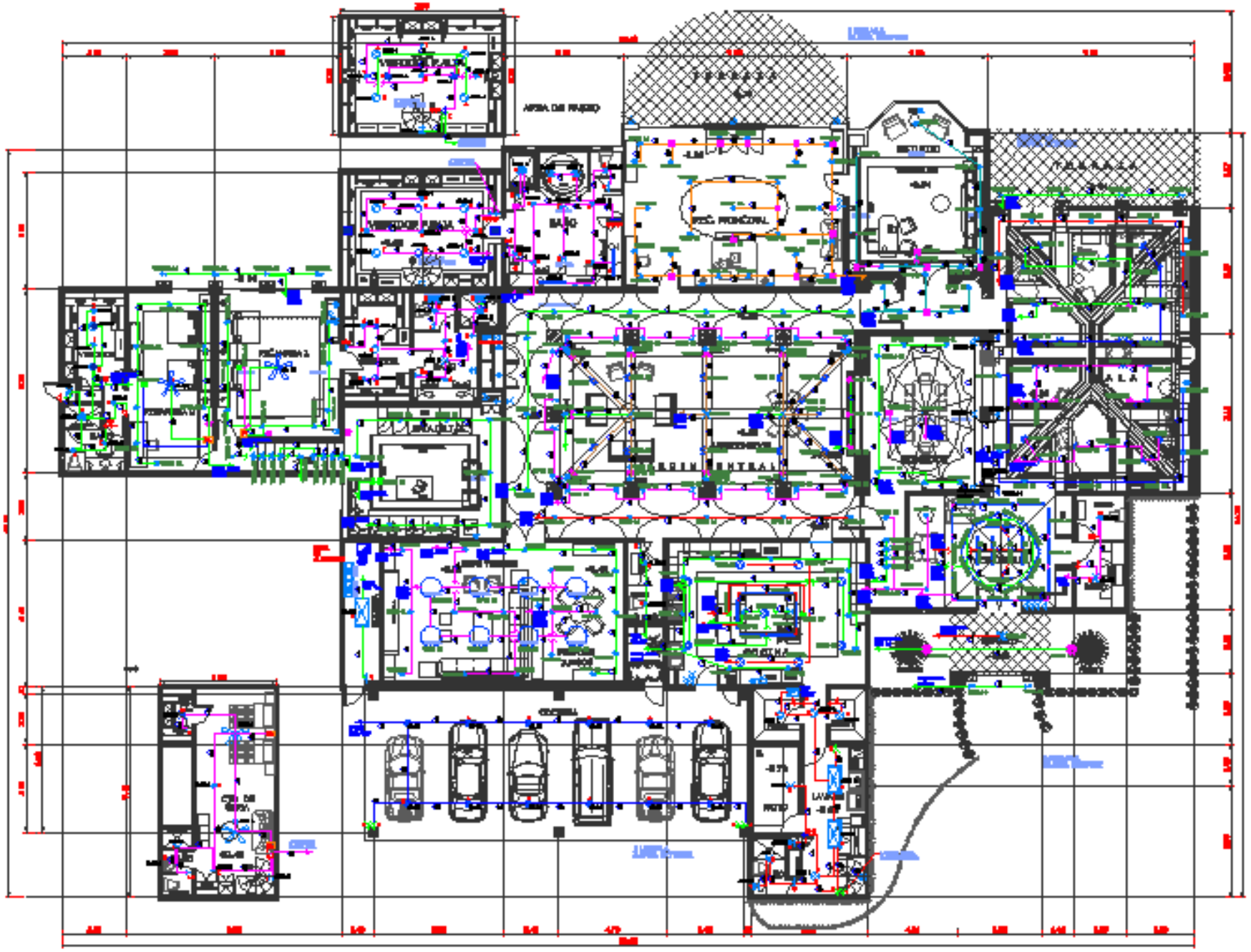
Scale 1:50

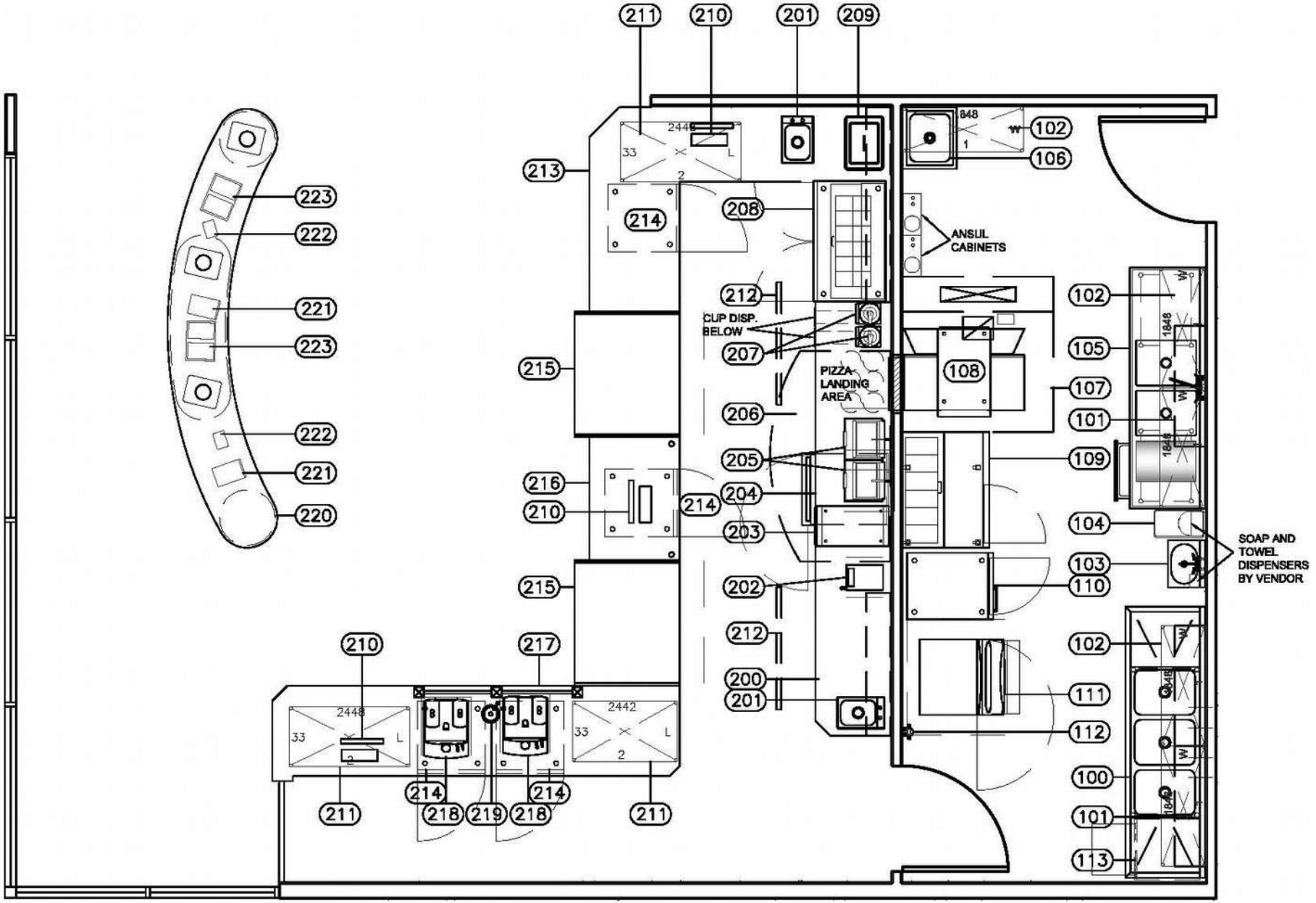


detail section of 'X' beam element

Scale 1:5


First design has changes save changes
2D
3D
240V
GFI
WP
duplex outlet
ceiling light


TERRAZA
TERRAZA
AREA DE RIEGO
VESTIDOR PISCINA
BAÑO
PISCINA PROYECTADA
ESTUDIO
SALA
COCINA
PATIO
LAVANDERIA
CTO. DE BOMBAS


211
210
201
209
213
214
208
212
CUP DISP. BELOW
207
215
PIZZA LANDING AREA
206
205
216
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214
211
223
222
221
223
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221
220
ANSUL CABINETS
102
106
102
105
108
107
101
109
104
103
110
102
111
112
100
101
113
SOAP AND TOWEL DISPENSERS BY VENDOR

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husband and carpenter architects ltd
13b St George Wharf, London. SW8 2LE
e-mail: admin@husbandandcarpenter.com
tel: 020 7793 3660 fax 020 7793 3661



drawing notes:

This Drawing must not be scaled, only figured dimensions should be used. Large scale Drawings are to be preferred to small scale Drawings in case of discrepancy. This Drawing should not be used for land transfer. This Drawing should not be used in isolation. If there are any discrepancies, or contradictions between this and other information, it should be brought to the Architect's attention. Any Adjoining Buildings/Landscape Features shown are for illustrative purposes only and have not been formally surveyed for this Drawing unless noted otherwise. This Drawing and the information contained therein are the copyright of the Architects, and must not be reproduced in part or whole, or disclosed to third parties, unless agreed in writing with Husband & Carpenter Architects Ltd.

project:

Falcon Road, London SW11

client:

Oak Trading Company Ltd

Revision notes:

drawing title:

Ground Floor Layout

date	scale	project no	drawing no	revision
MARCH 2010	1:100	2122	H/PL_01	

N

GENERAL NOTES

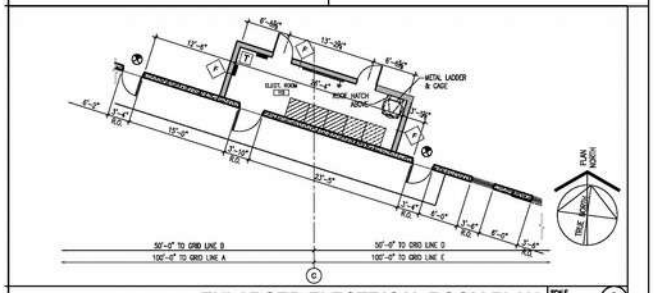
1. THE PROJECT SHALL COMPLY WITH ALL THE SECURITY PROVISIONS CONTAINED IN THE CITY ORDINANCE.
2. ANY HAZARDOUS MATERIALS STORED OR USED WITHIN THE BUILDING SHALL NOT EXCEED THE QUANTITIES LISTED IN CBC TABLES 307.1(1) & 307.1(2).
3. WALL AND CEILING MATERIALS SHALL NOT EXCEED THE FLAME SPREAD CLASSIFICATION IN CBC TABLE 803.9
4. EXIT DOORS MUST BE OPERABLE FROM ESGRESS SIDE WITHOUT THE USE OF A KEY, SPECIAL KNOWLEDGE, OR EFFORT. - CBC 1008.1.9.
5. EXIT SIGNS SHALL BE LOCATED AS NECESSARY TO CLEARLY INDICATE THE DIRECTION OF EGRESS TRAVEL. NO POINT SHALL BE MORE THAN 100 FEET FROM THE NEAREST VISIBLE SIGN.
6. EXIT SIGNS SHALL BE READILY VISIBLE FROM ANY DIRECTION OF APPROACH.
7. WHERE DOOR HAS LATCH AND CLOSERS 12" REQUIRED ON PUSH SIDE.
   OPERATIONAL PARTS SHALL BE OPERABLE WITH ONE HAND AND SHALL NOT REQUIRE TIGHT GRASPING, PINCHING, OR TWISTING OF THE WRIST.
   THE FORCE REQUIRED TO ACTIVATE OPERABLE PARTS SHALL BE 5 POUNDS . (110.306.4)

FLOOR PLAN NOTES

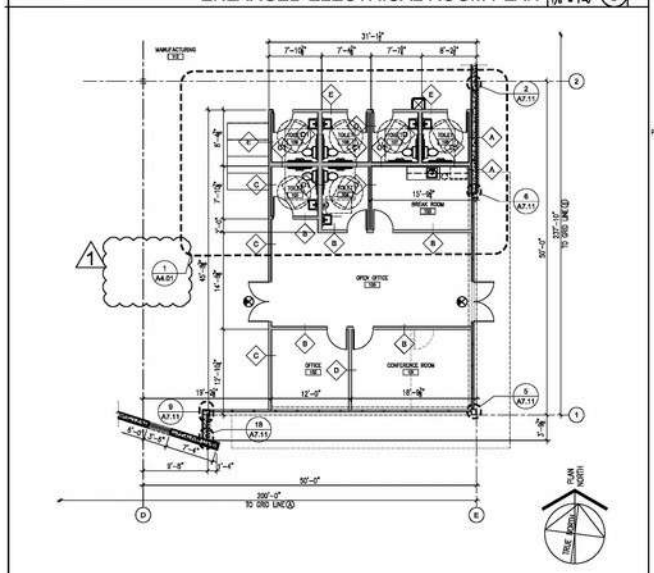
1. ALL DIMENSIONS ARE TO BE VERIFIED IN THE FIELD.
2. INTERIOR DIMENSIONS ARE TO FACE OF STUD OR GRID LINE, OR UNLESS NOTED OTHERWISE.
3. EXTERIOR DIMENSIONS ARE TO, FACE OF STUD, FACE OF CONC. PANEL, GRID LINES, OR UNLESS NOTED OTHERWISE.
4. SEE SHEET A0.01 FOR WALL TYPES, THUS MARKED IN PLAN.
5. SEE SHEET A3.10 FOR DOOR & WINDOW SCHEDULES.

SYMBOL LEGEND

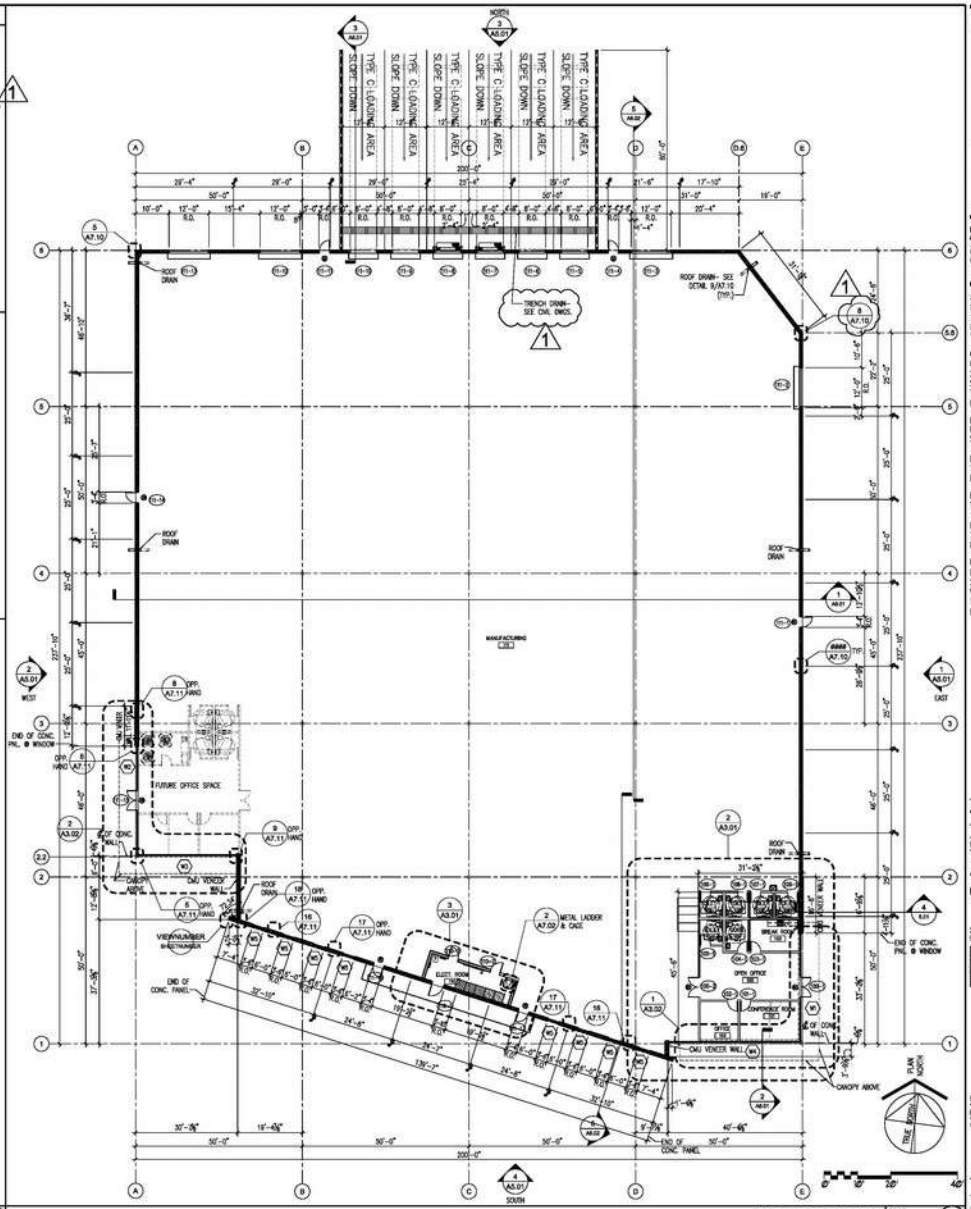
V.I.F.	VERIFY IN THE FIELD		CEILING MOUNTED/WALL MOUNTED SELF - ILLUMINATED EXIT SIGN
N.I.C.	NOT IN CONTRACT	ROOM LABEL 1	ROOM LABEL- ROOM NO & ROOM CEILING HEIGHT
	CONCRETE PANEL	D1	DOOR TAG - SEE SHEET A3.10 FOR SCHEDULE
	INTERIOR WALL PARTITION	XX	WINDOW TYPE - SEE SHEET A3.10 FOR SCHEDULE
	DOOR		ELEVATION REFERENCE
1	WALL TYPE, SEE SHEET A0.01	1	GRID LINE



ENLARGED ELECTRICAL ROOM PLAN SCALE 1/8" = 1'-0" (3)



ENLARGED FLOOR PLAN SCALE 1/16" = 1'-0" (2)



FLOOR PLAN SCALE 1/16" = 1'-0" (1)

COMBS + MIGUEL

ARCHITECTURE

MEMBERS OF THE
AMERICAN INSTITUTE OF ARCHITECTS
31294 THE OLD ROAD SUITE 220
CASTAIC, CALIFORNIA 91384
(661) 259-7600

CONSULTANTS:

STRUCTURAL ENGINEERING
K&C STRUCTURAL ENGINEERS
25142 HICHSPRING AVENUE
NEWHALL CA 91321
TEL:(661) 803-2092

CIVIL ENGINEERING
CRC ENTERPRISES
2700 BOUQUET CANYON ROAD SUITE 200
SANTA CLARITA, CA. 91350
TEL: (661) 297-2336
FAX: (661) 297-2331

MECHANICAL & PLUMBING ENGINEERING
SPEC GROUP INC.
16830 VENTURA BLVD. SUITE #326
ENCINO, CA. 91436
TEL: (818) 783-6965
FAX: (818) 783-6996

ELECTRICAL ENGINEERING
TEK ENGINEERING
16830 VENTURA BLVD. SUITE #326
ENCINO, CA. 91436
TEL: (818) 783-6965
FAX: (818) 783-6996

LANSCAPE ARCHITECT
LANDSCAPE DEVELOPMENT, INC.
28447 WITHERSPOON PARKWAY
VALENCIA, CA. 91355
TEL: (661) 295-1970
FAX: (661) 295-1969

PLAN CHECK RESUBMITTAL	03.18.2016
PLAN CHECK SUBMITTAL	11.16.2015

AVALON BUSINESS CENTER

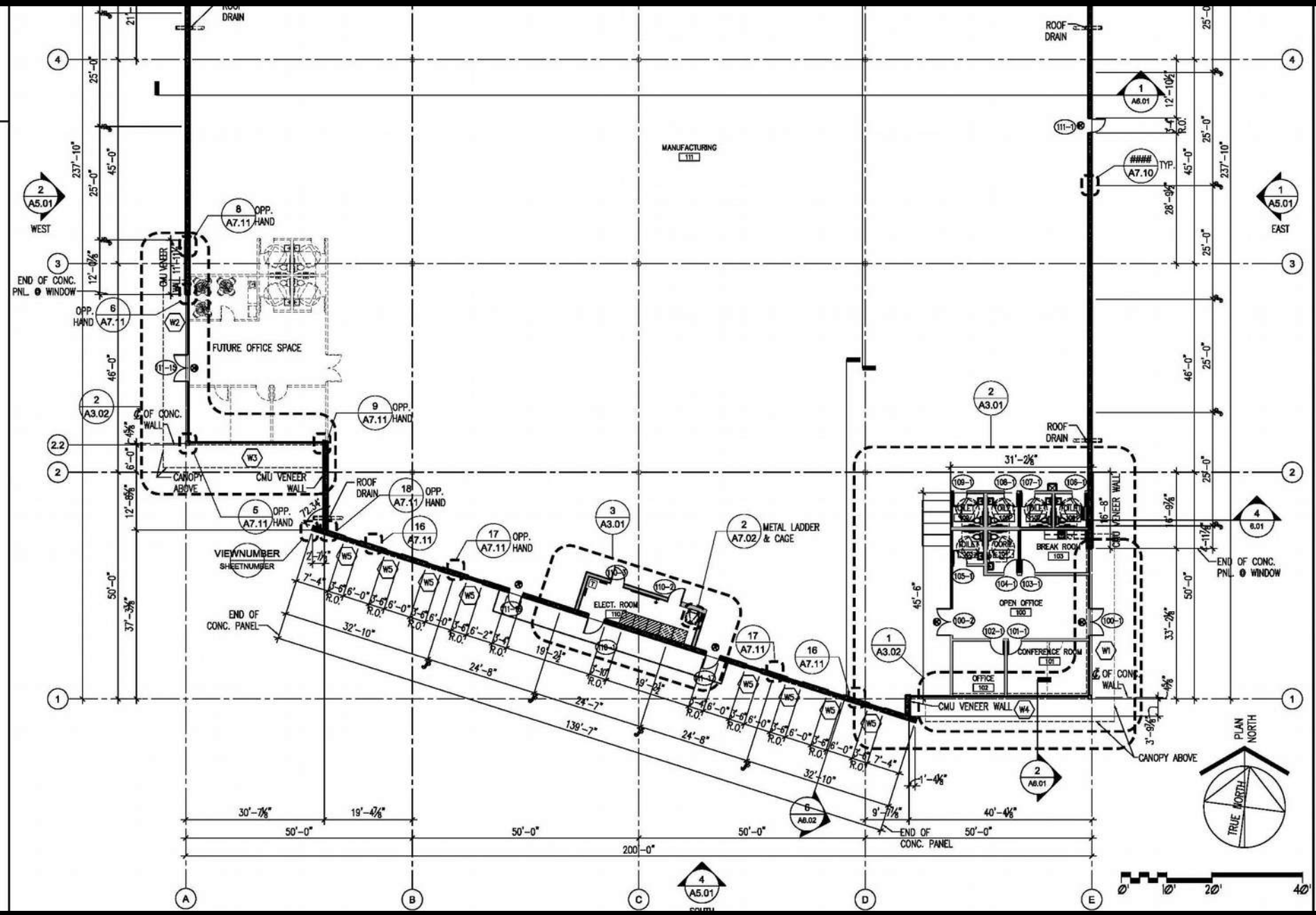
28606 HASLEY CANYON ROAD
CASTAIC, CA. 91384

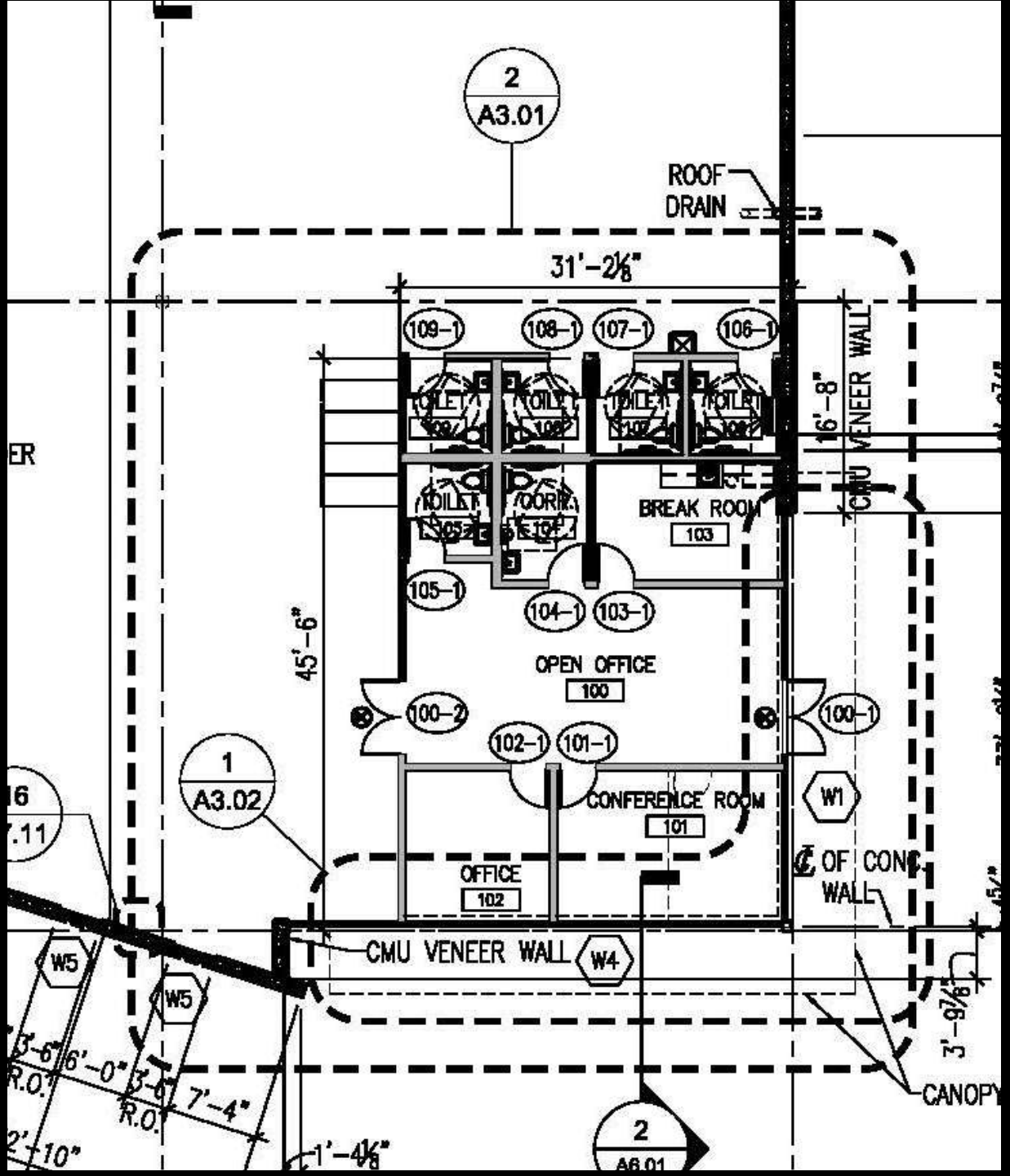
PLAN CHECK NO:
BL-1511170080

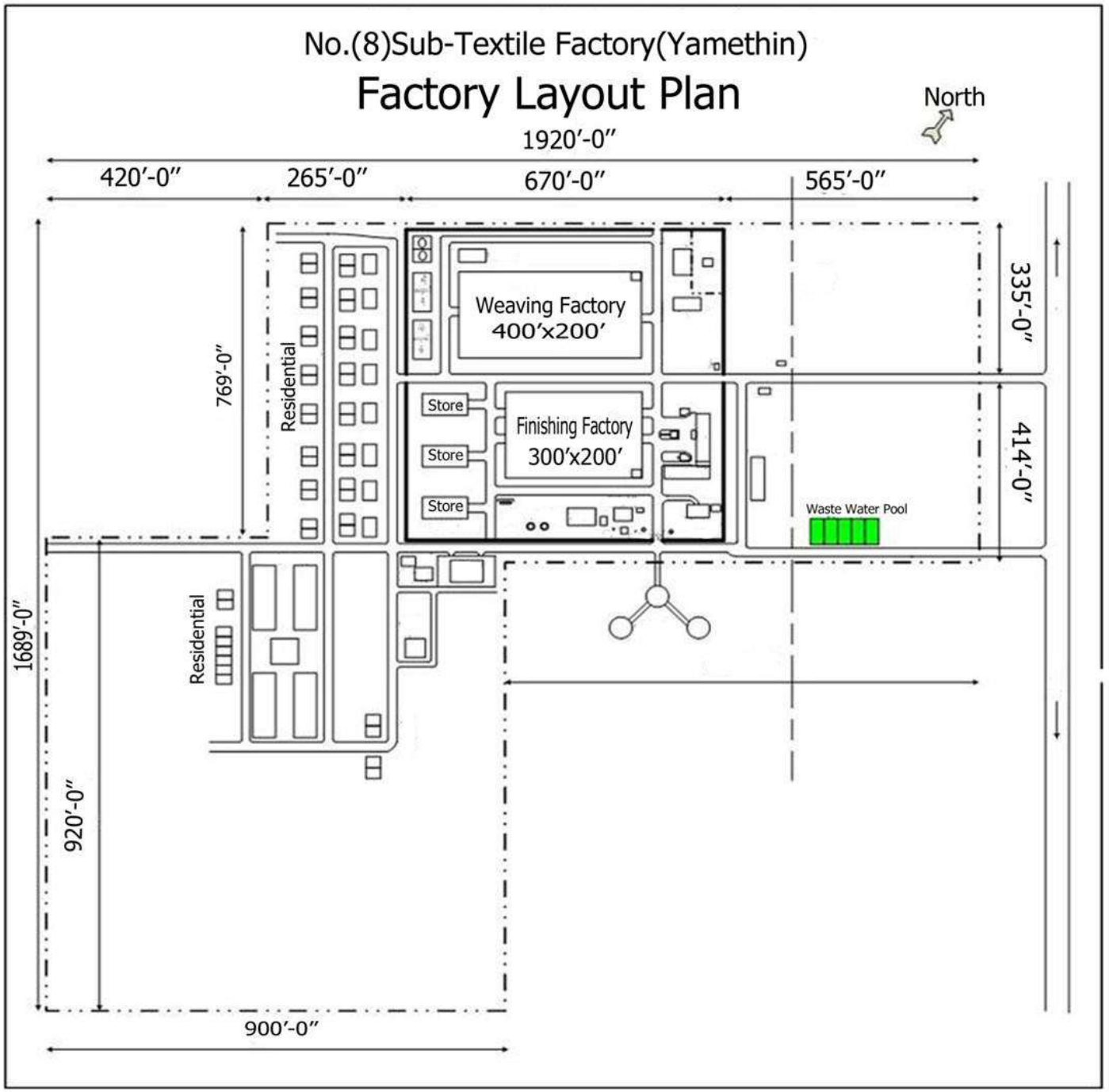
LICENSED ARCHITECT JAMES M. COMBS C-11381 RENEWAL 08/31/2017 STATE OF CALIFORNIA

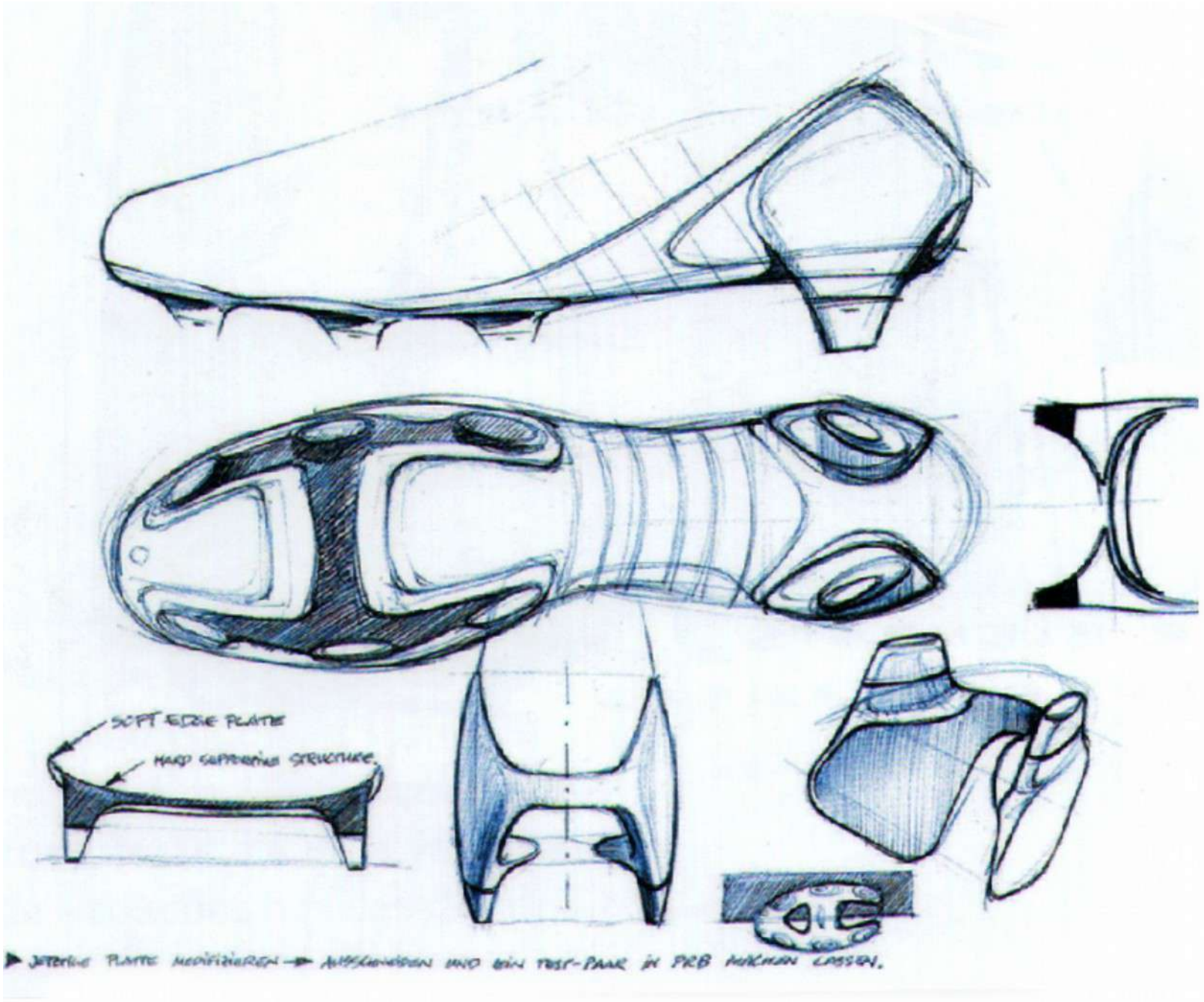
PLOT DATE: 02.23.2016
PROJECT NO. 1508
14 OF 68

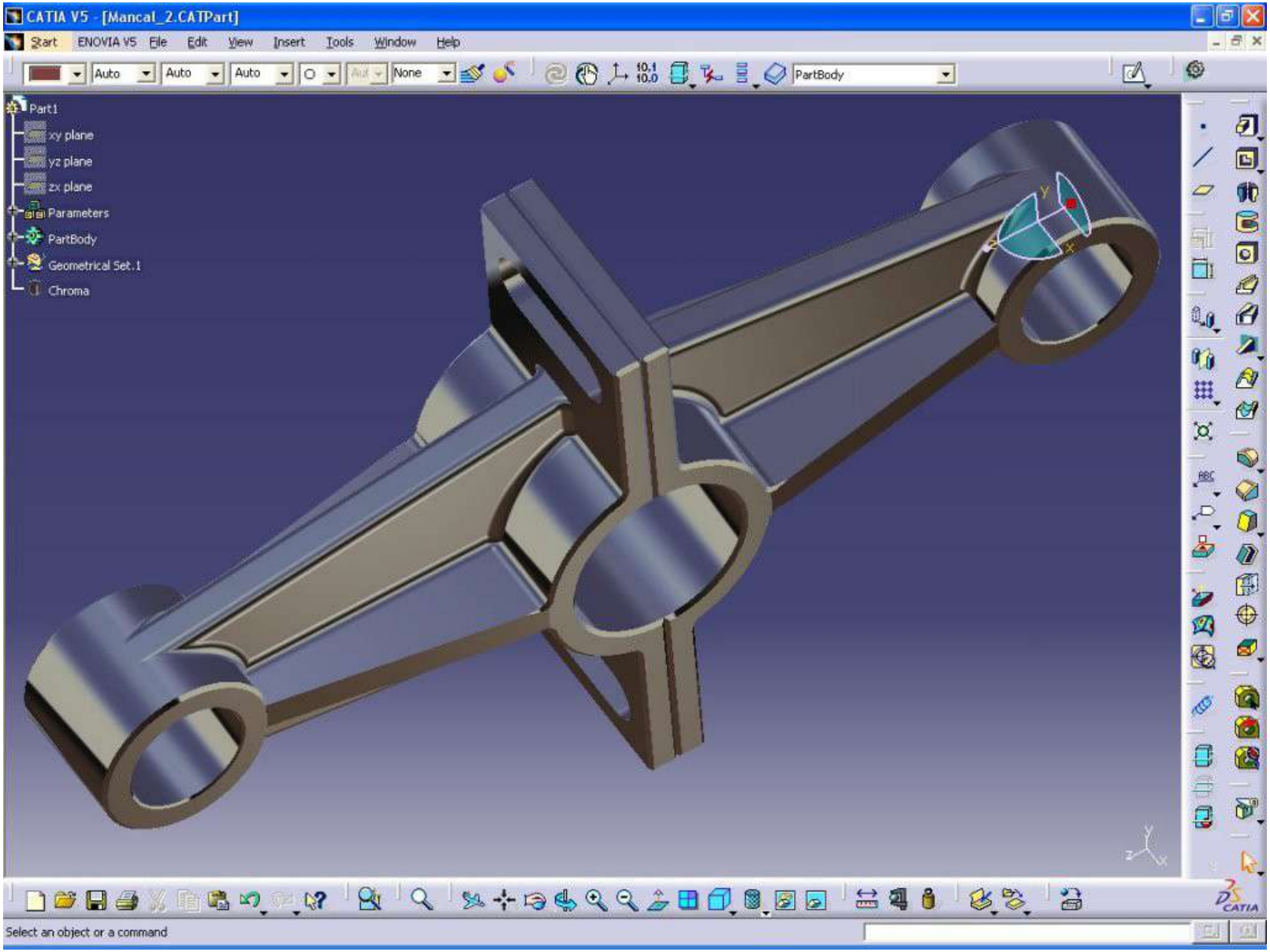
A3.01
FLOOR PLAN


ROOF DRAIN
MANUFACTURING
111
FUTURE OFFICE SPACE
CMU VENEER WALL
CANOPY ABOVE
ELECT. ROOM
METAL LADDER & CAGE
BREAK ROOM
OPEN OFFICE
CONFERENCE ROOM
OFFICE
END OF CONC. PANEL
END OF CONC. PNL. @ WINDOW
CMU VENEER WALL
CANOPY ABOVE
PLAN NORTH
TRUE NORTH
WEST
EAST
VIEWNUMBER
SHEETNUMBER
200'-0"
0' 10' 20' 40'

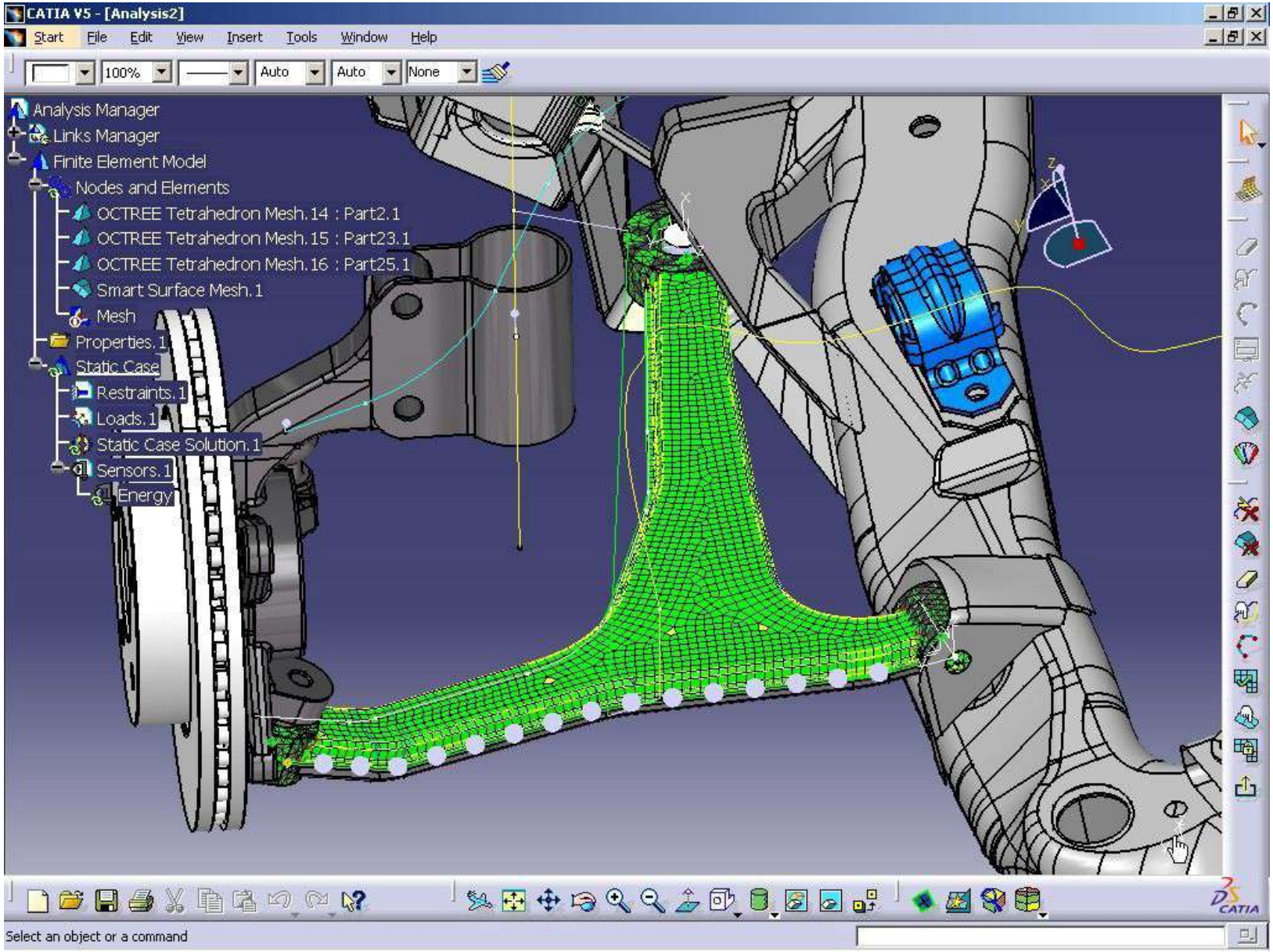

2
A3.01
ROOF
DRAIN
31'-2⅛"
109-1
108-1
107-1
106-1
TOILET
TOILET
TOILET
TOILET
16'-8"
CMU VENEER WALL
TOILET
CORR.
BREAK ROOM
103
105-1
104-1
103-1
45'-6"
OPEN OFFICE
100
100-2
100-1
102-1
101-1
1
A3.02
CONFERENCE ROOM
101
W1
OFFICE
102
℄ OF CONC
WALL
CMU VENEER WALL
W4
W5
W5
3'-6"
6'-0"
3'-6"
R.O.
R.O.
7'-4"
2'-10"
1'-4⅛"
3'-9⅞"
CANOPY
2
A6.01


No.(8)Sub-Textile Factory(Yamethin)
Factory Layout Plan
North
1920′-0″
420′-0″
265′-0″
670′-0″
565′-0″
335′-0″
414′-0″
769′-0″
1689′-0″
920′-0″
900′-0″
Residential
Residential
Weaving Factory
400′x200′
Store
Store
Store
Finishing Factory
300′x200′
Waste Water Pool


SOFT EDGE PLATTE
HARD SUPPORTING STRUCTURE



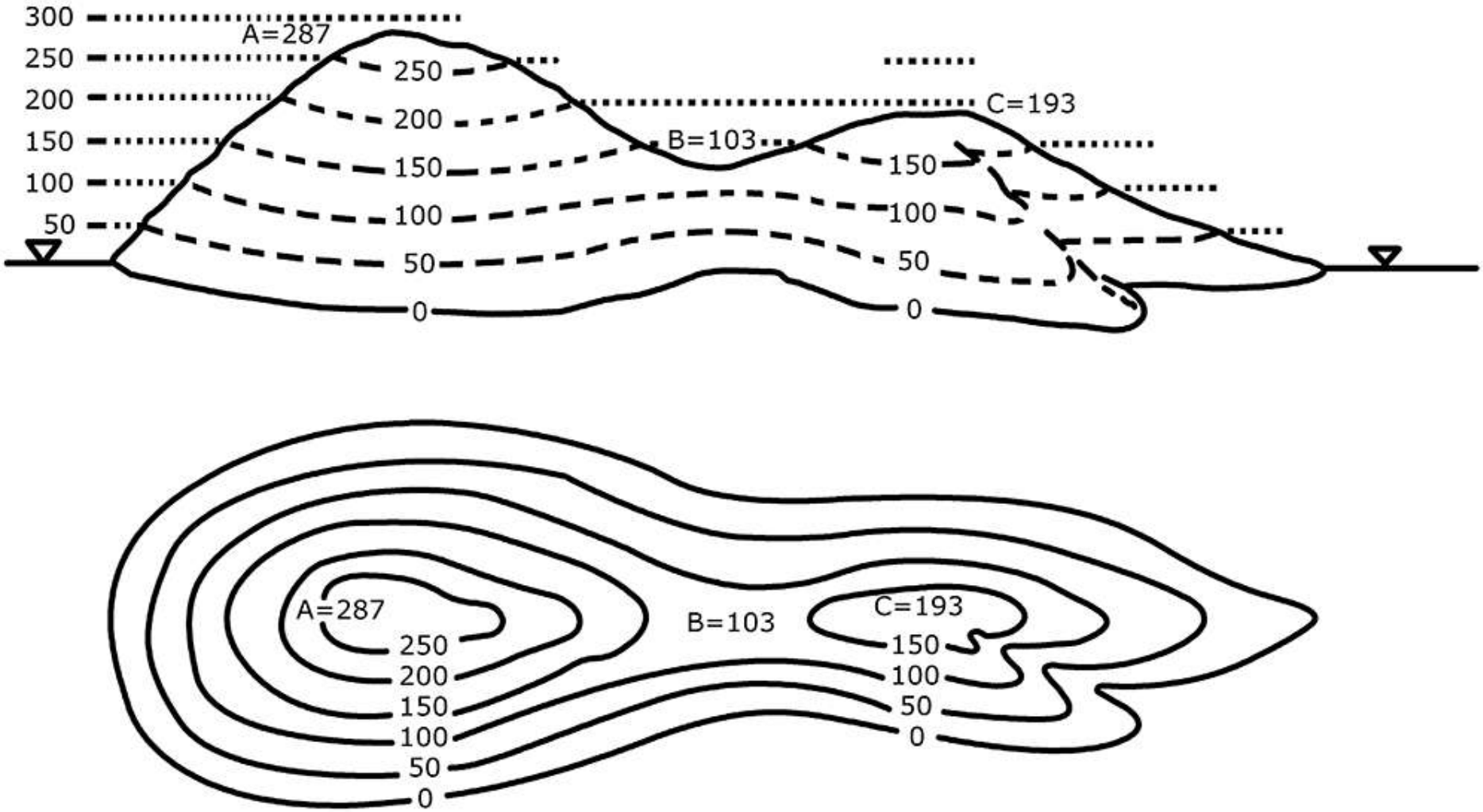
3. Pensamento tridimensional do objeto.

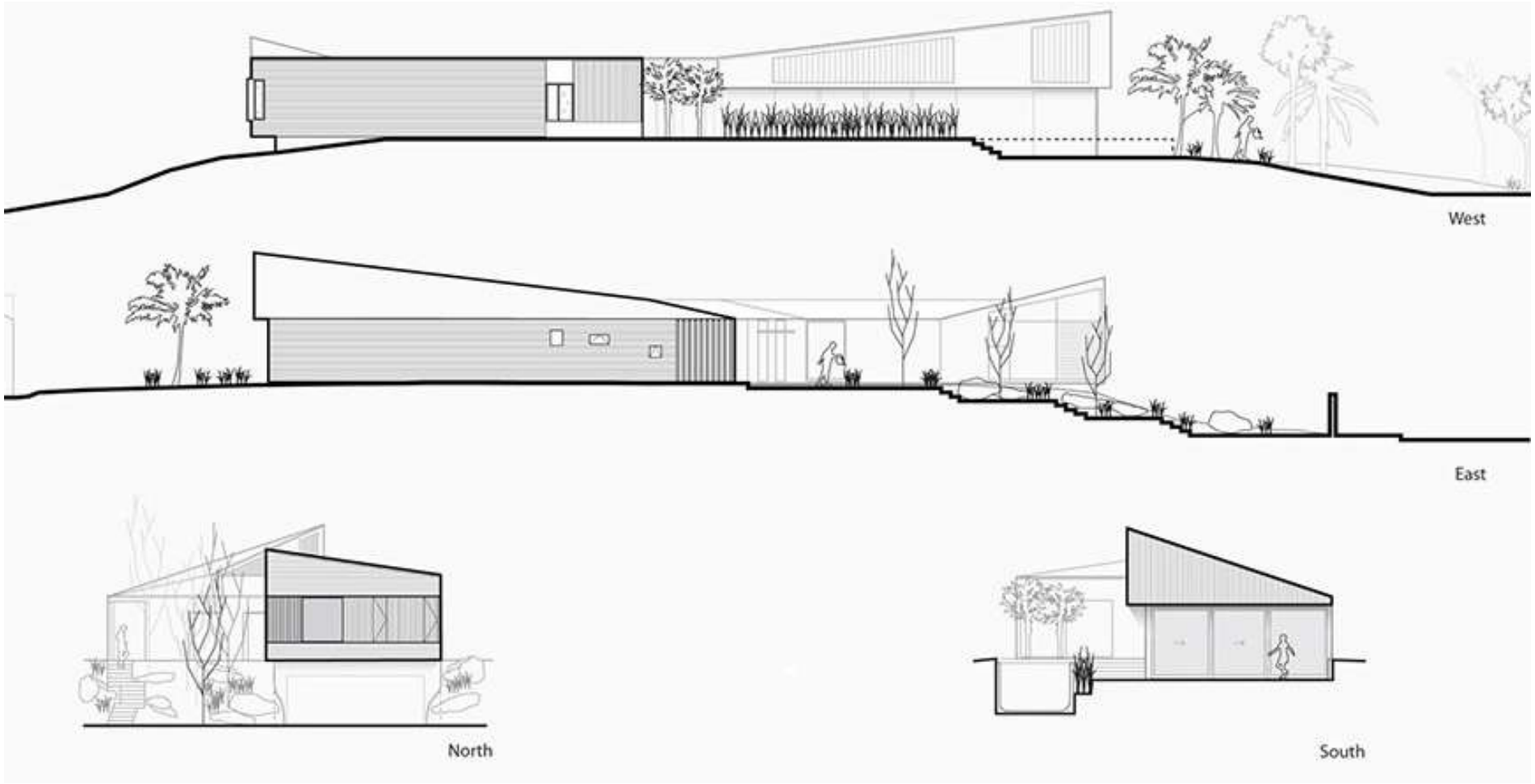


3. Pensamento tridimensional do objeto.

ADM	ARQ	BAN	BIO	BQI	BVE
CCO	CEL	CIS	CIV	COM	DAD
DAN	DIR	DPE	EAM	ECD	ECO
EDU	EFG	EFI	EIN	ELT	ENA
ENF	ENG	ENQ	ENT	EPR	ERU
ESM	EST	FIP	FIS	FIT	GEO
HIS	INF	LET	MAT	MBI	MEC
MED	MOB	NUT	QUI	SEC	SOL
TAL	VER	VET	ZOO		

4. Suporte para outras disciplinas de graduação
(CIV, EAL, ARQ, EAM, MEC, etc...)


300
250
200
150
100
50
A=287
250
200
150
100
50
0
B=103
C=193
150
100
50
0
A=287
250
200
150
100
50
0
B=103
C=193
150
100
50
0


West
East
North
South

ARQ 201 – Representação Gráfica para Engenharias

TEORIA DA REPRESENTAÇÃO

- METODOLOGIA DE REPRESENTAÇÃO NO ESPAÇO;
- ELEMENTOS DA REPRESENTAÇÃO;
- PROBLEMAS ENVOLVENDO A GEOMETRIA E NOÇÕES DE PROJETO.

PRÁTICA DE PRANCHETA

- NORMATIZAÇÃO TÉCNICA PARA DESENHO DE PEÇAS;
- METODOLOGIA TRADICIONAL DE DESENHO;
- PRANCHETA, LAPISEIRA E ESQUADROS;

PRÁTICA DE COMPUTADOR

- NORMATIZAÇÃO TÉCNICA PARA DESENHO DE CONSTRUÇÕES;
- USO DO SOFTWARE AUTOCAD®;
- COMPUTADORES PESSOAIS OU DO LABORATÓRIO;

Monitoria

Objetivos

- Auxílio no desenvolvimento de trabalhos práticos
- Auxílio no desenvolvimento de exercícios teóricos
- Terapia ocasional

Horários

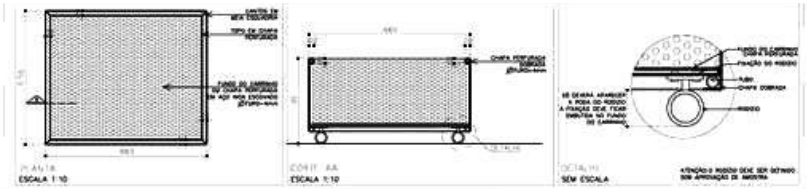
(a confirmar)

PLANO DE ENSINO/ Conteúdo programático/ Programa da disciplina

(Disponível no PVAnet)

ARQ 201

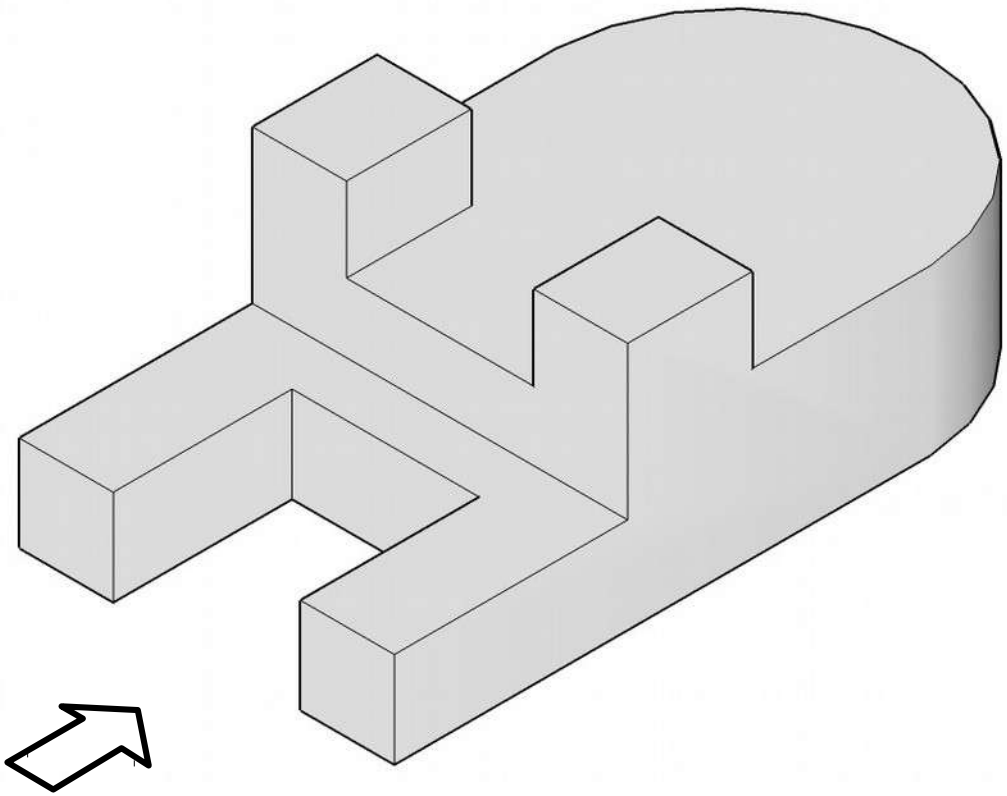
Perspectiva isométrica

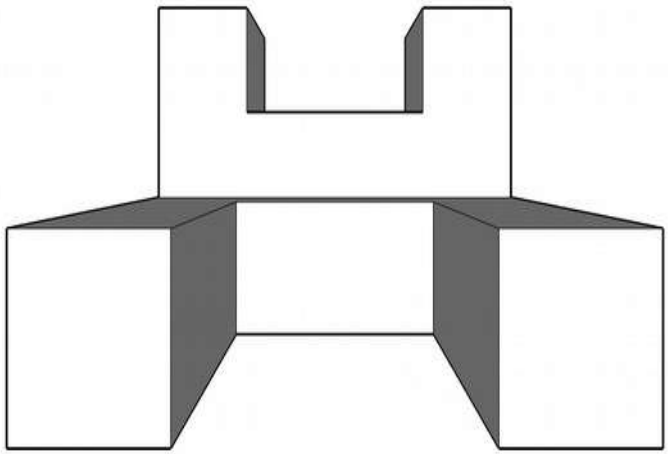


CONCEITOS INICIAIS

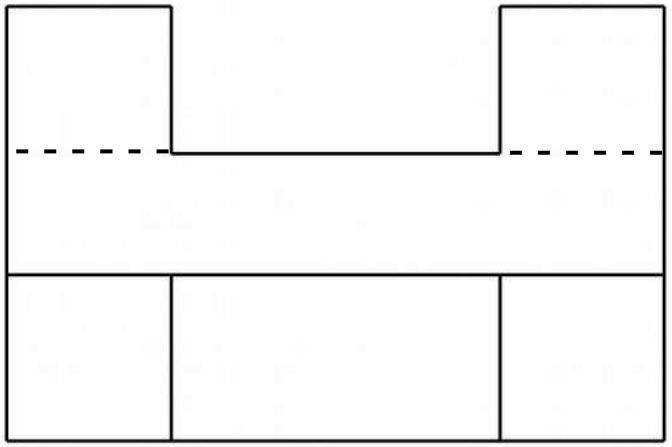
Perspectiva Isométrica

Diferença entre projeção e perspectiva

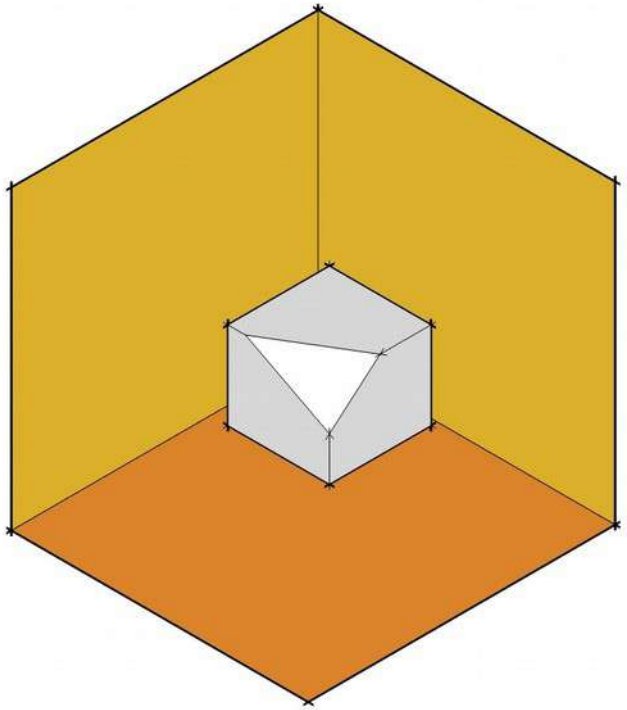




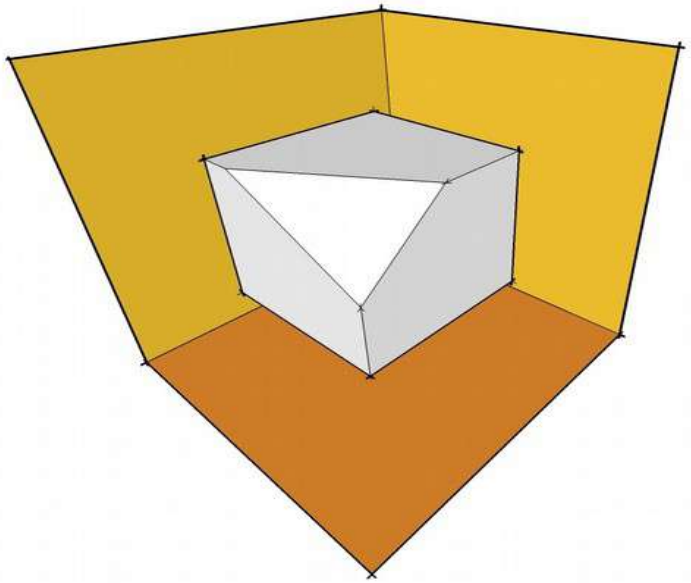
Perspectiva simples



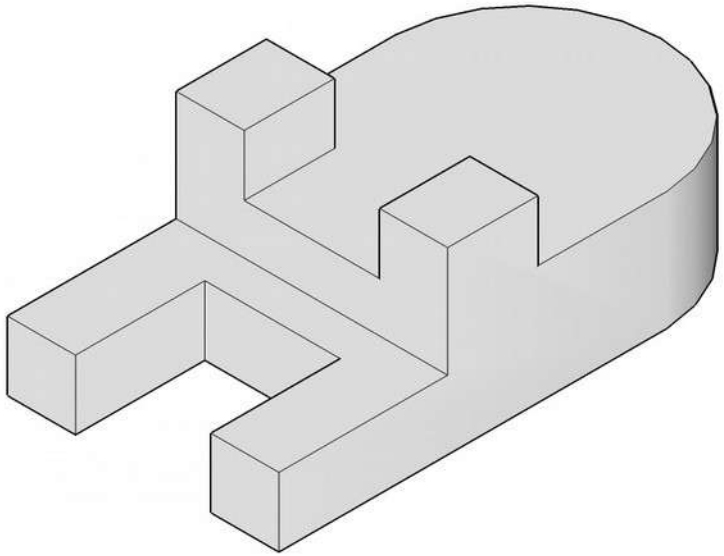
Projeção cilíndrica ortogonal

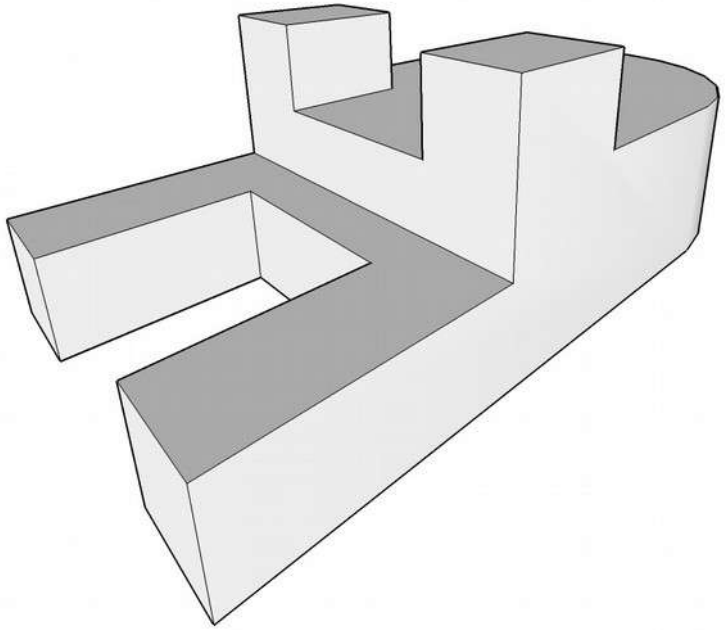


Perspectiva Isométrica

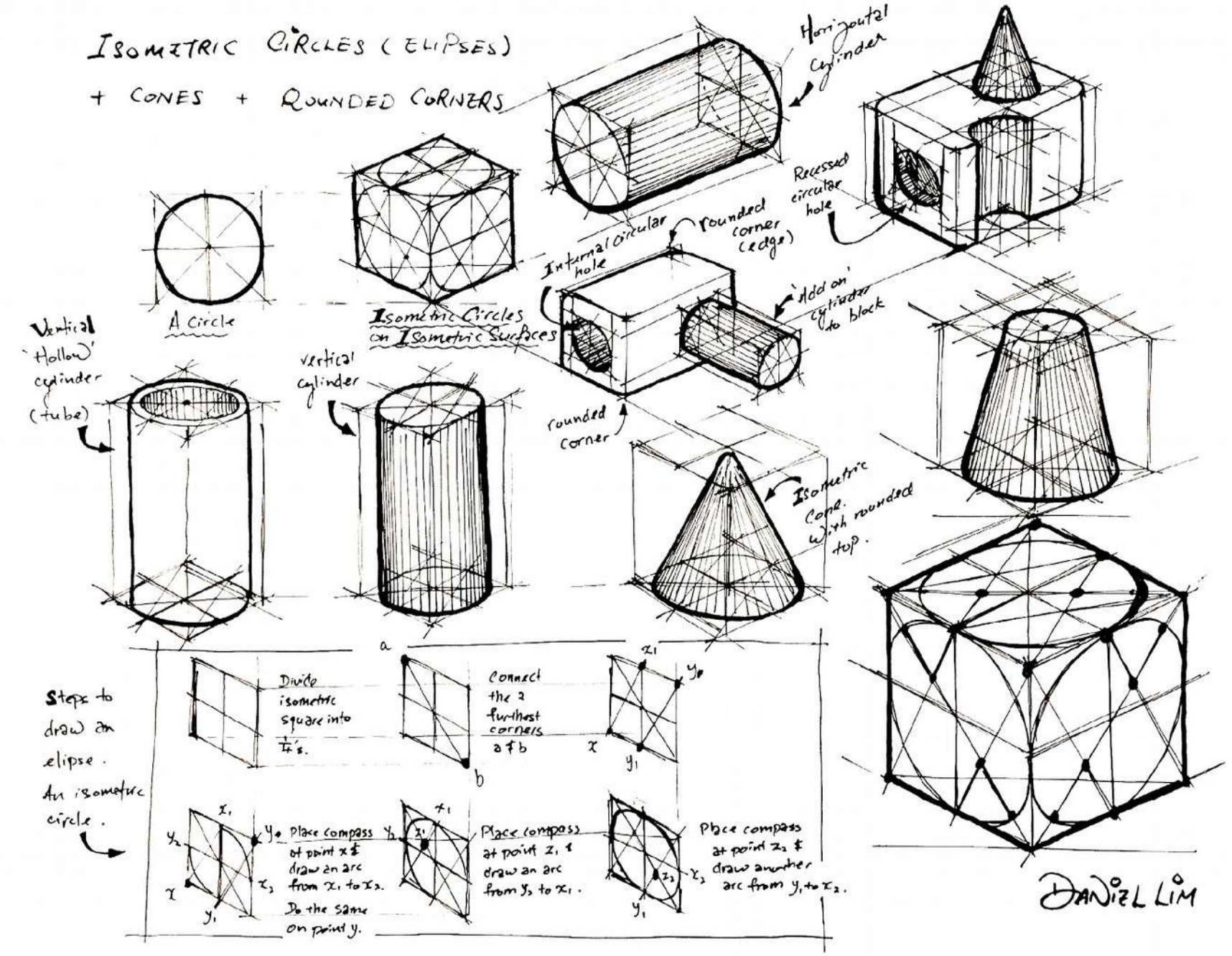


Perspectiva Axonométrica

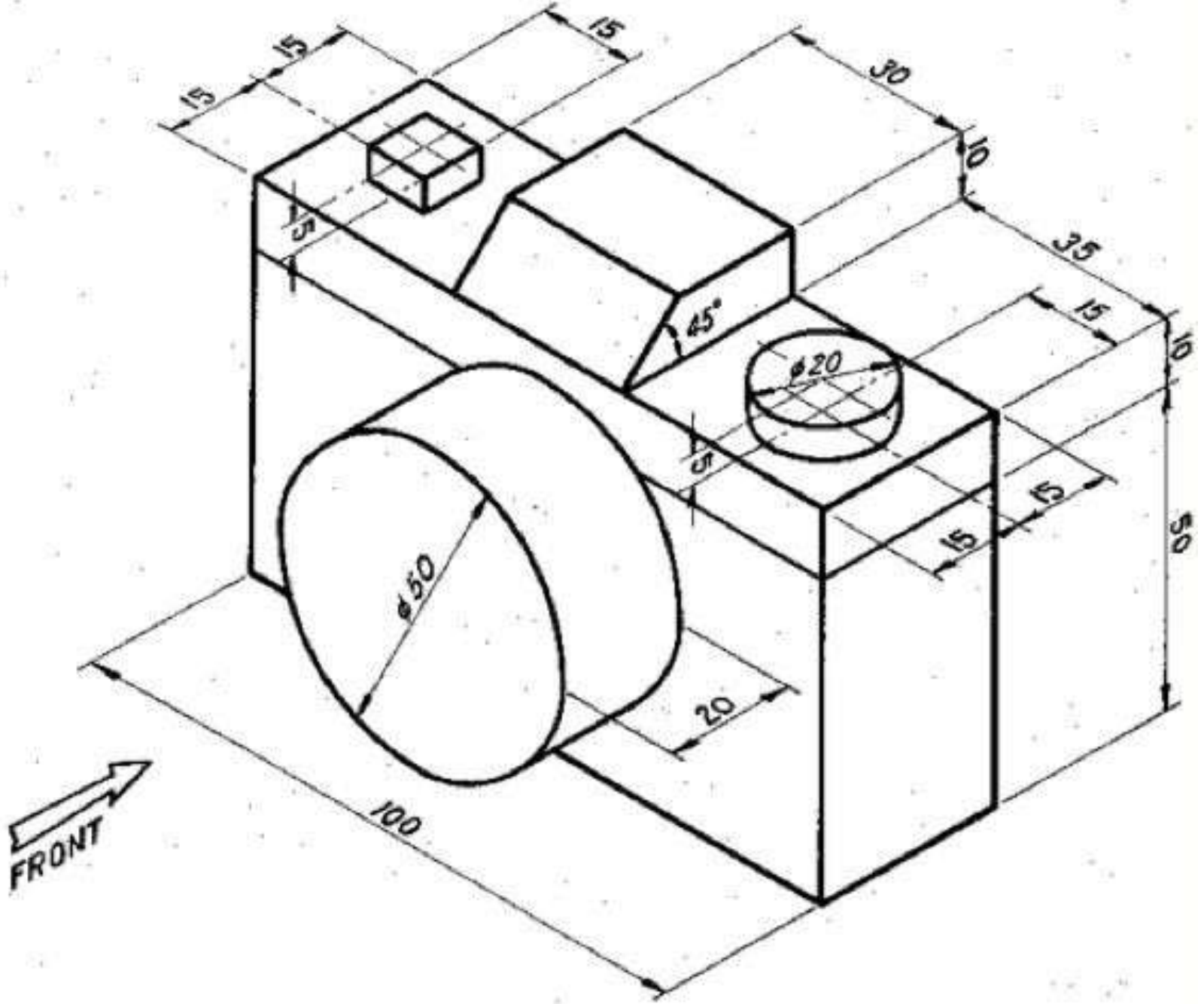




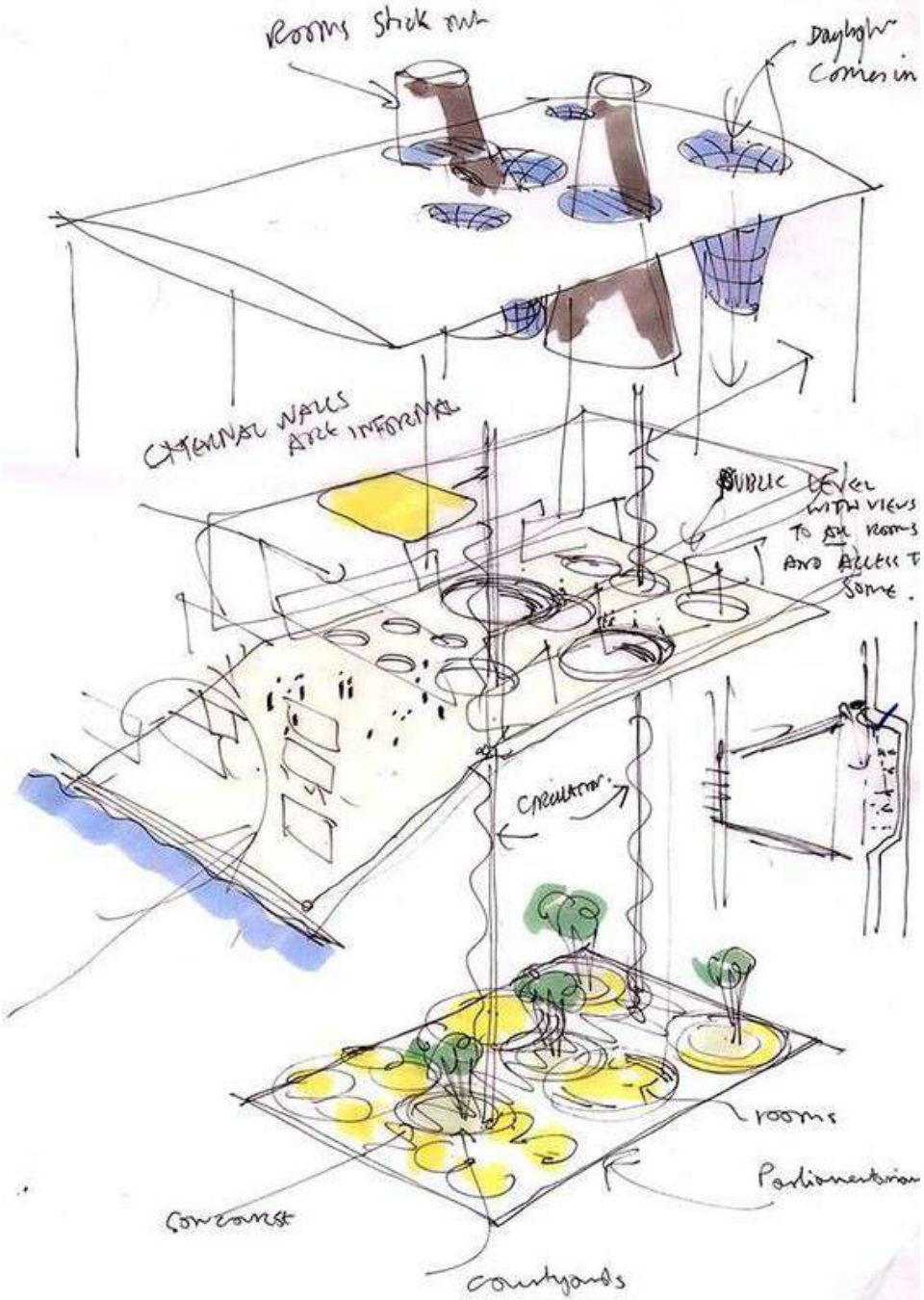
Perspectiva isométrica: estudos de geometrias



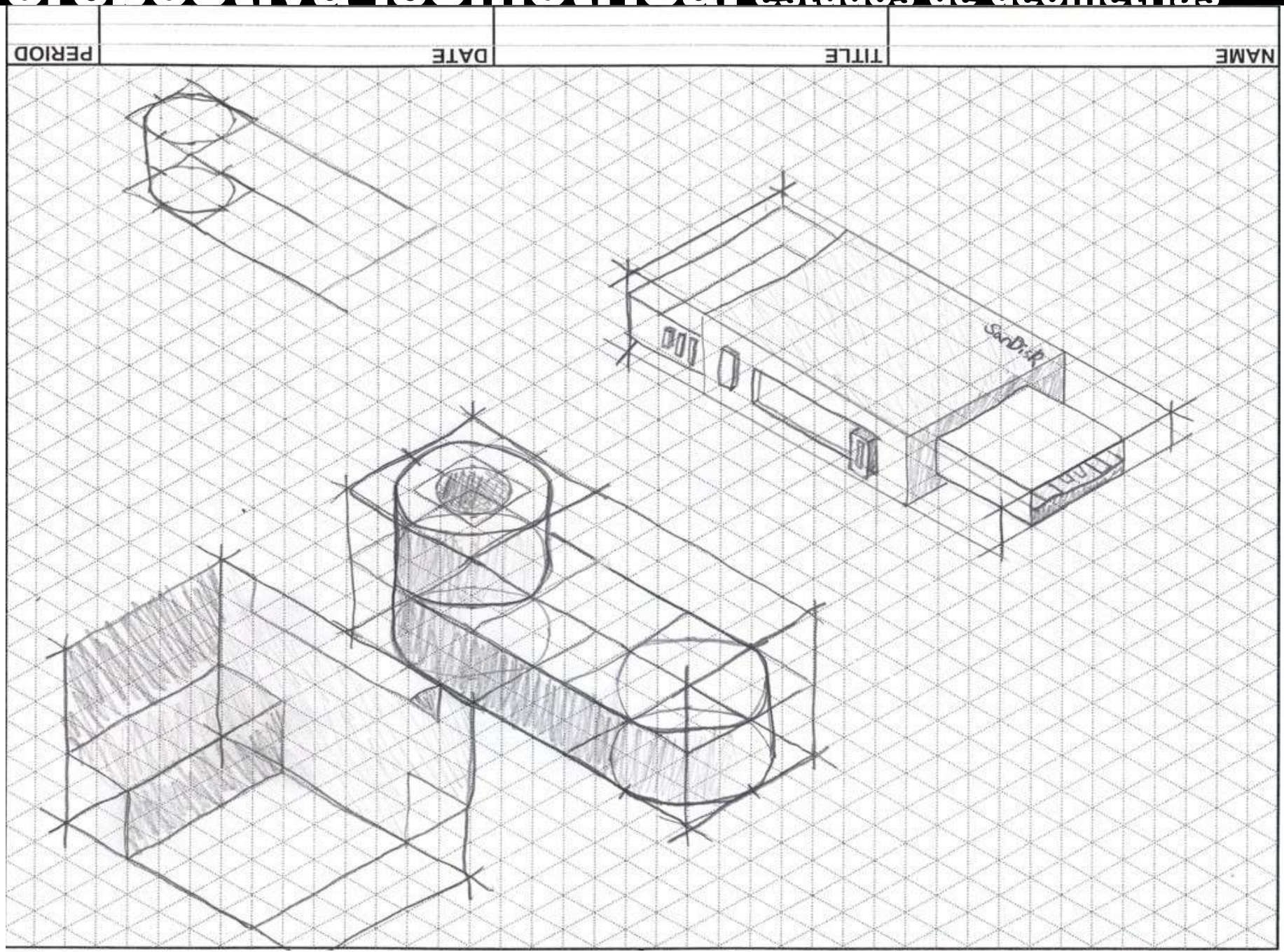
Perspectiva isométrica: estudos de geometrias



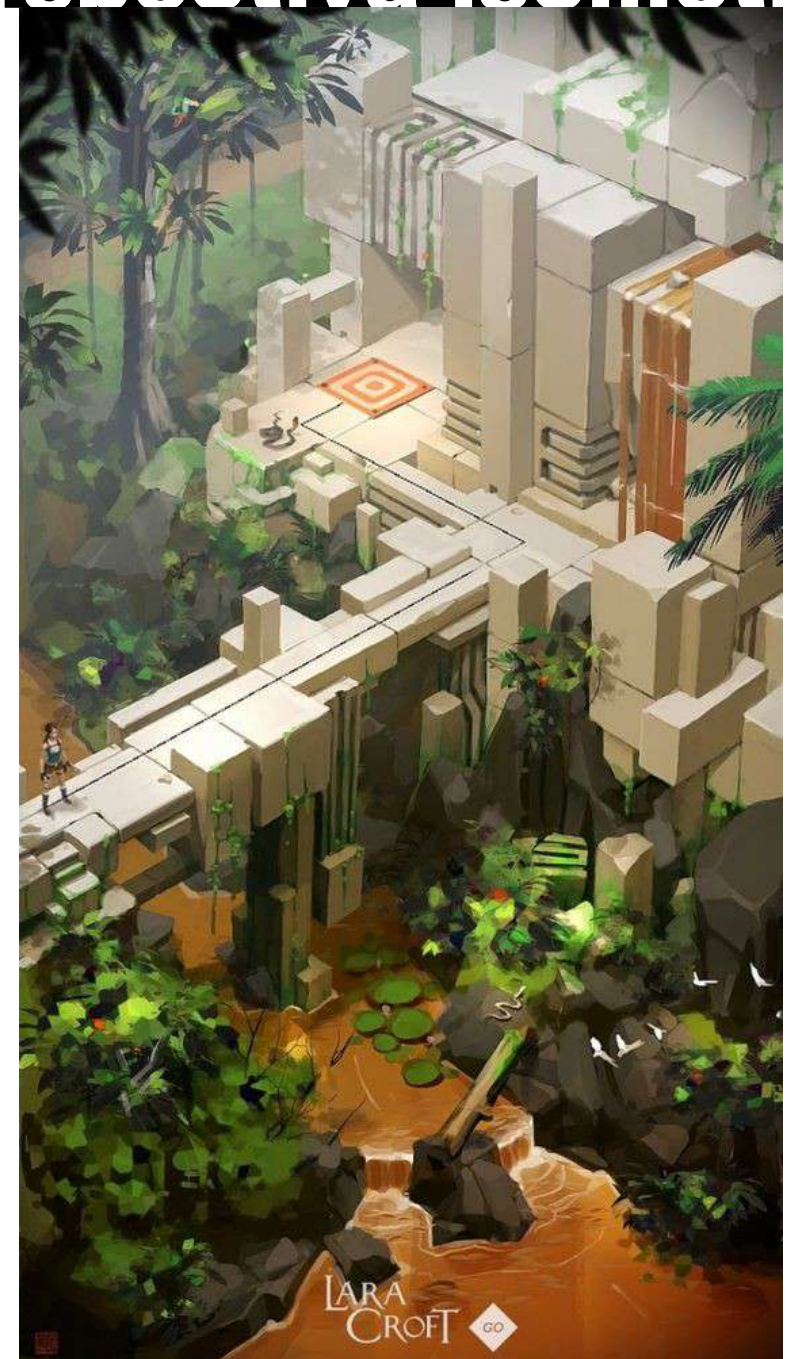
Perspectiva isométrica: estudos de geometrias



Perspectiva isométrica: estudos de geometrias

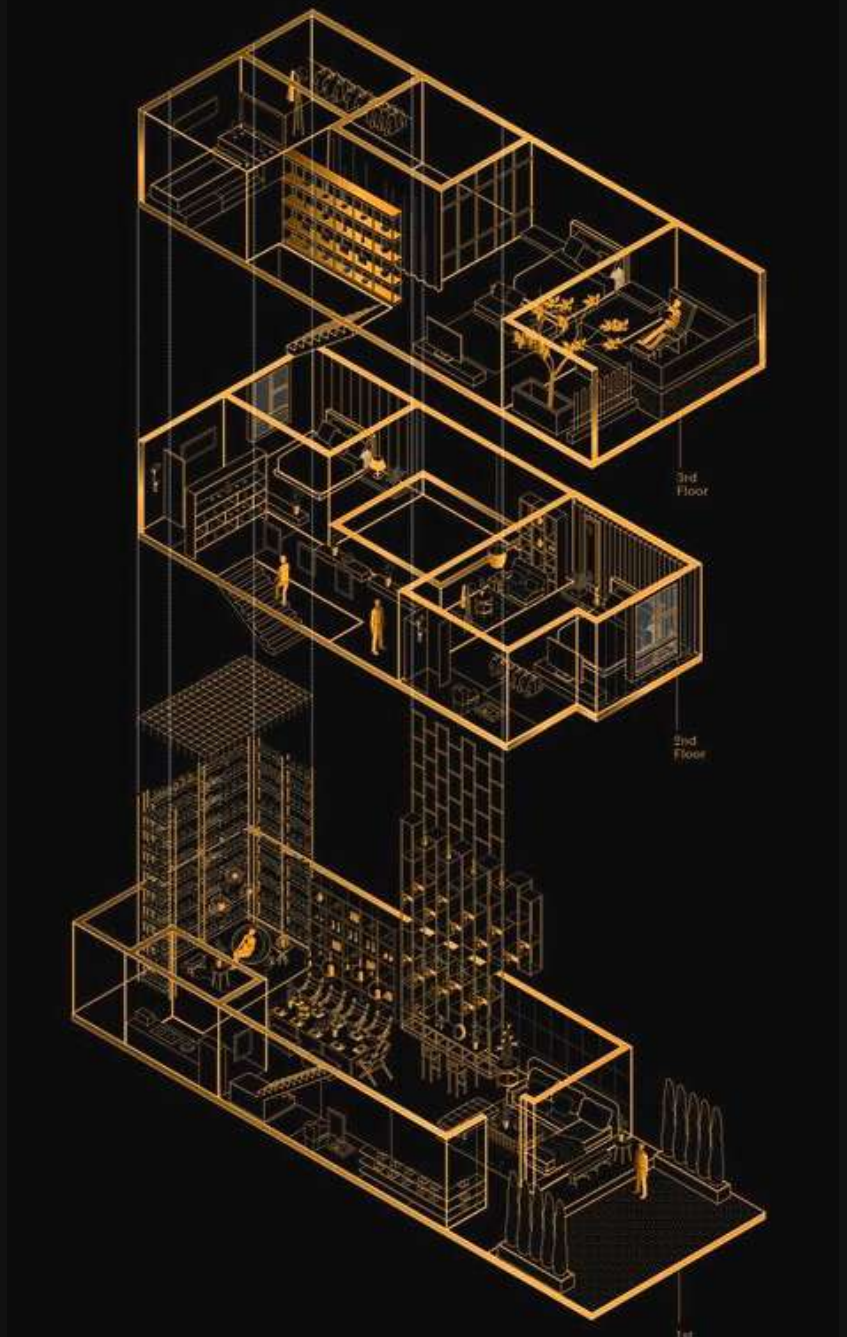


Perspectiva isométrica: estudos de geometrias

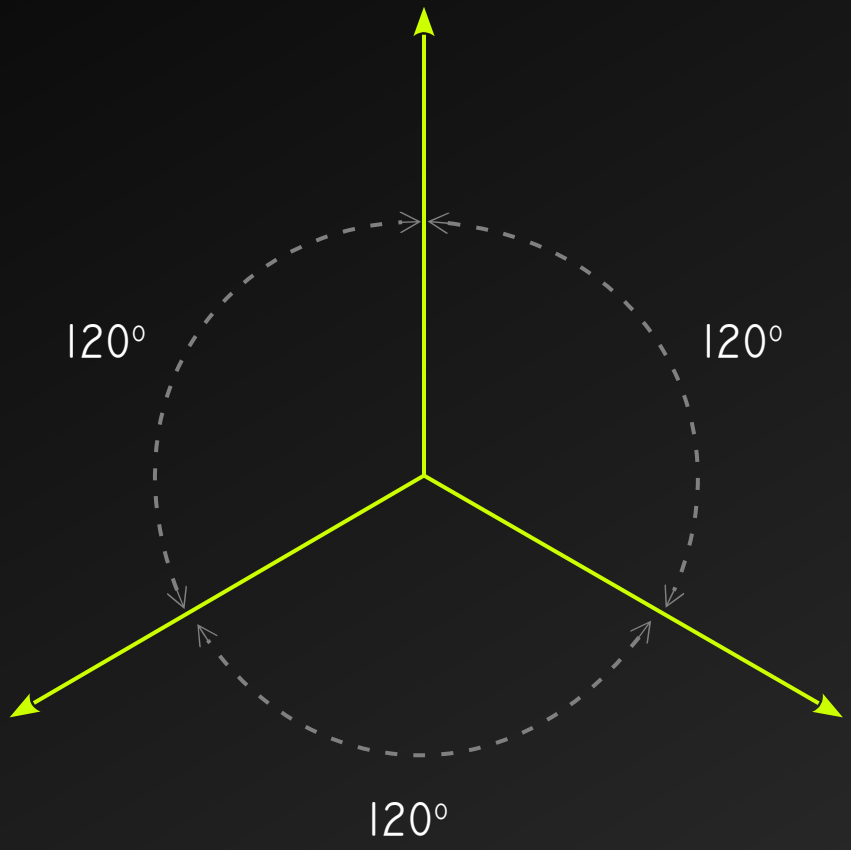


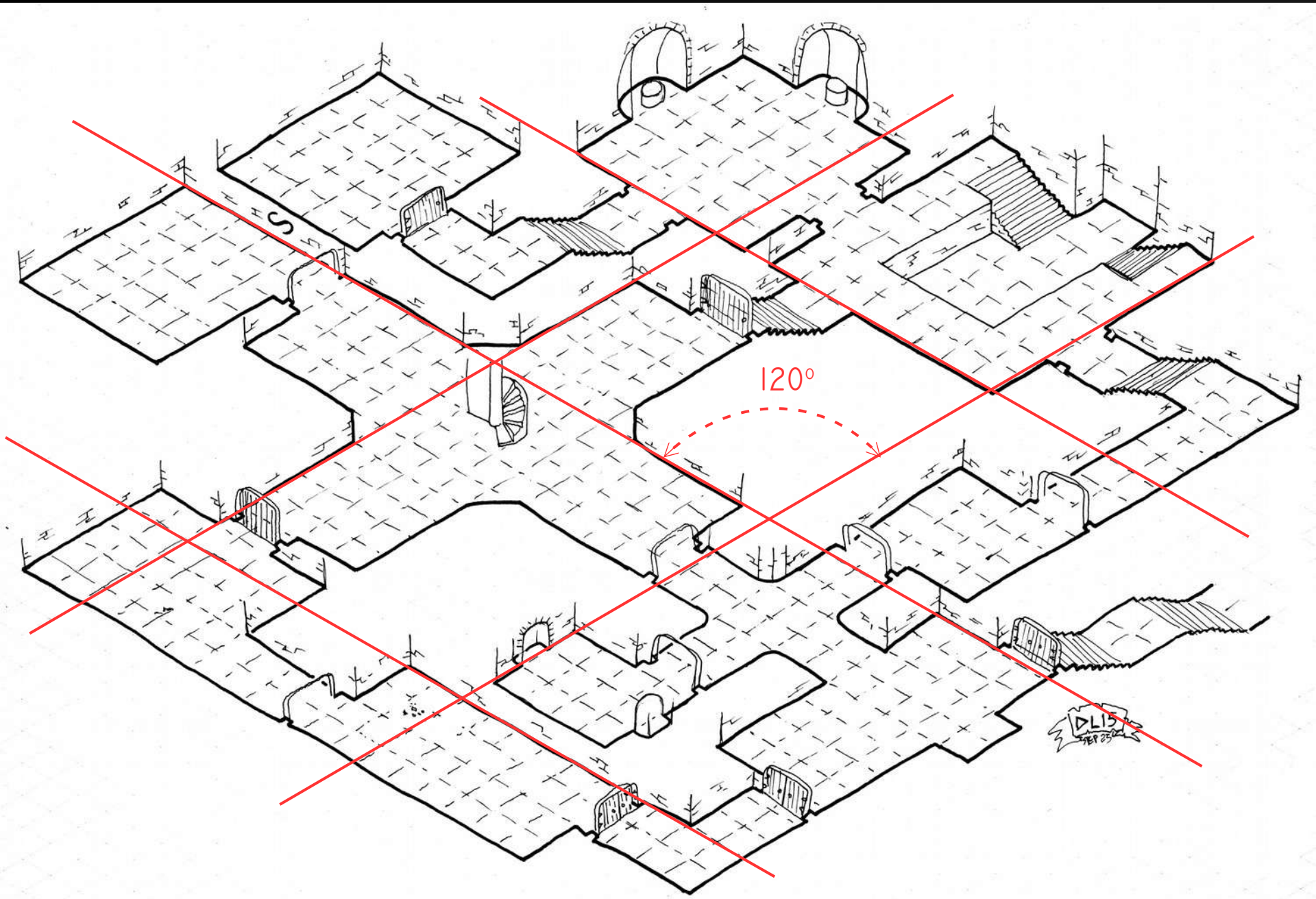


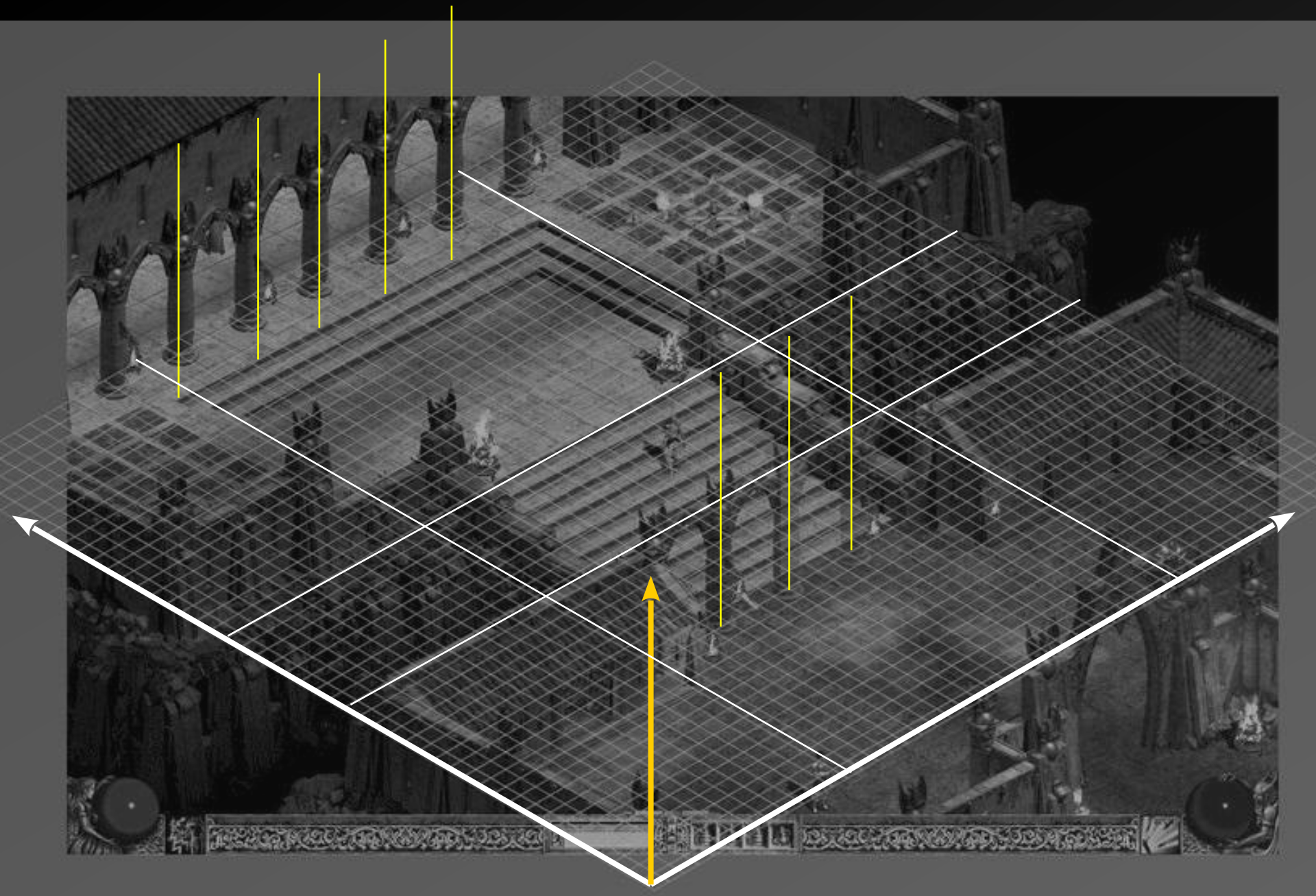
Perspectiva isométrica: estudos de geometrias



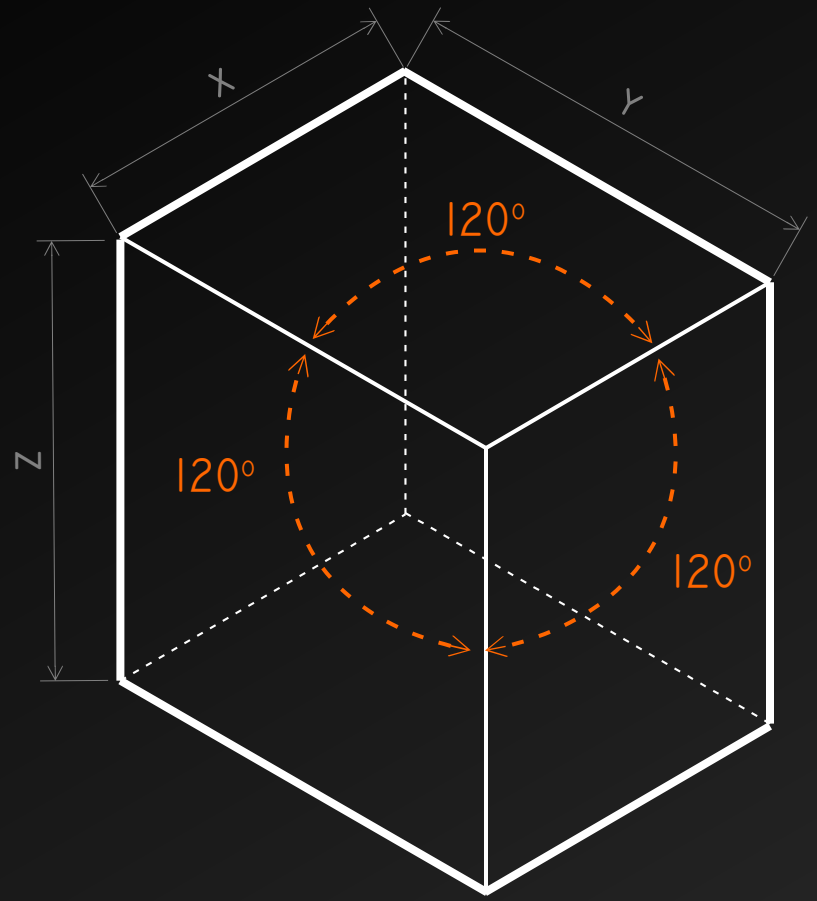
Perspectiva isométrica



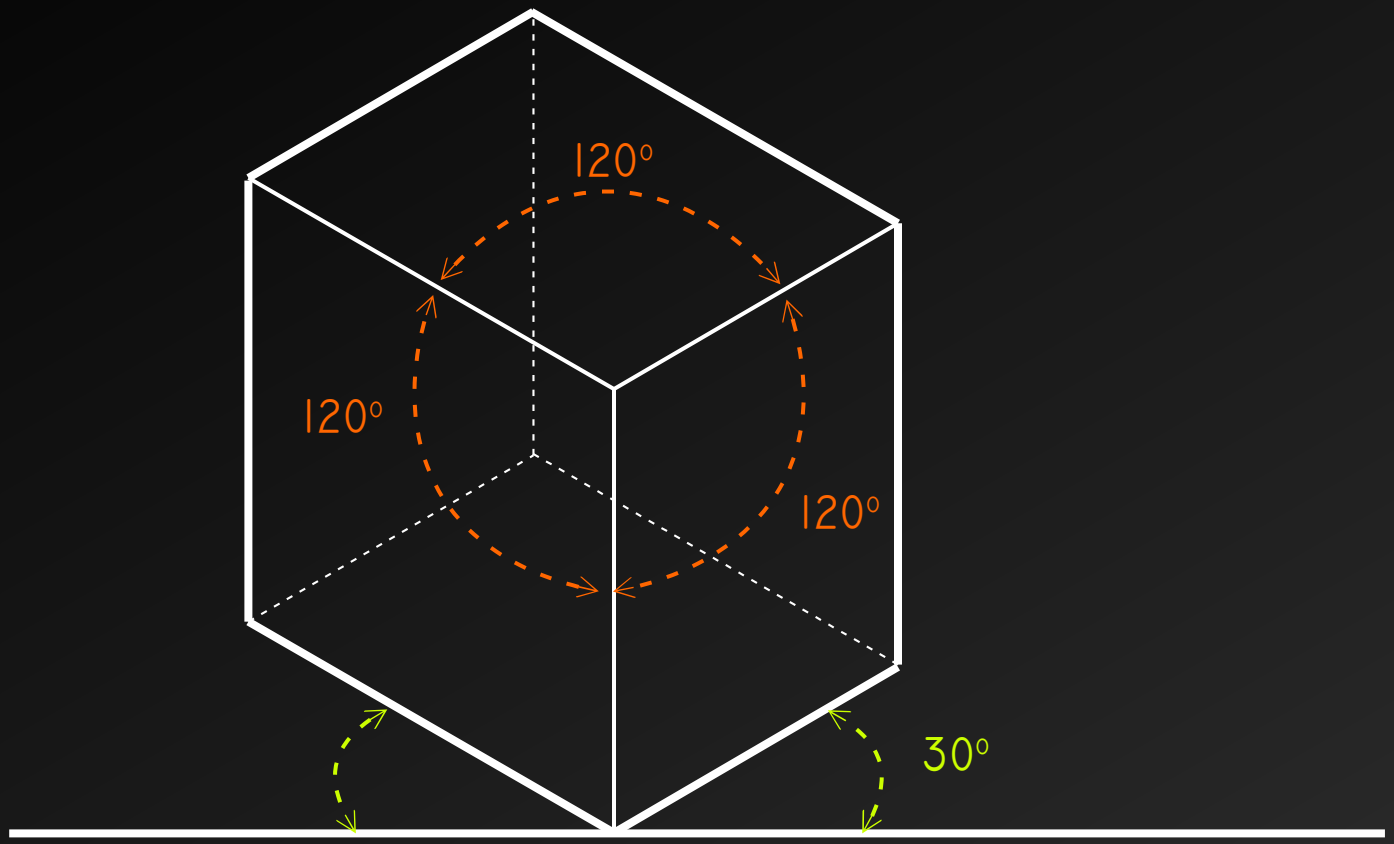

120°



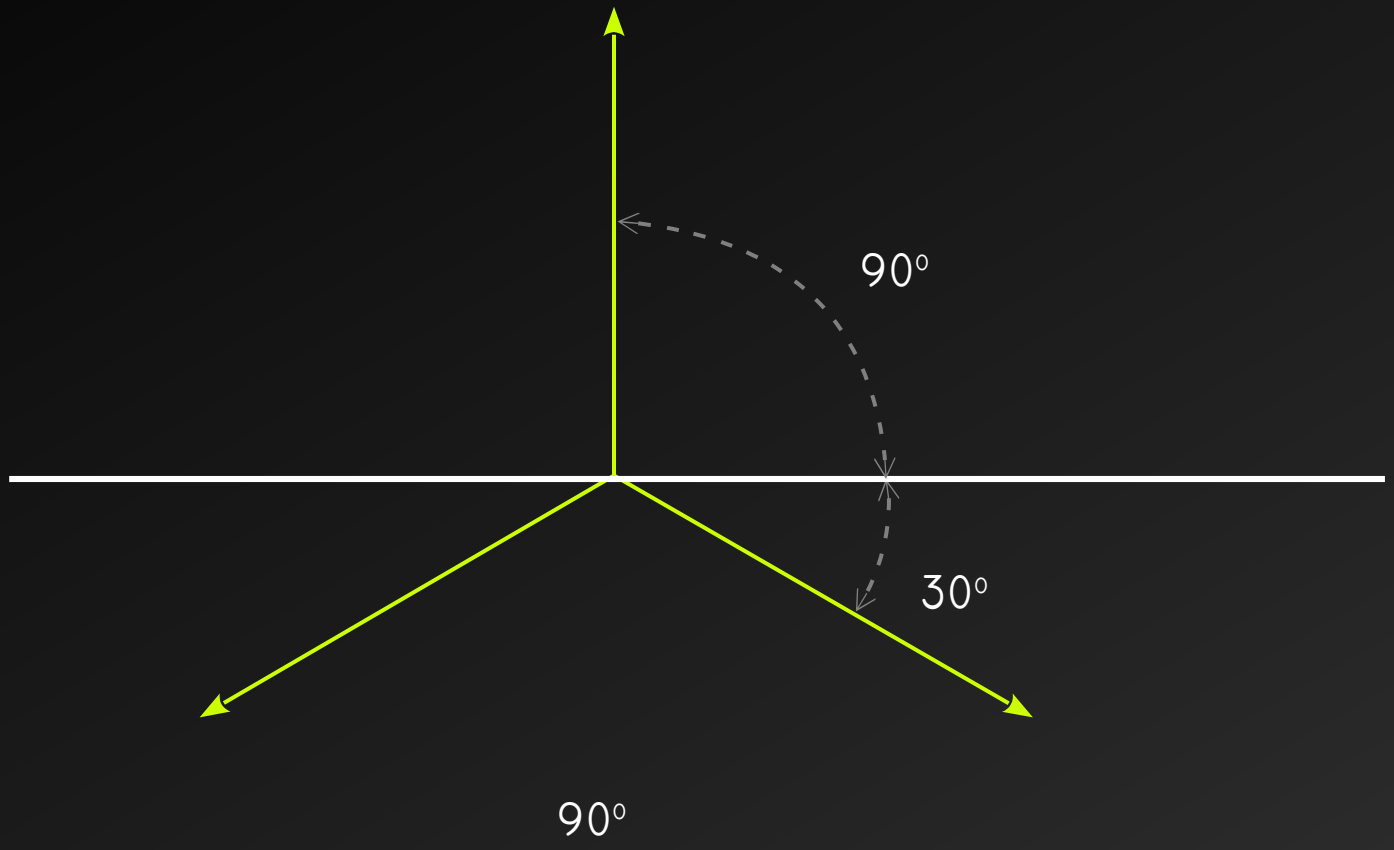
Representação em Perspectiva Isométrica



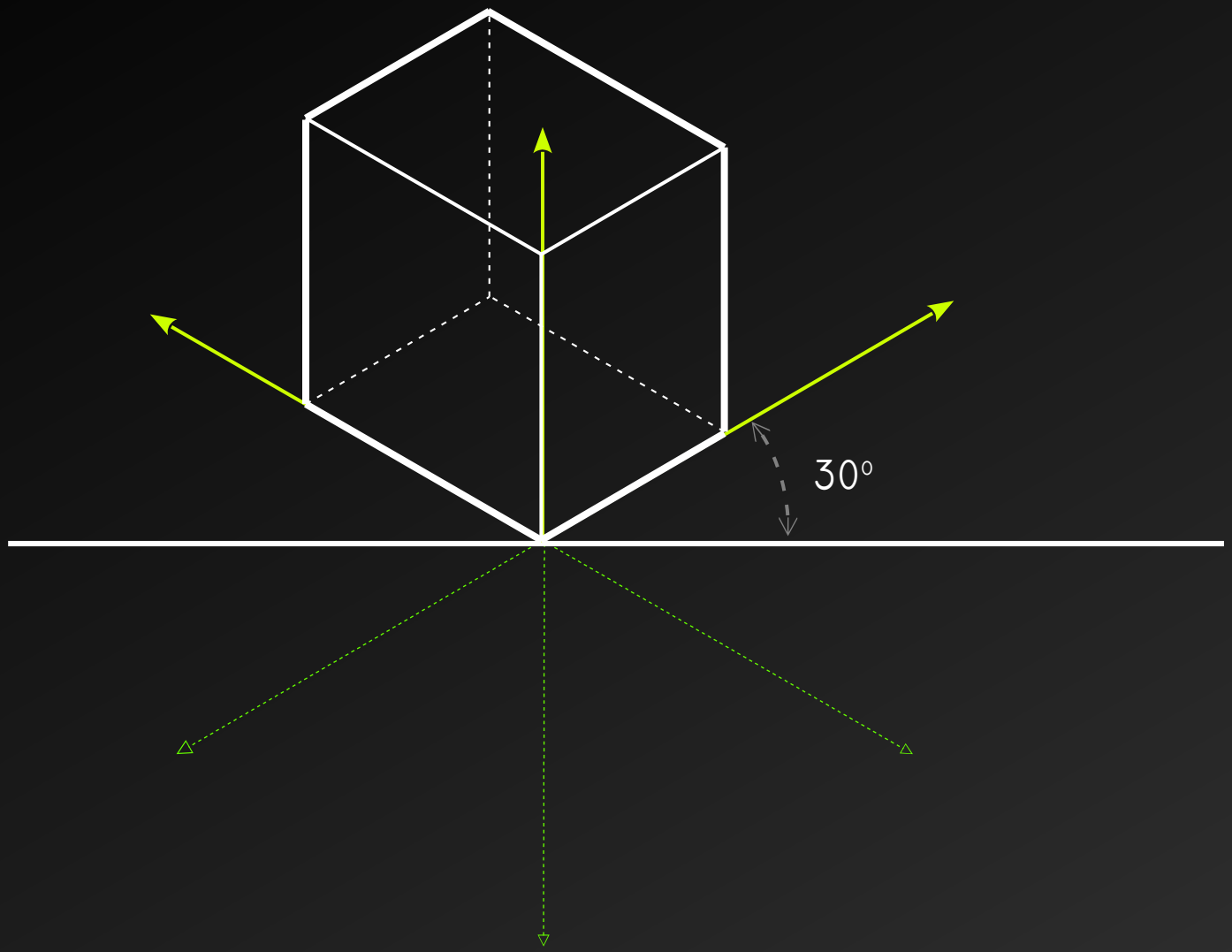
Representação em Perspectiva Isométrica

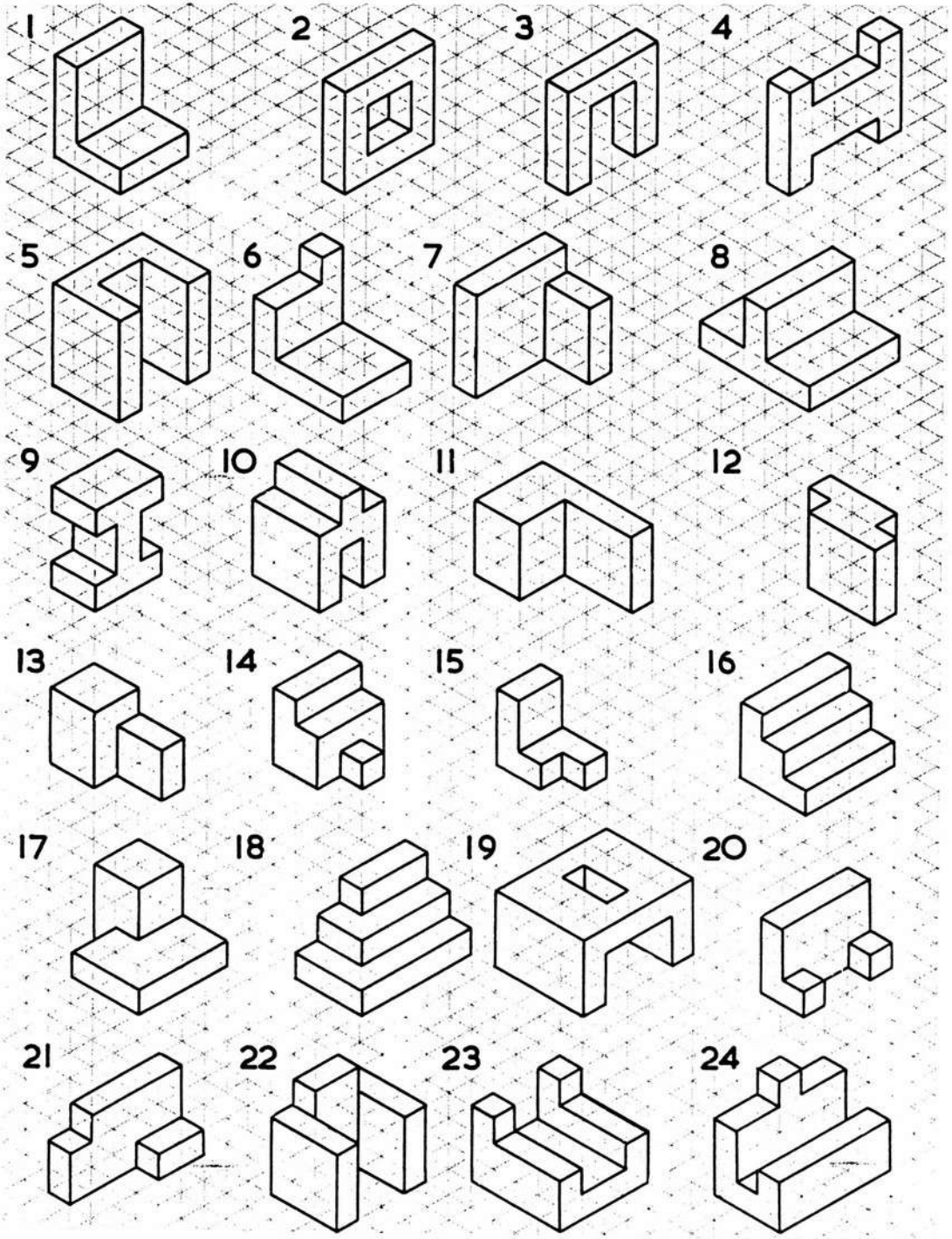


Representação em Perspectiva Isométrica

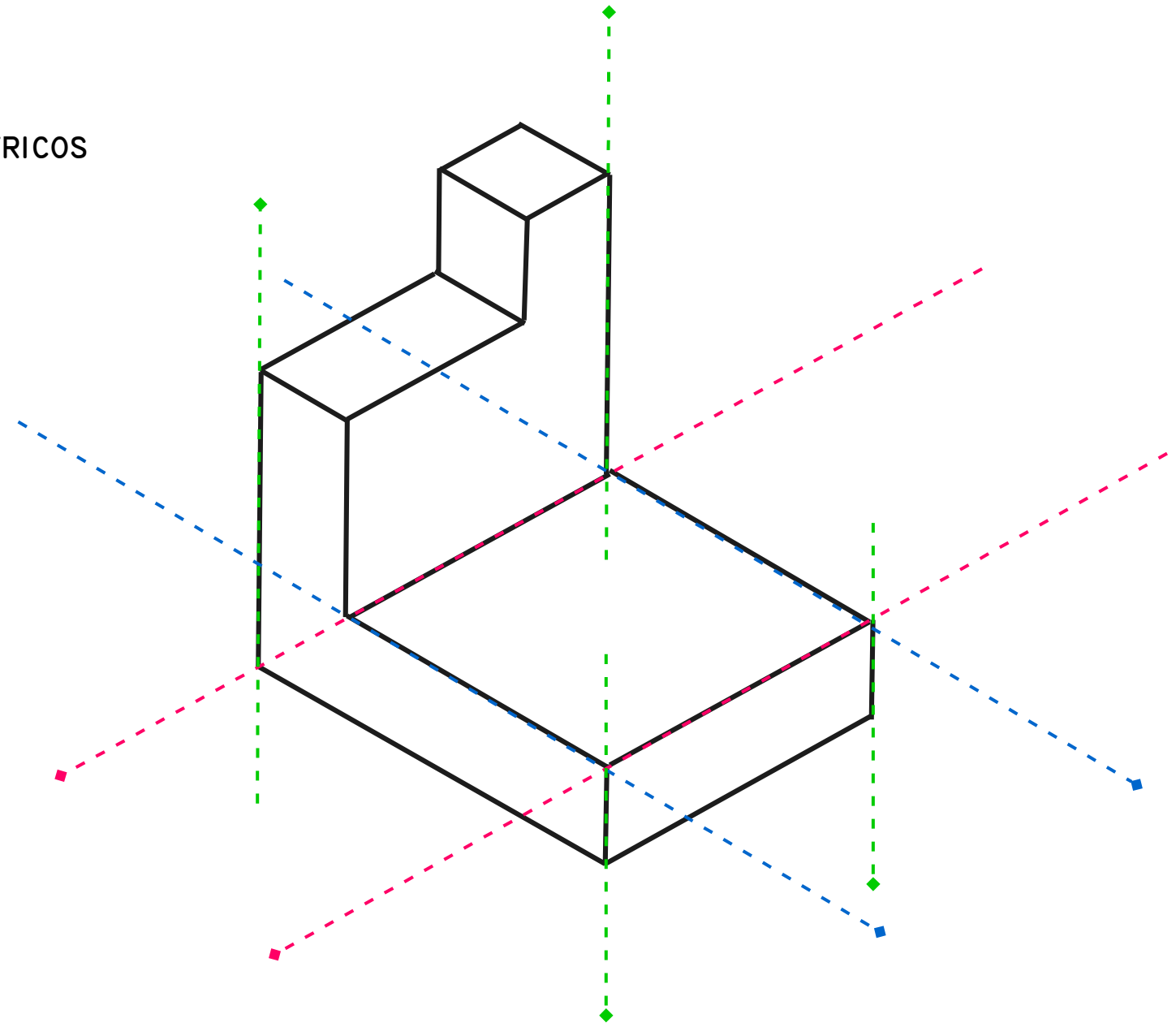


Representação em Perspectiva Isométrica



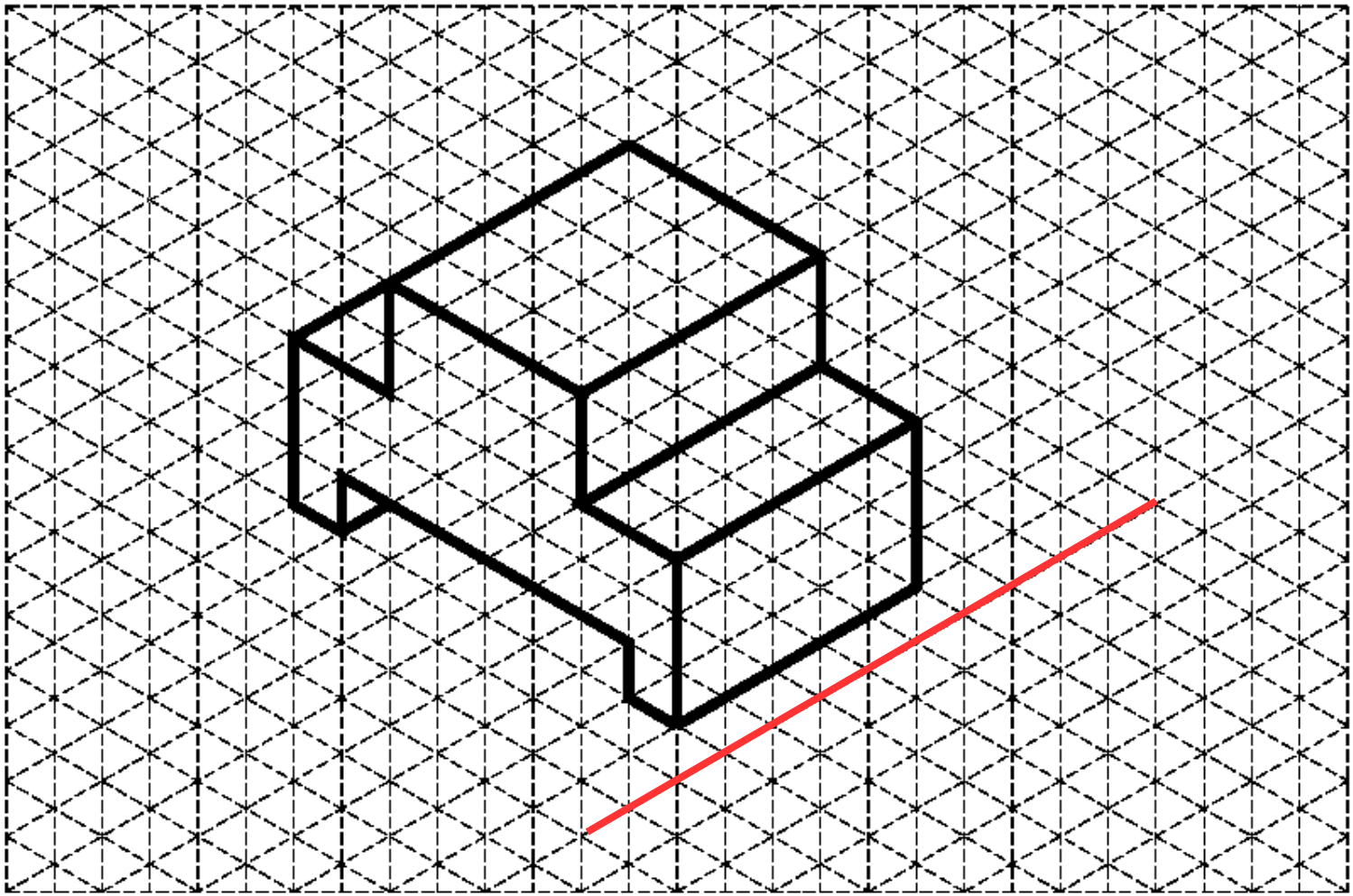

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Eixos Isométricos



Acrobat Reader - [Activity3_1_1Mass_Properties_Isometric_2.pdf]
File Edit Document Tools View Window Help
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Bookmarks
Thumbnails

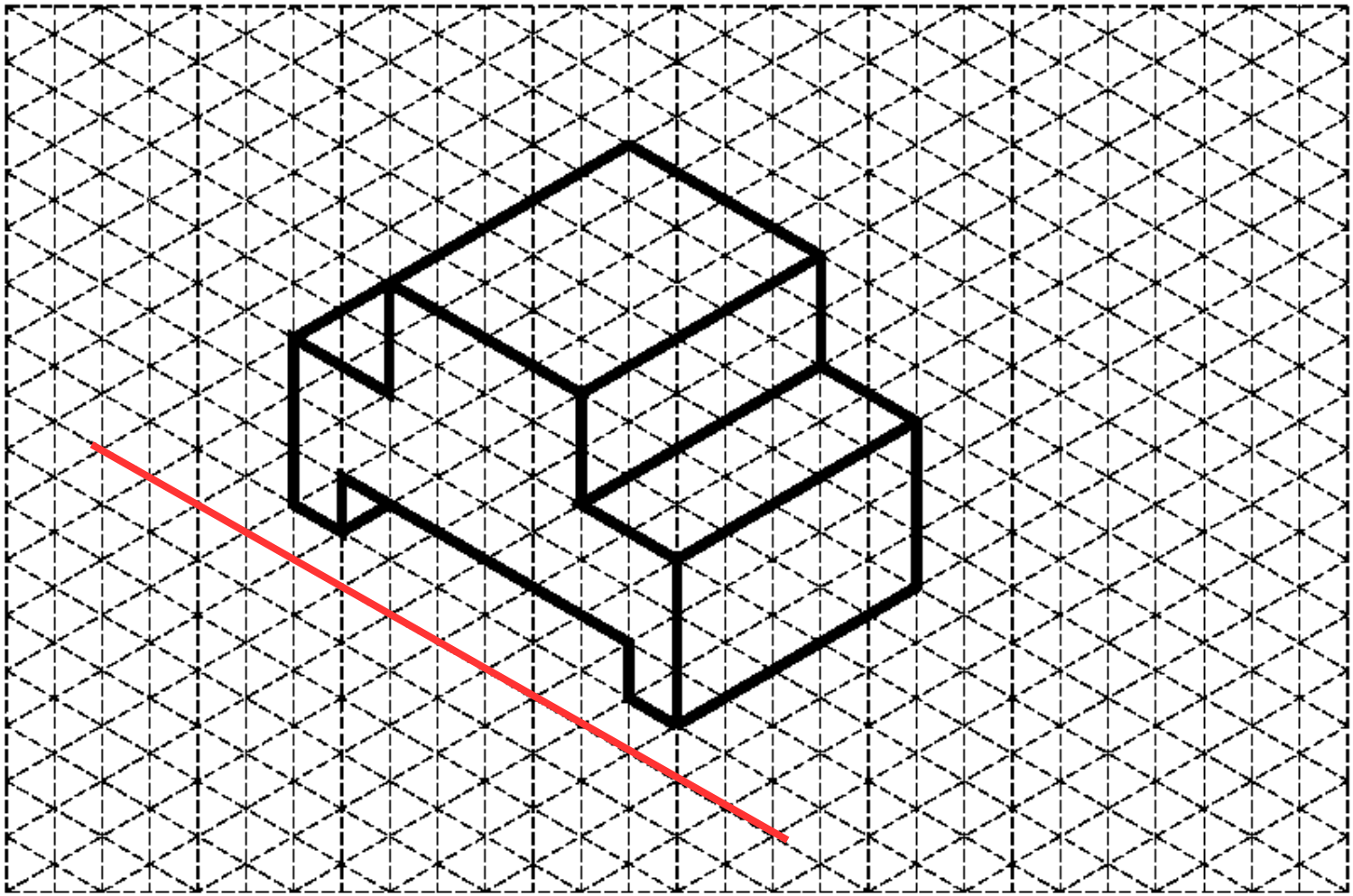
MALHA ISOMÉTRICA



1 of 2 8.5 x 11 in
start Unit 3.1 Activities Acrobat Reader - ... 6:17 PM

Acrobat Reader - [Activity3_1_1Mass_Properties_Isometric_2.pdf]
File Edit Document Tools View Window Help
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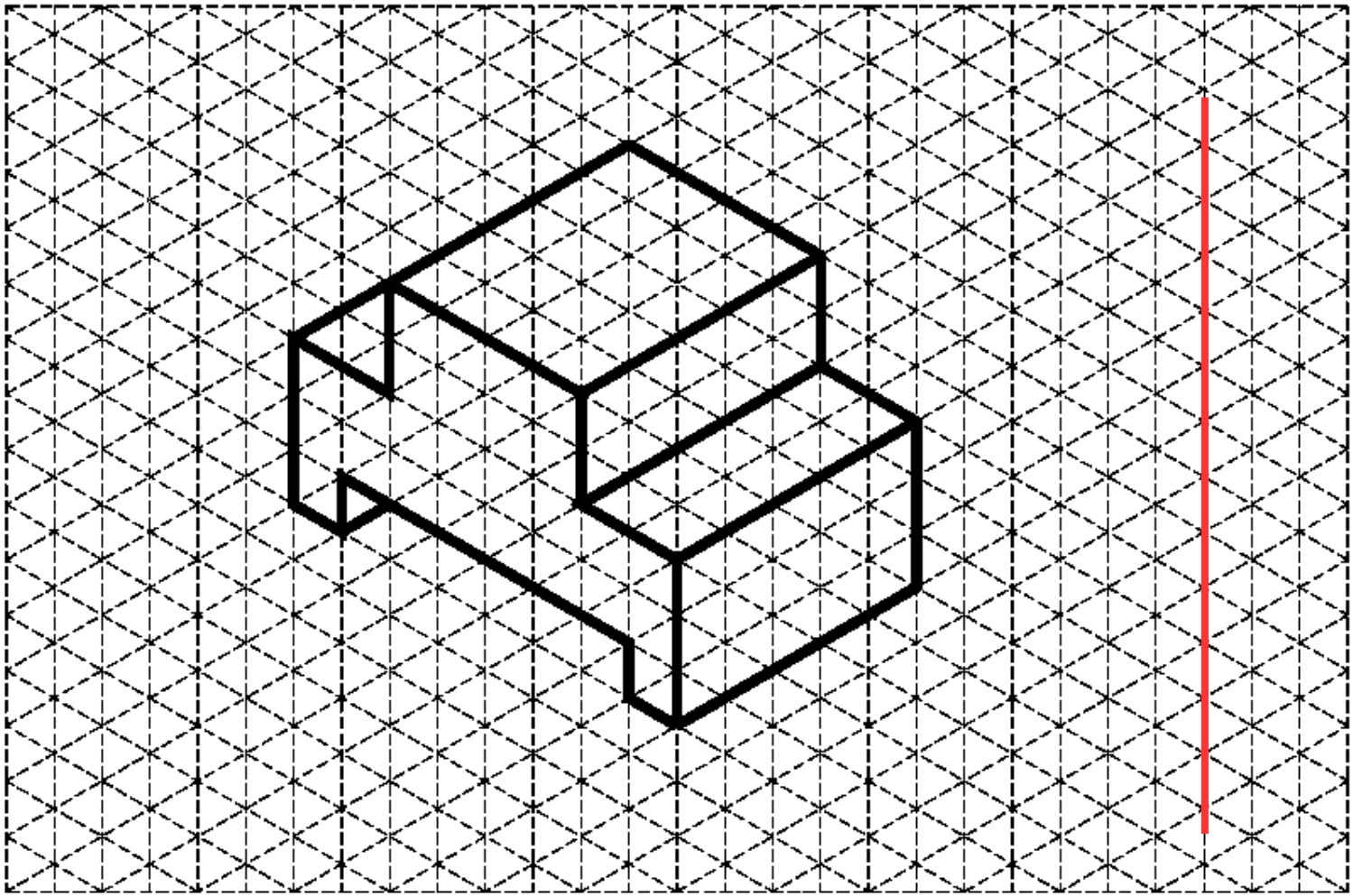
MALHA ISOMÉTRICA



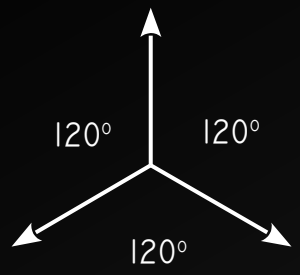
1 of 2 8.5 x 11 in
start Unit 3.1 Activities Acrobat Reader - ... 6:17 PM

Acrobat Reader - [Activity3_1_1Mass_Properties_Isometric_2.pdf]
File Edit Document Tools View Window Help
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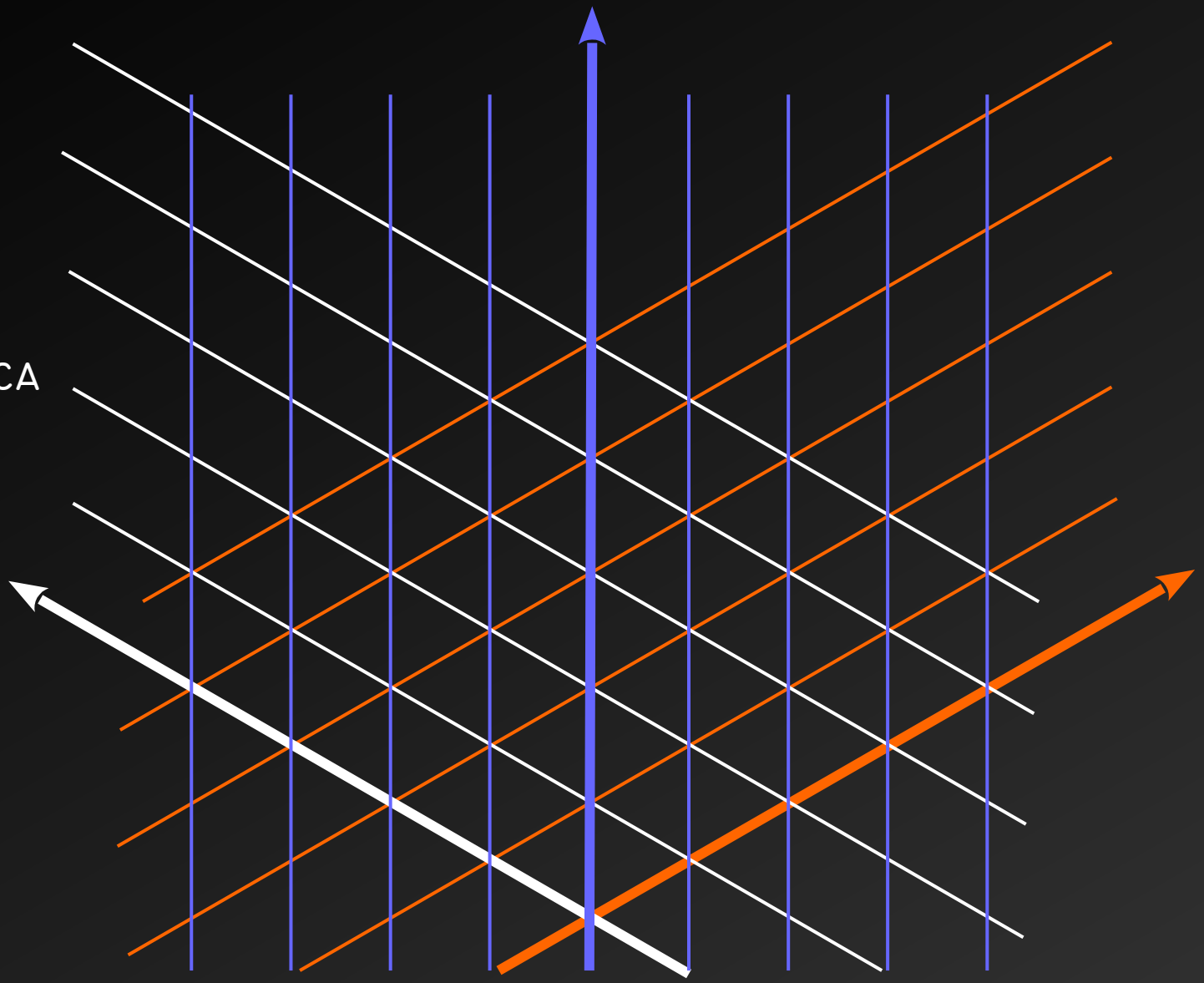
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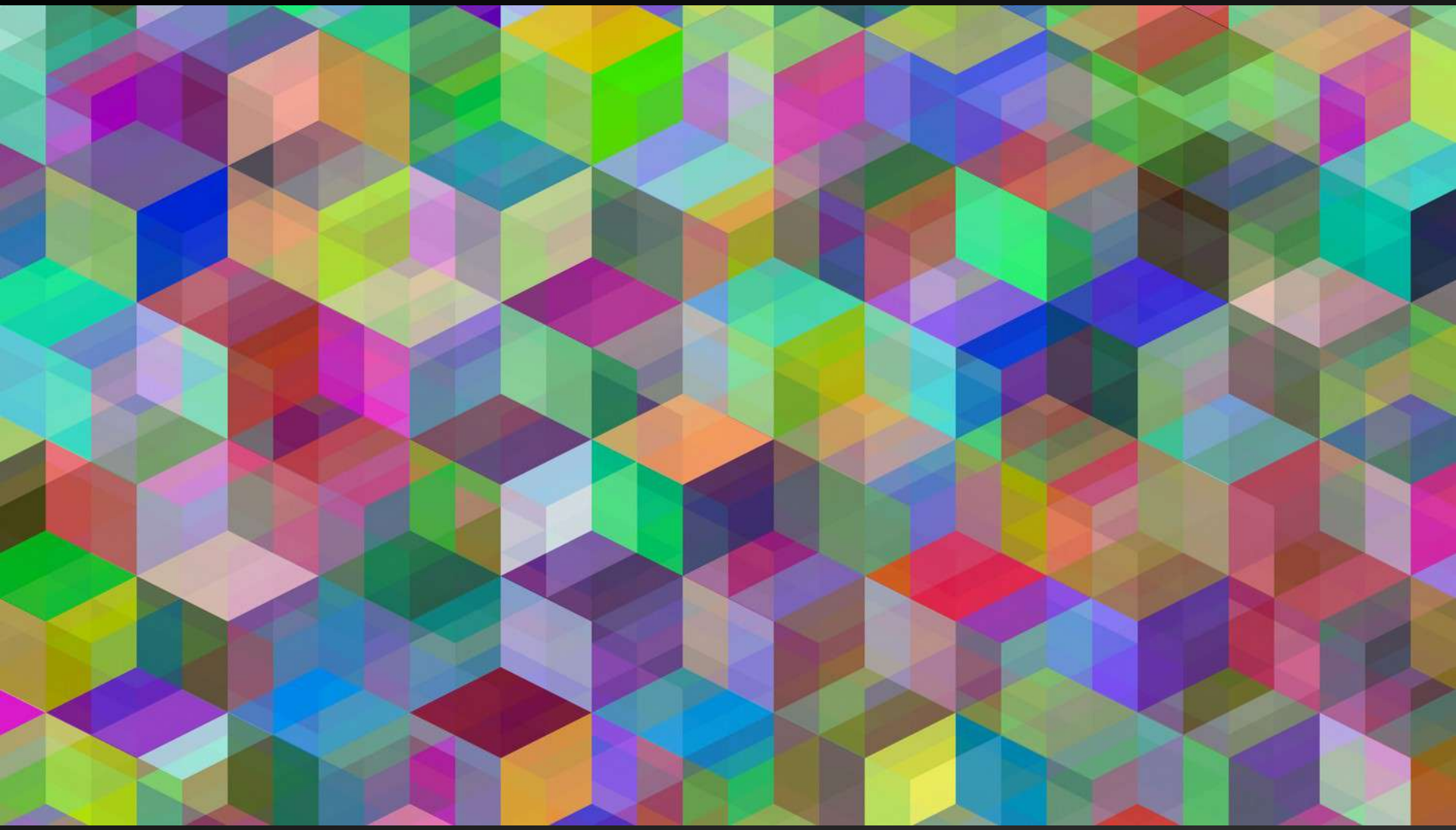


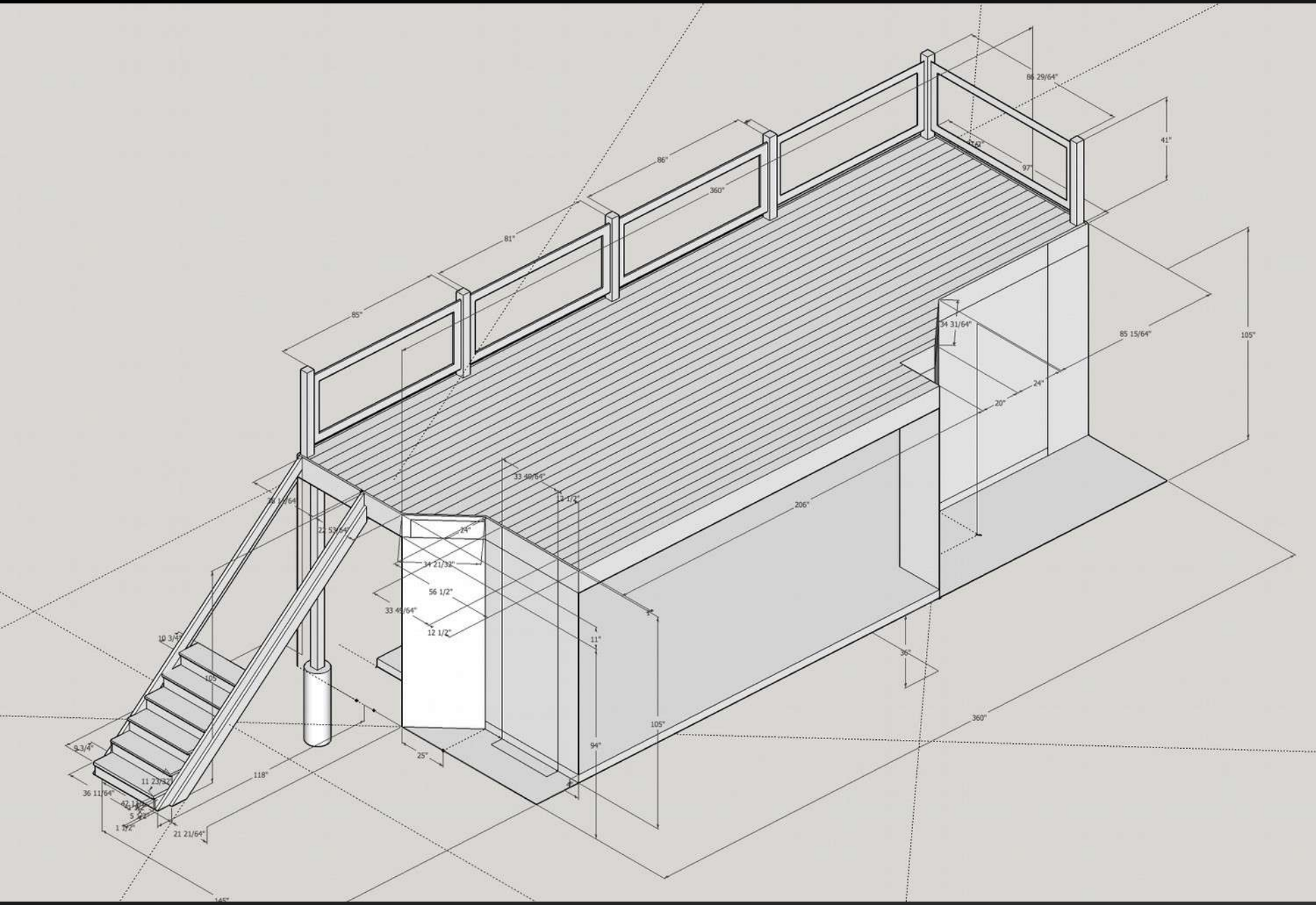
1 of 2 8.5 x 11 in
start Unit 3.1 Activities Acrobat Reader - ... 6:17 PM

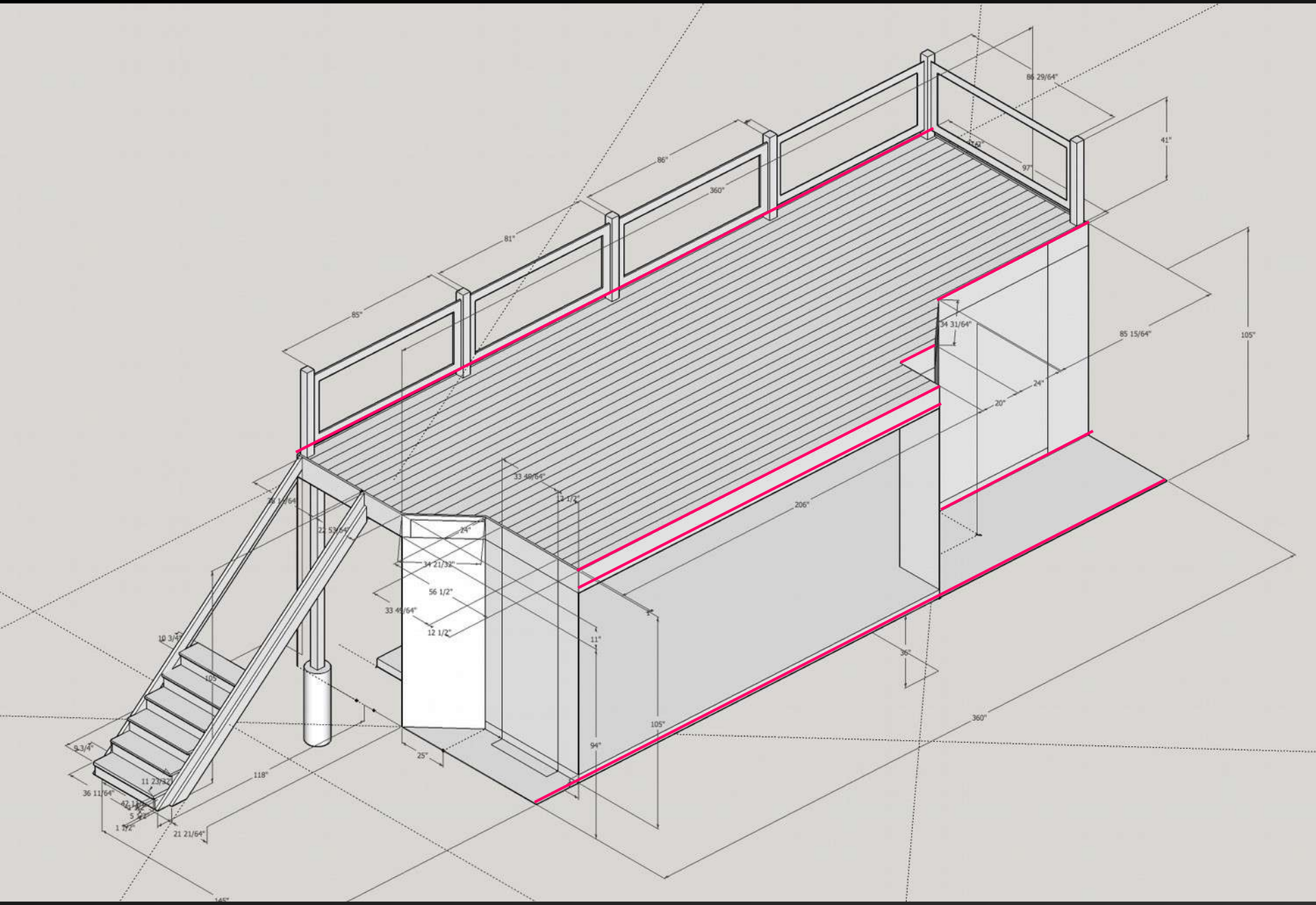


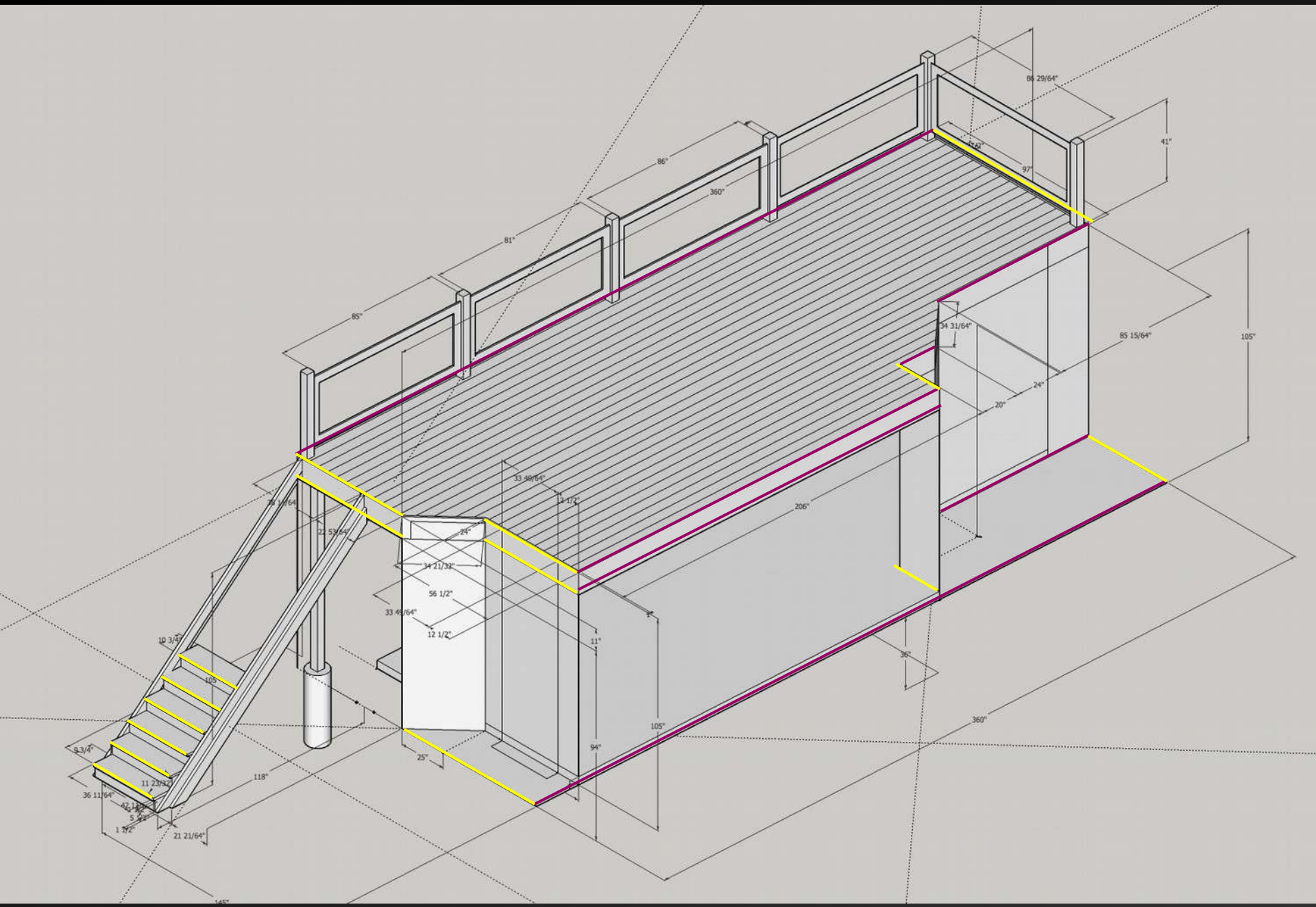
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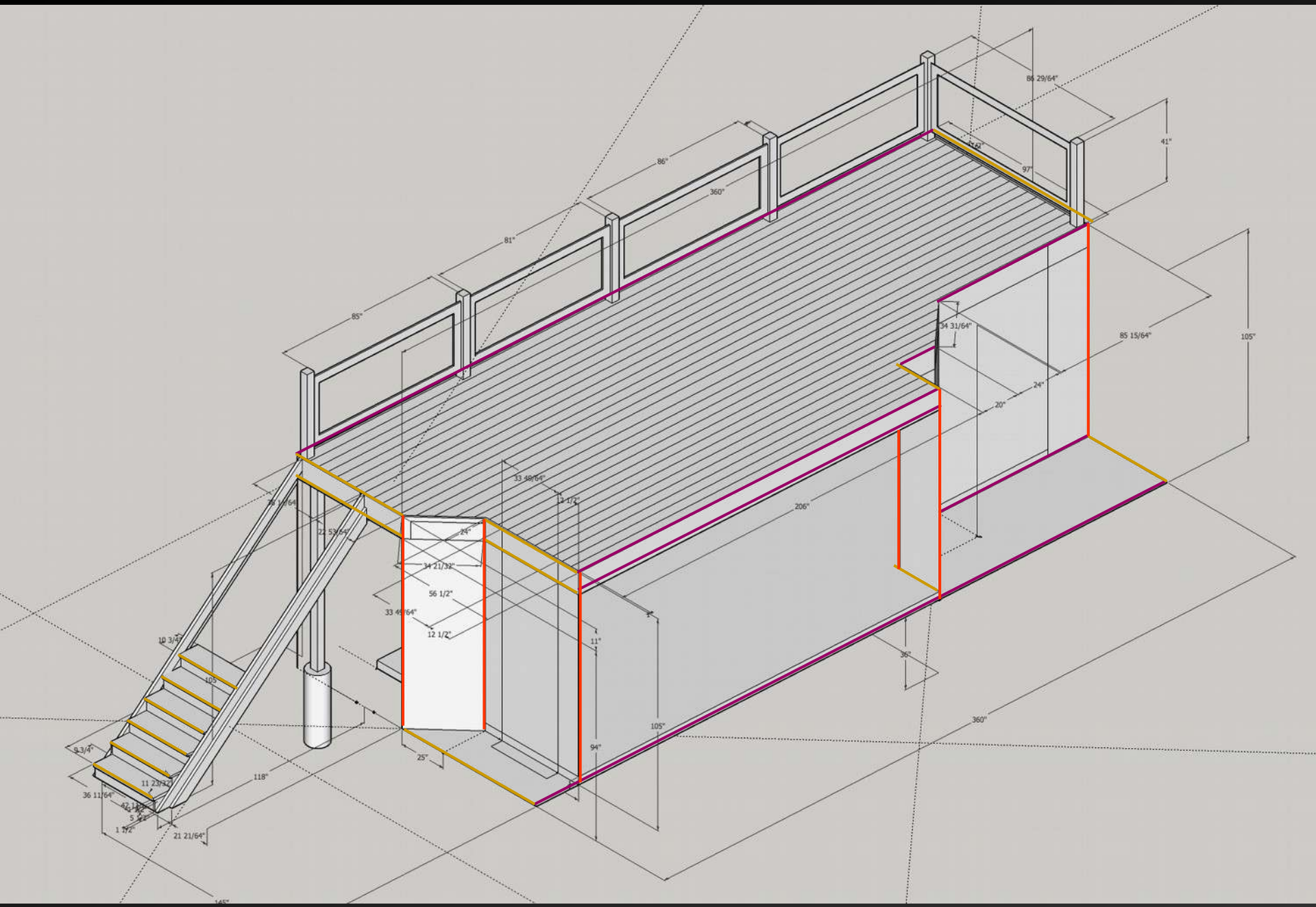


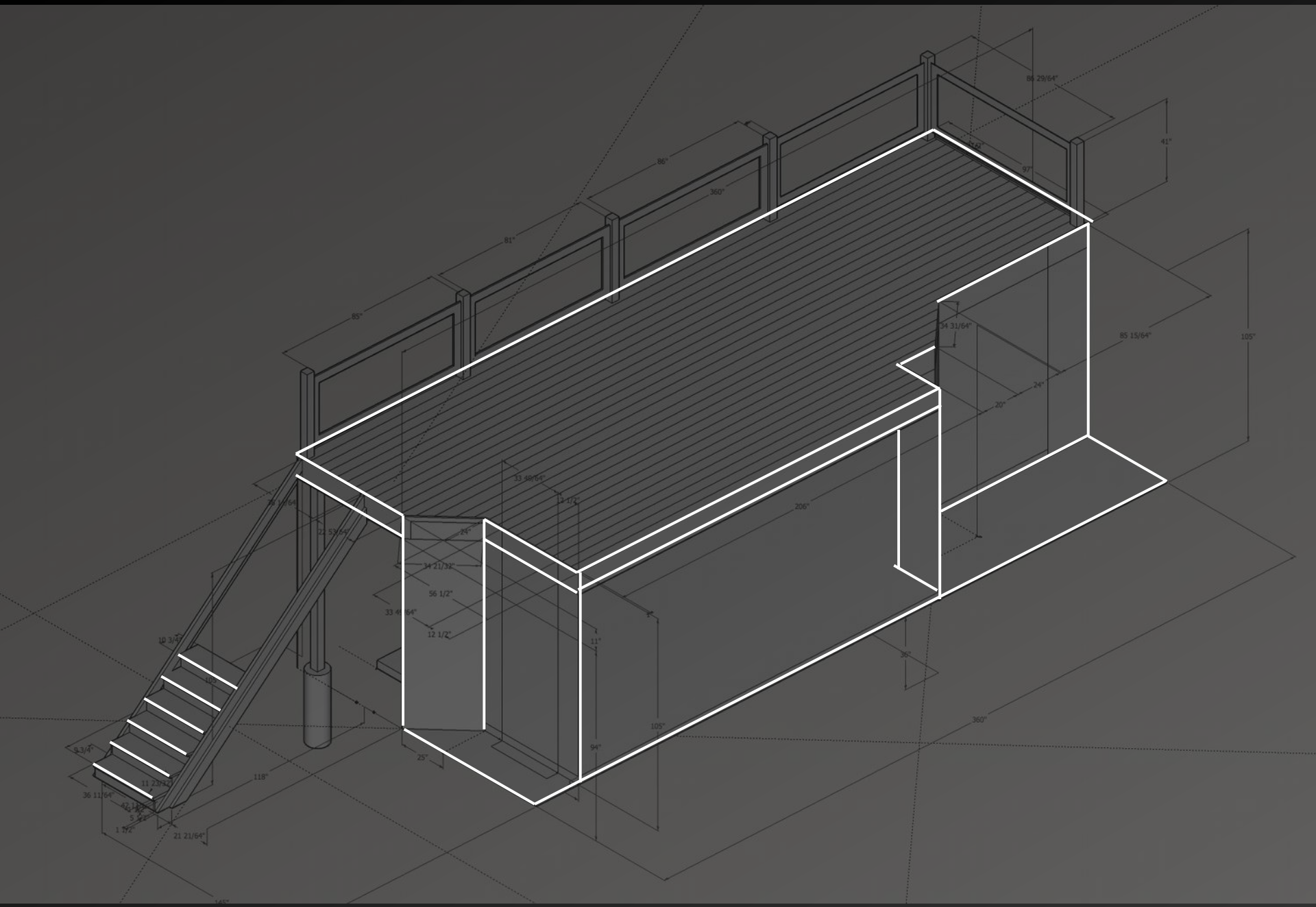


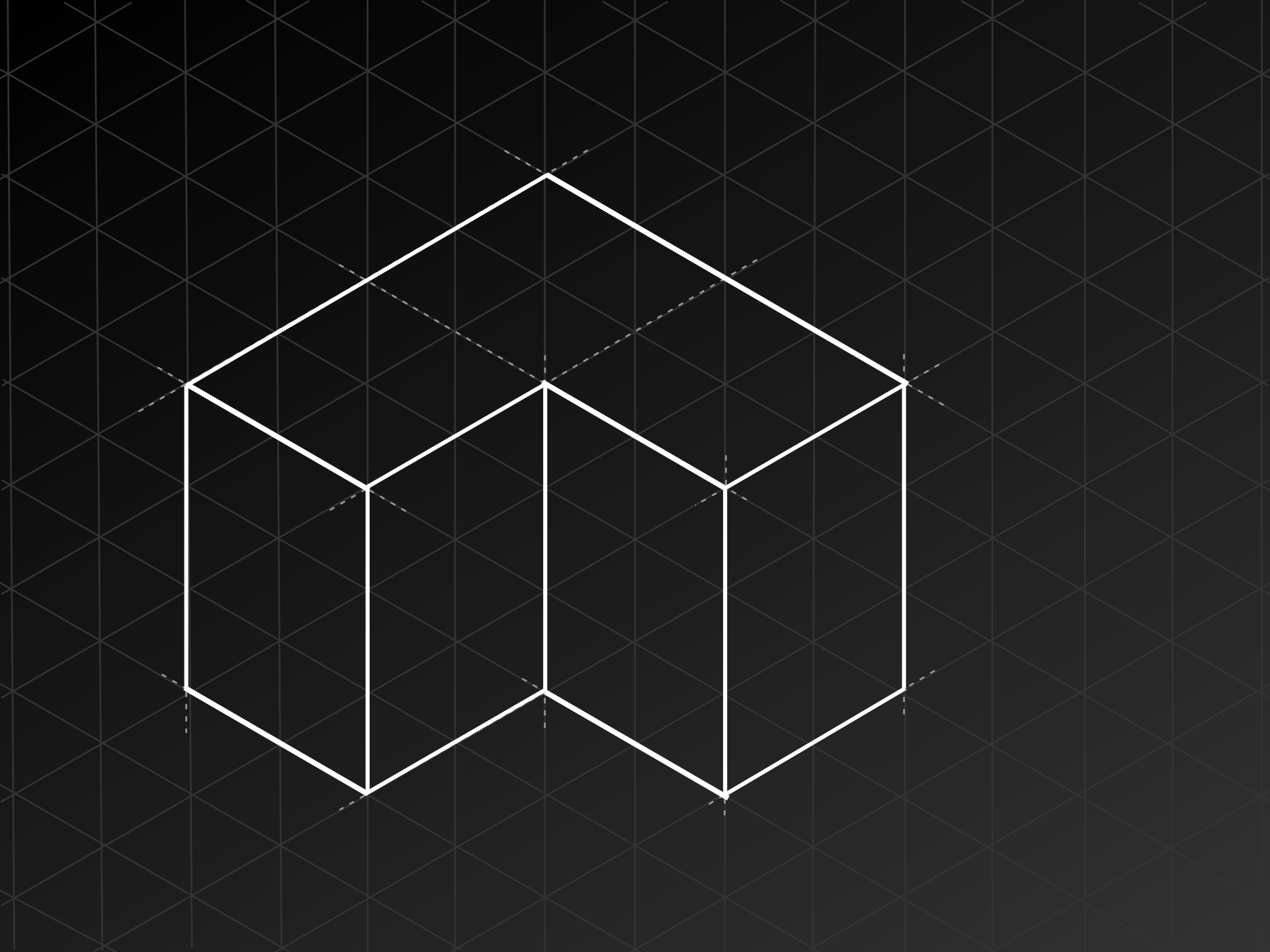

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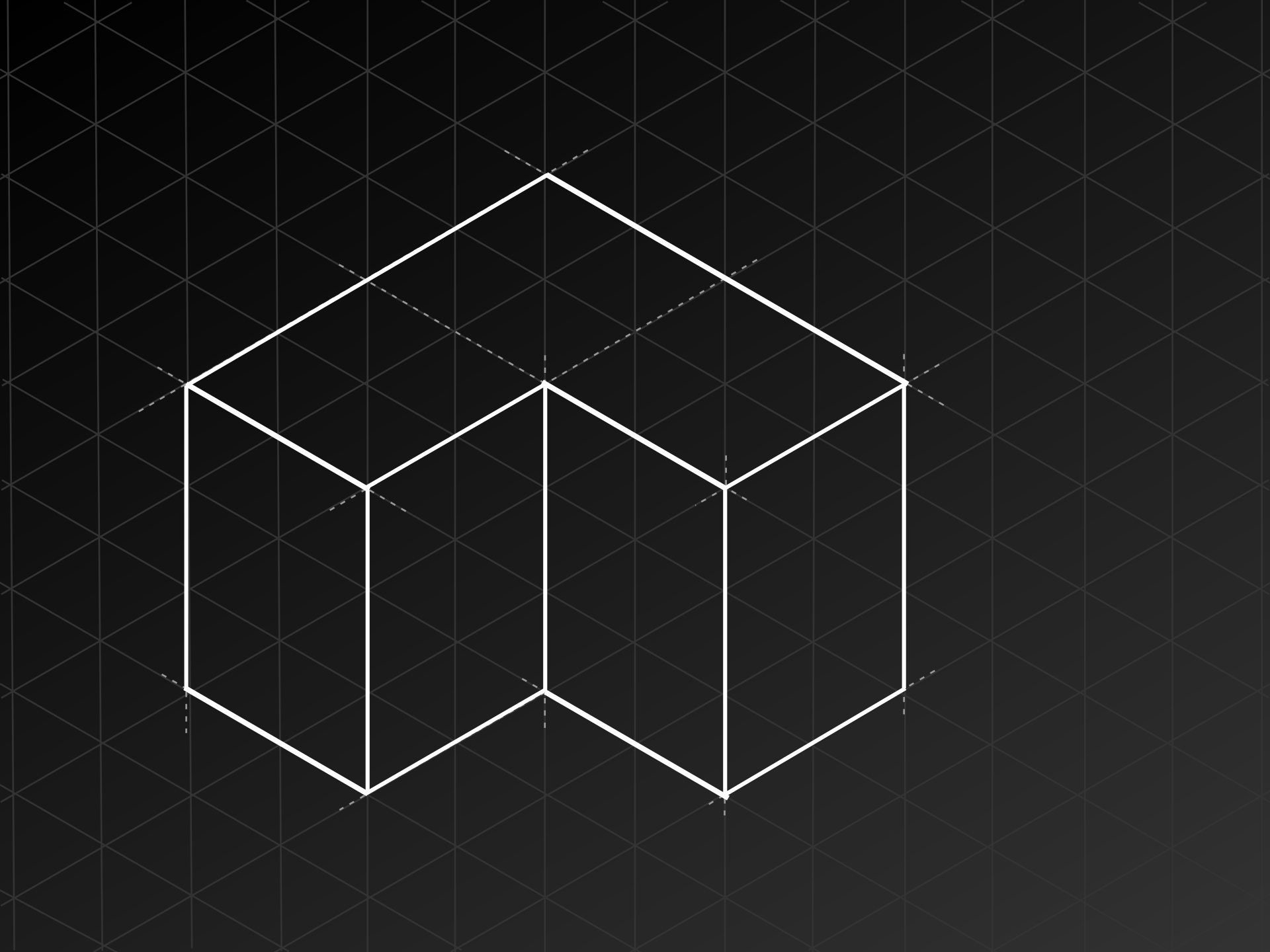

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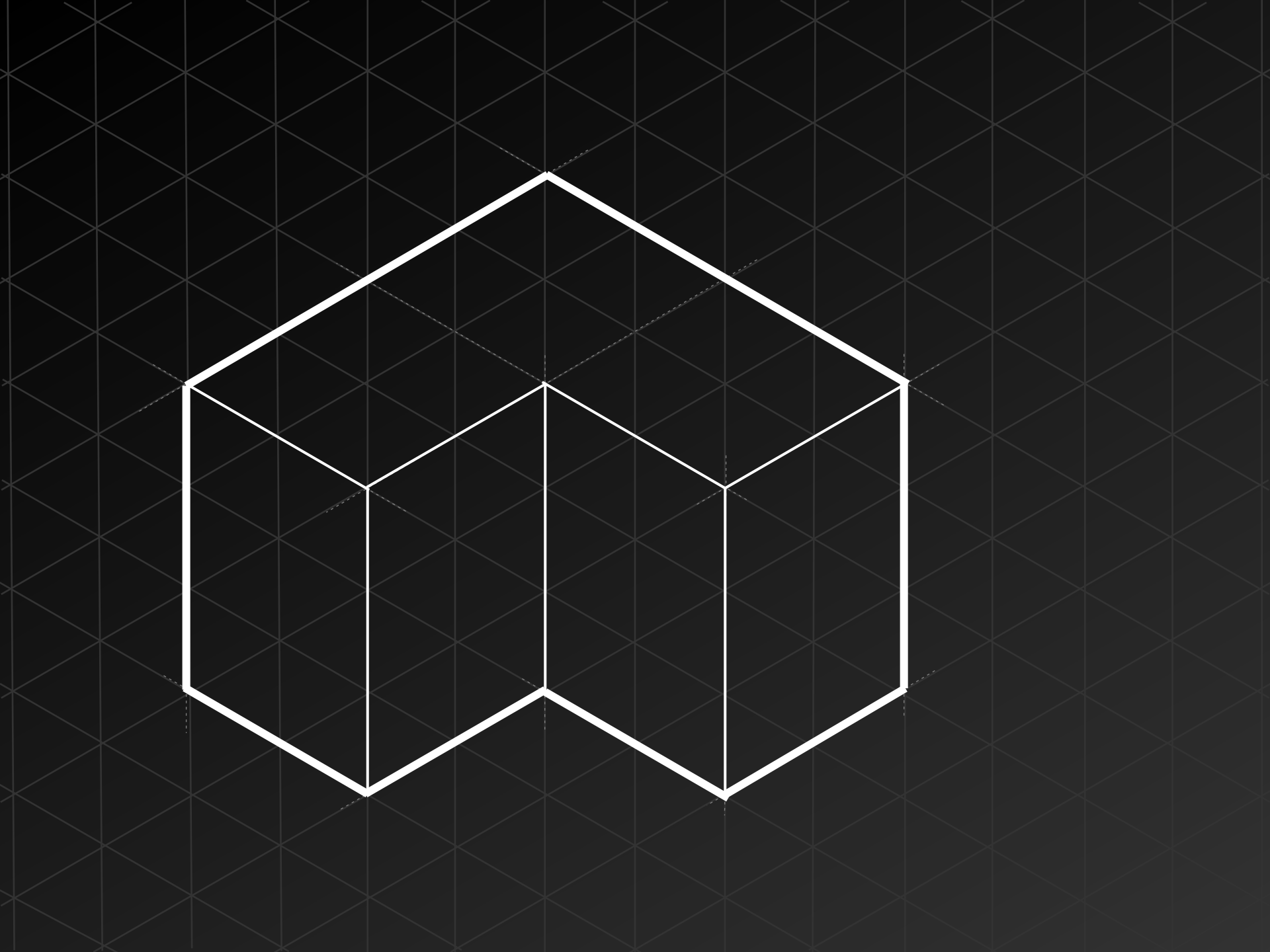

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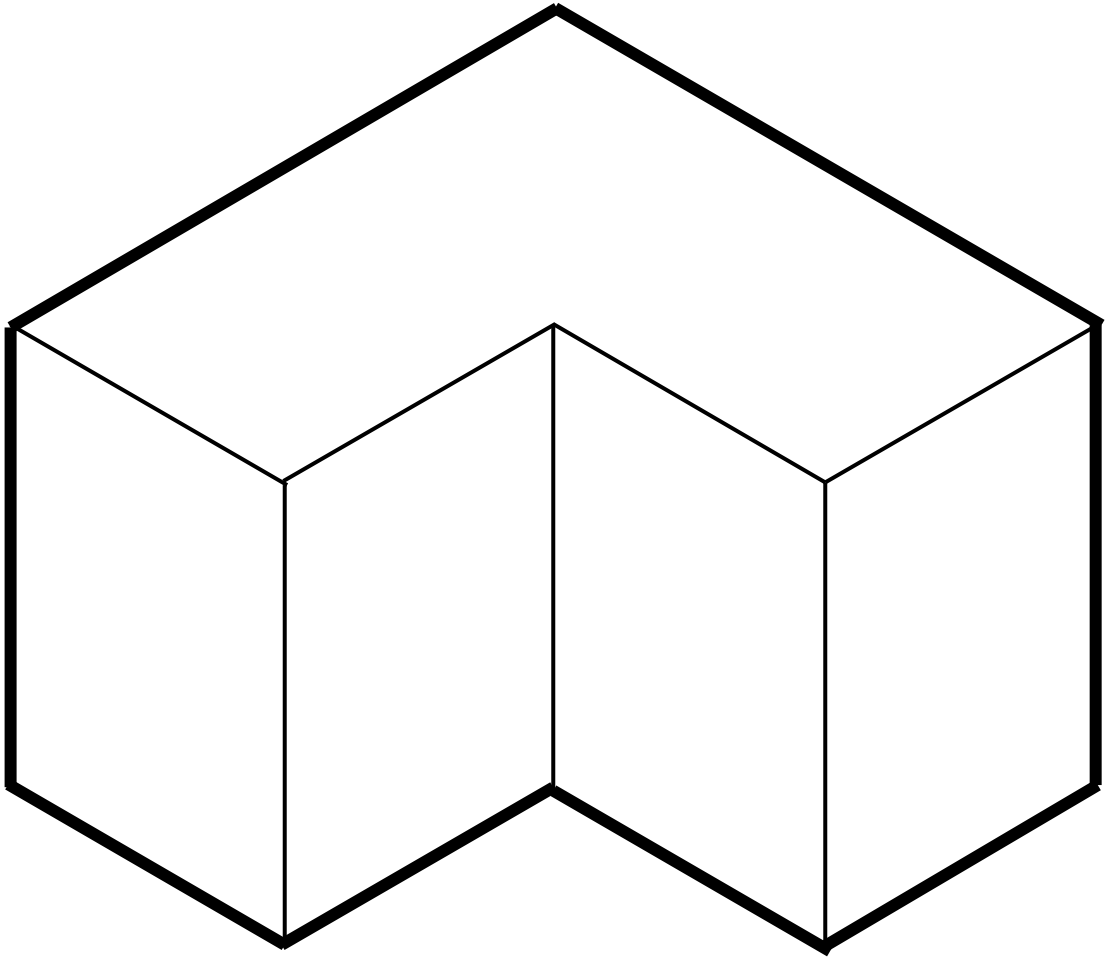






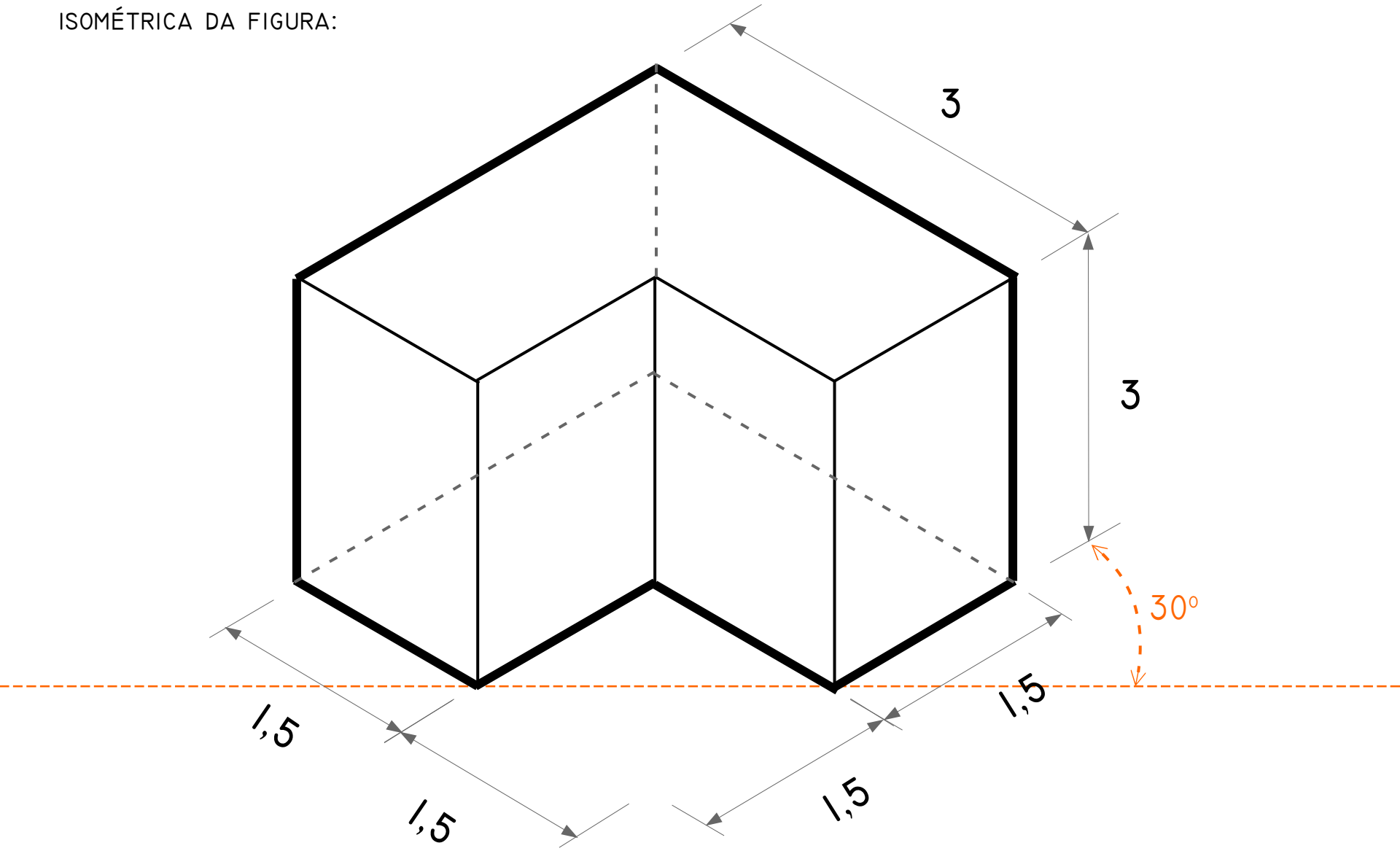






Exercício

Crie a perspectiva isométrica da figura:



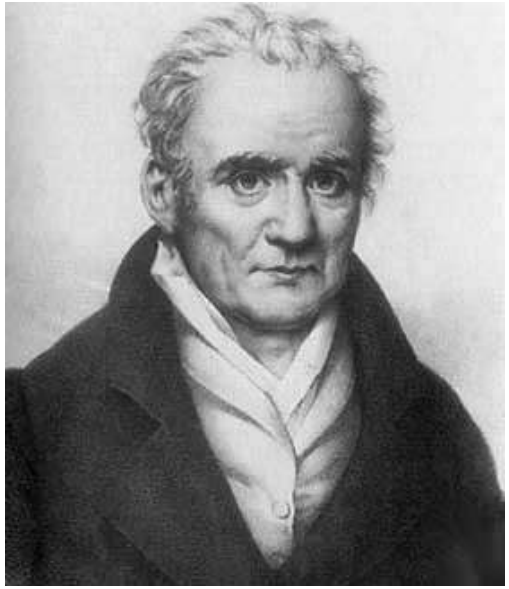
CONCEITOS INICIAIS

SISTEMAS DE PROJEÇÕES

Gaspard Monge 1746 – 1818

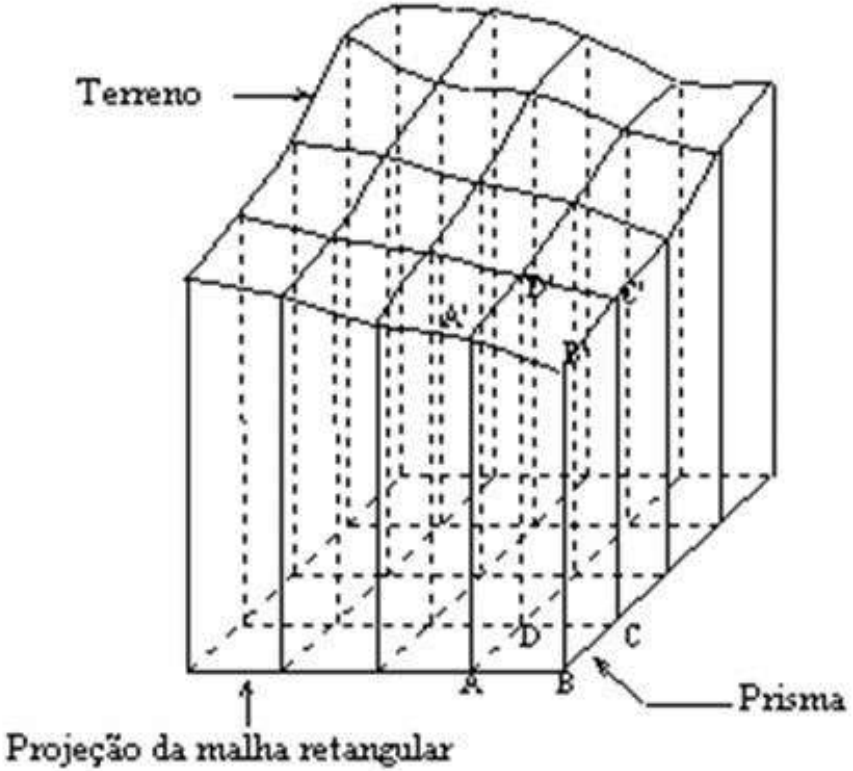
Iluminismo, Revolução Francesa
(era das revoluções burguesas)

- Desenvolvimento do Capitalismo
- Desenvolvimento da Indústria
- Saída do artesanato
- Surgimento das máquinas

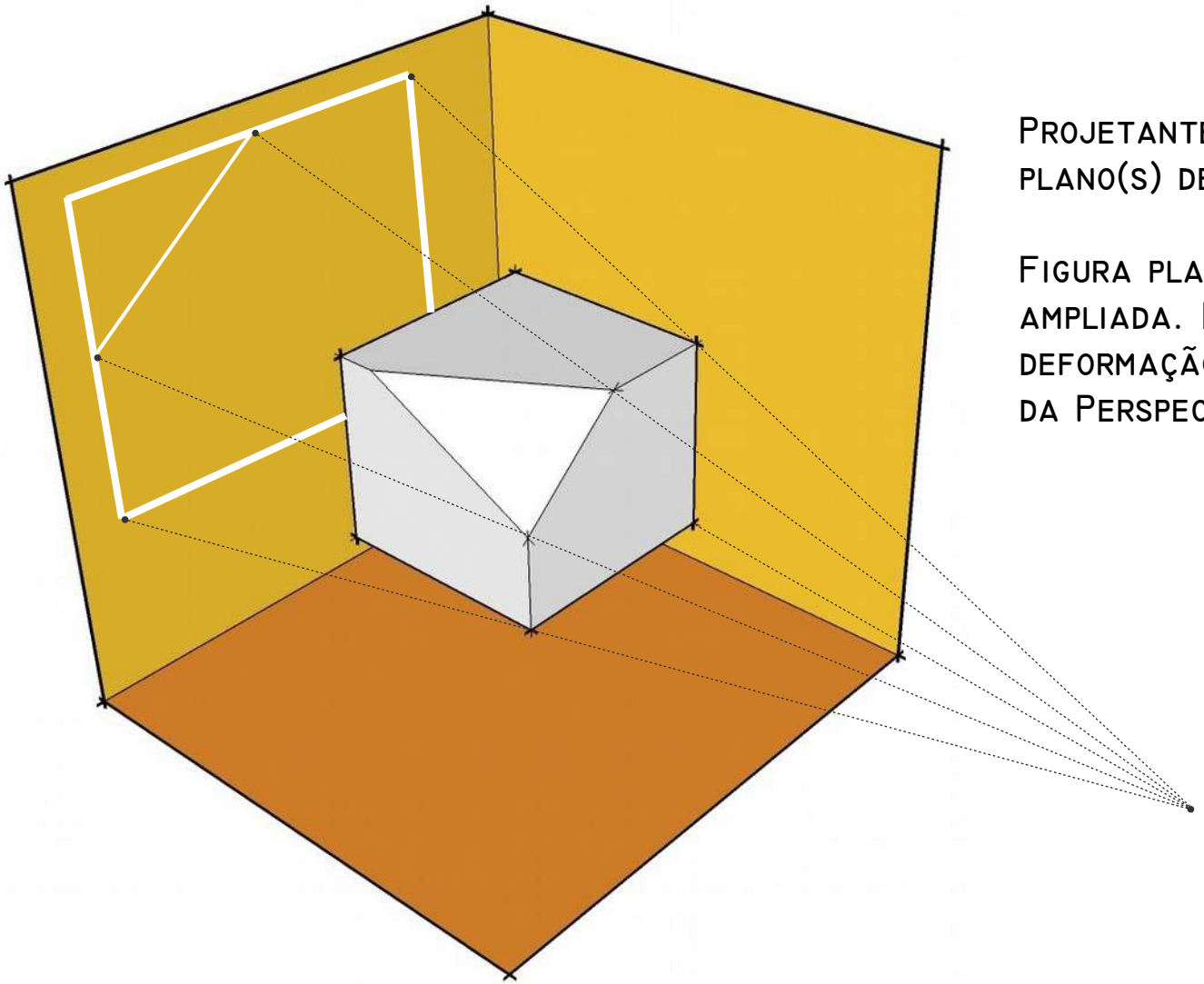







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Projeção da malha retangular

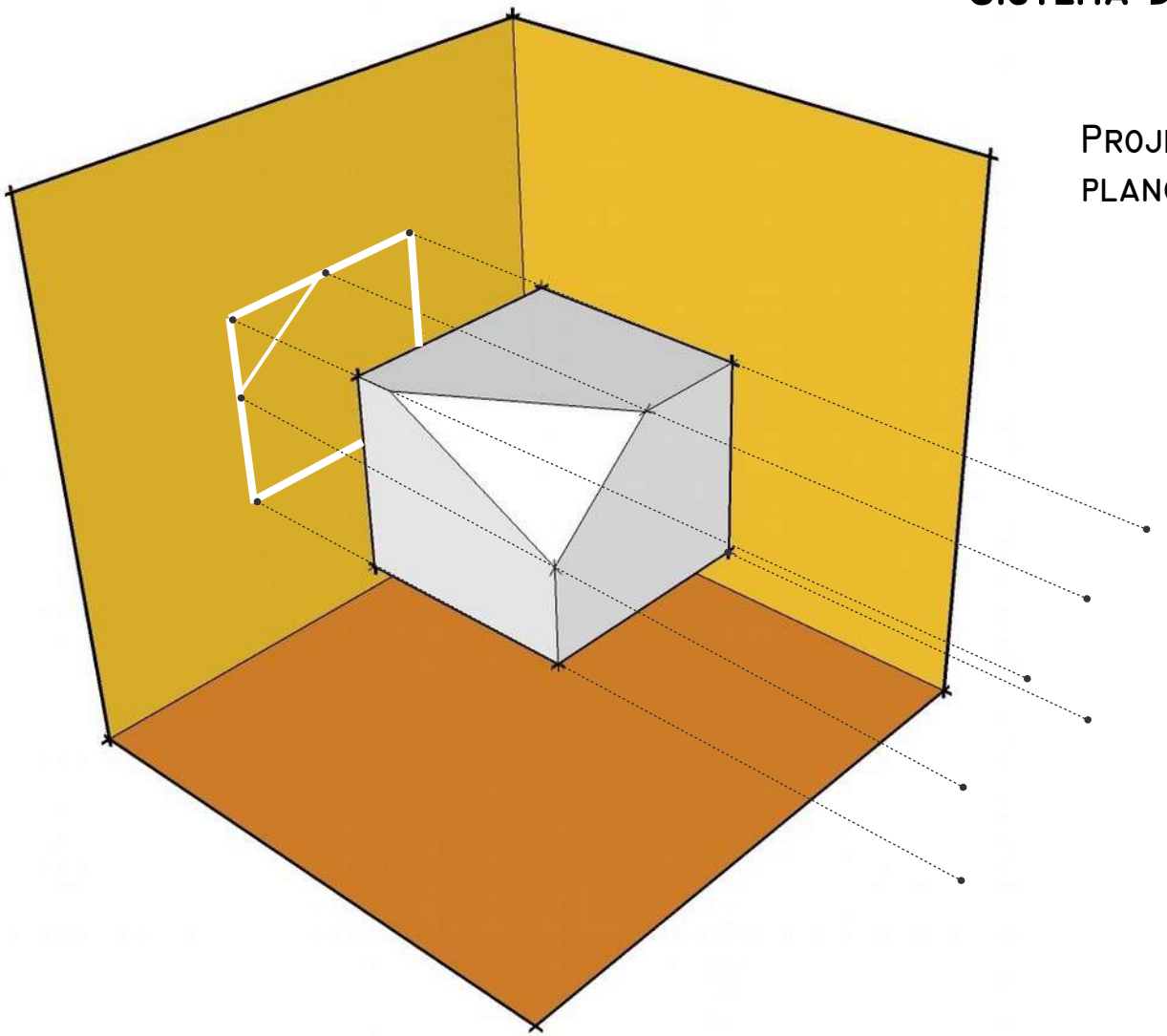
Sistema de projeção cônica



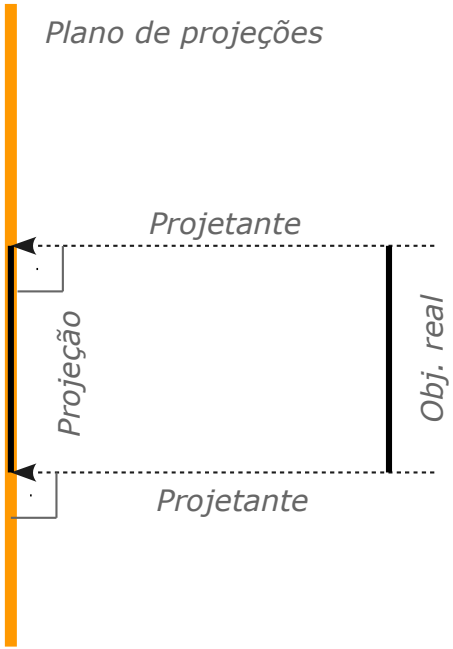
Projetantes oblíquas ao(s) plano(s) de projeções.

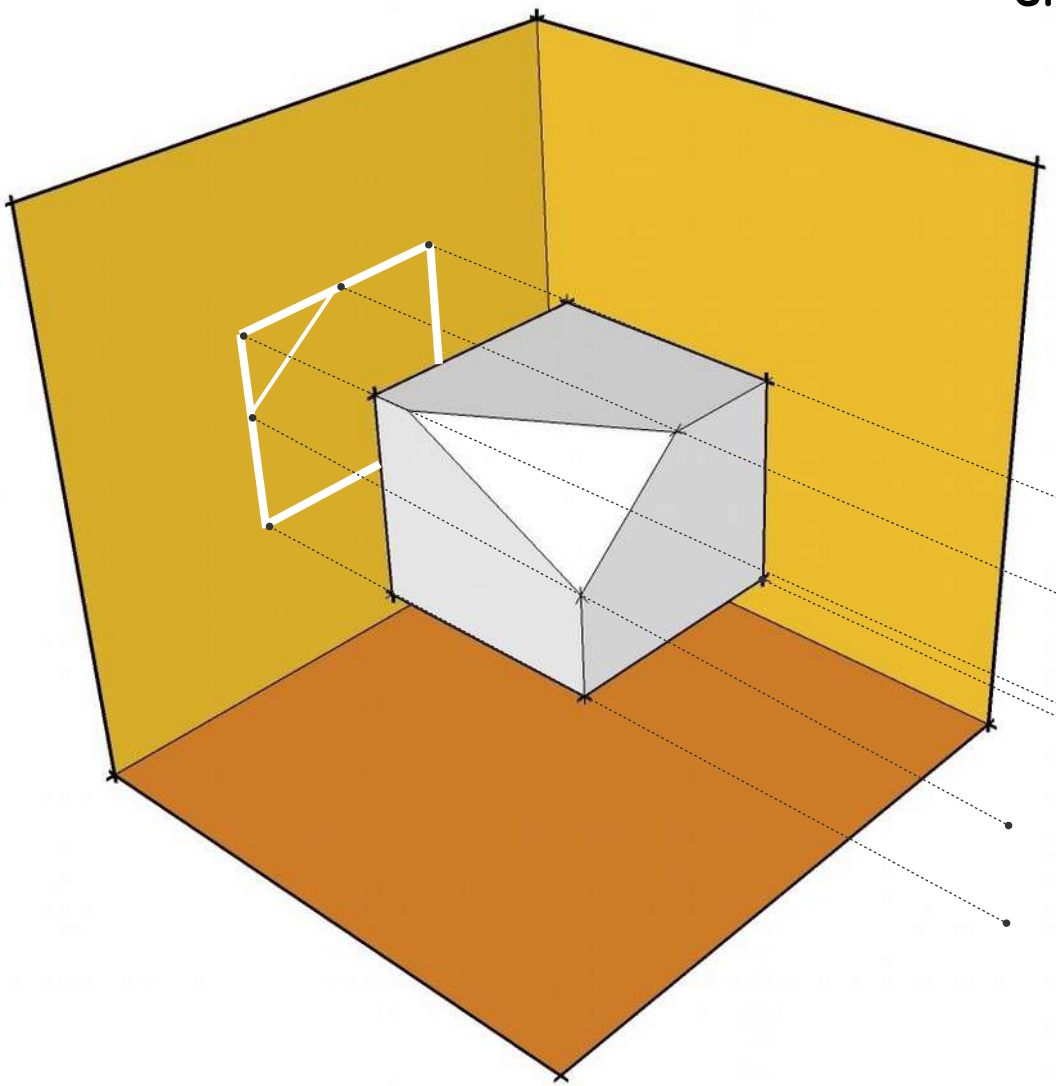
Figura plana projetada é ampliada. Existe uma deformação da imagem (Caso da Perspectiva).

Sistema de projeção cilíndrica



Projetantes ortogonais ao(s) plano(s) de projeções.





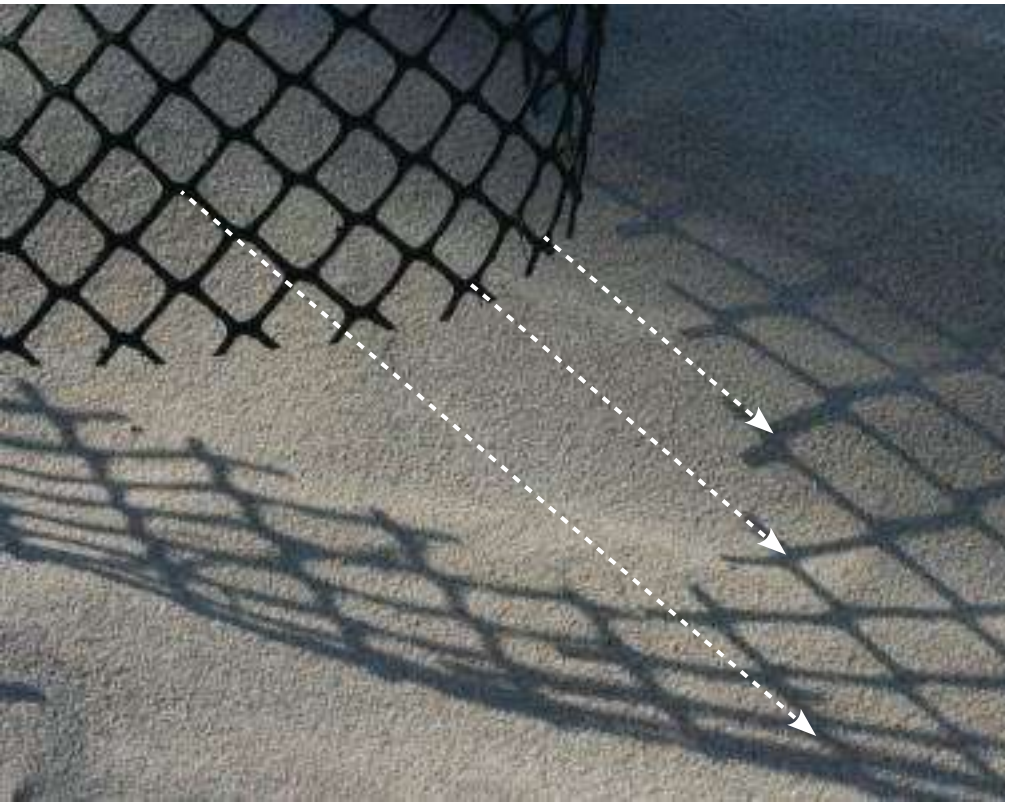
Sistema de projeção cilíndrica

Projetantes ortogonais ao(s) plano(s) de projeções.

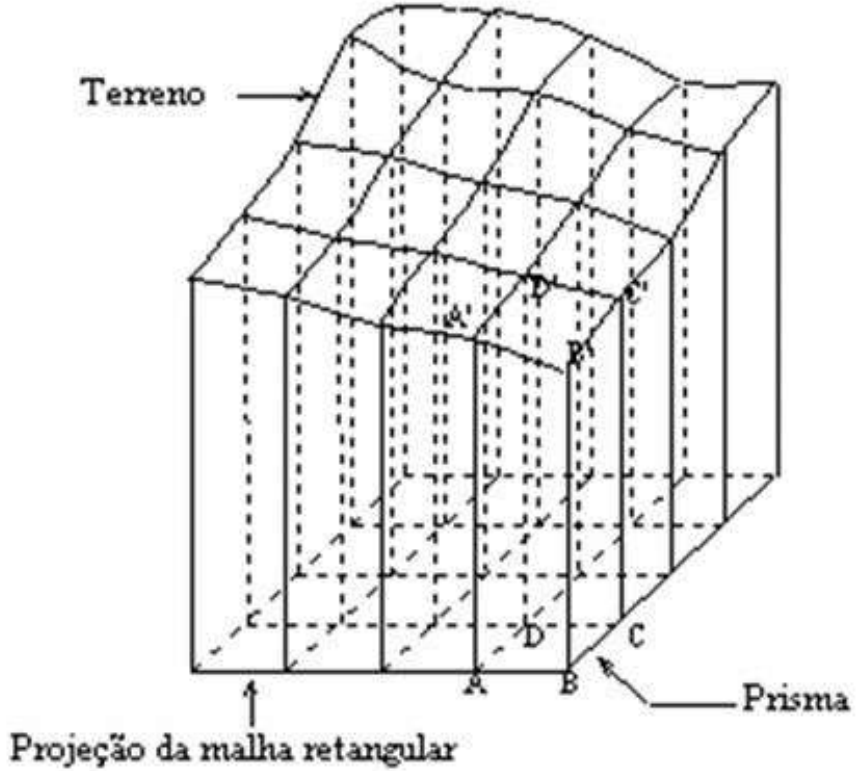
Figura plana projetada é fiel à real, se estiver paralela ao plano de projeções (aula futura).

É o sistema utilizado pela Geometria Descritiva, Desenho Técnico, Desenho Mecânico, Desenho Topográfico e Desenho Arquitetônico.

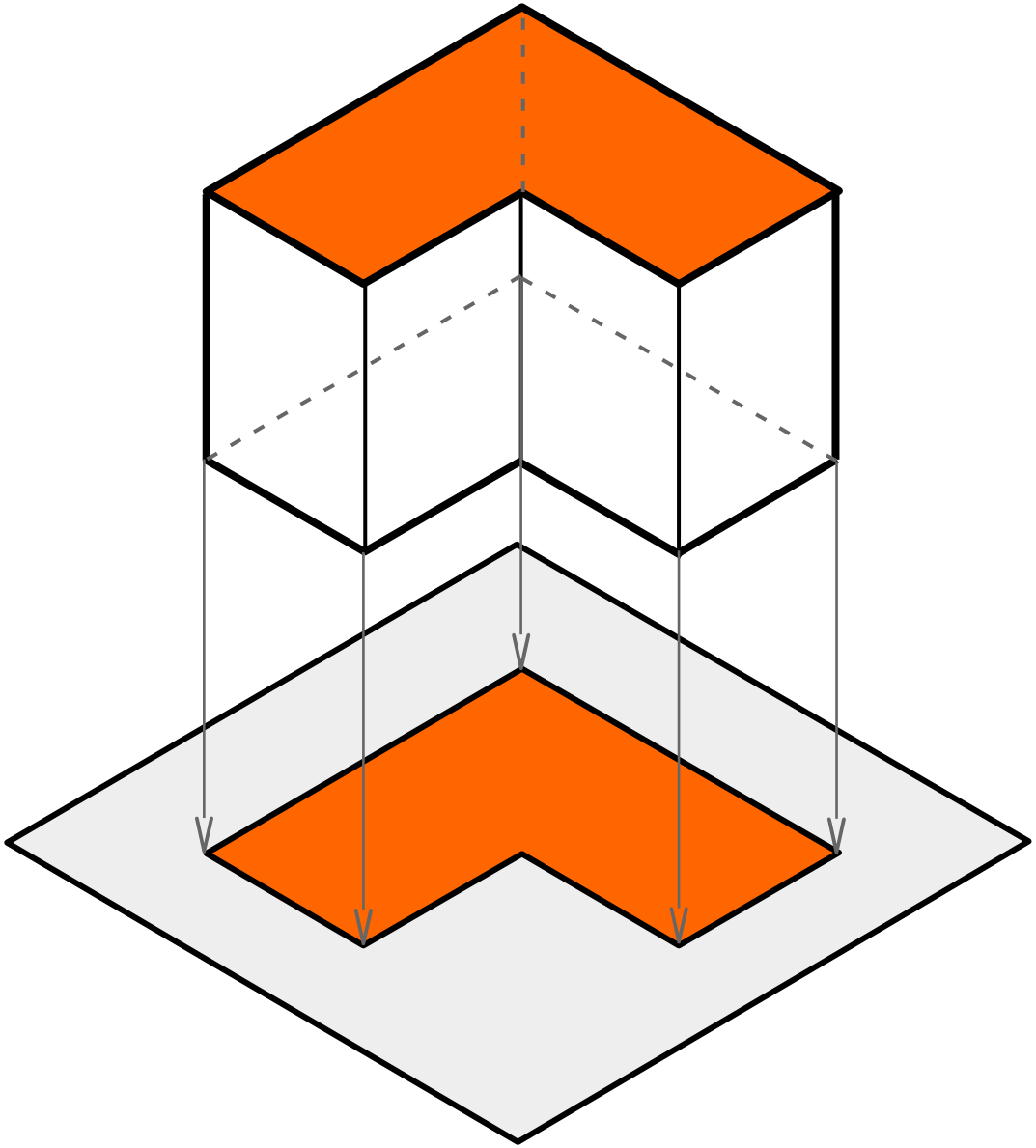
PROJEÇÃO CILÍNDRICA OBLÍQUA



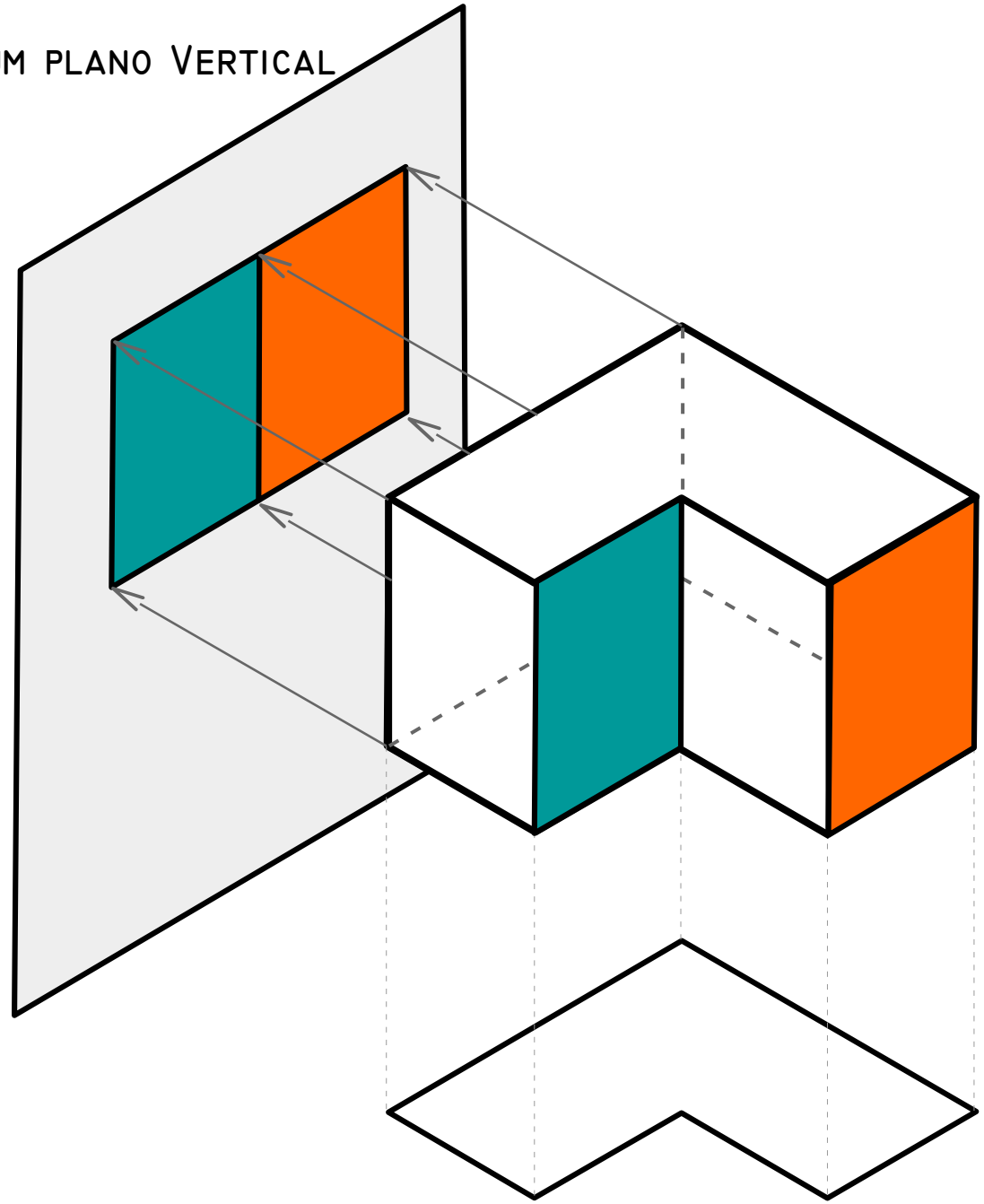
PROJEÇÃO CILÍNDRICA ORTOGONAL



Projeção da figura sobre um plano Horizontal



Projeção da figura sobre um plano Vertical



Sistemas de Projeção

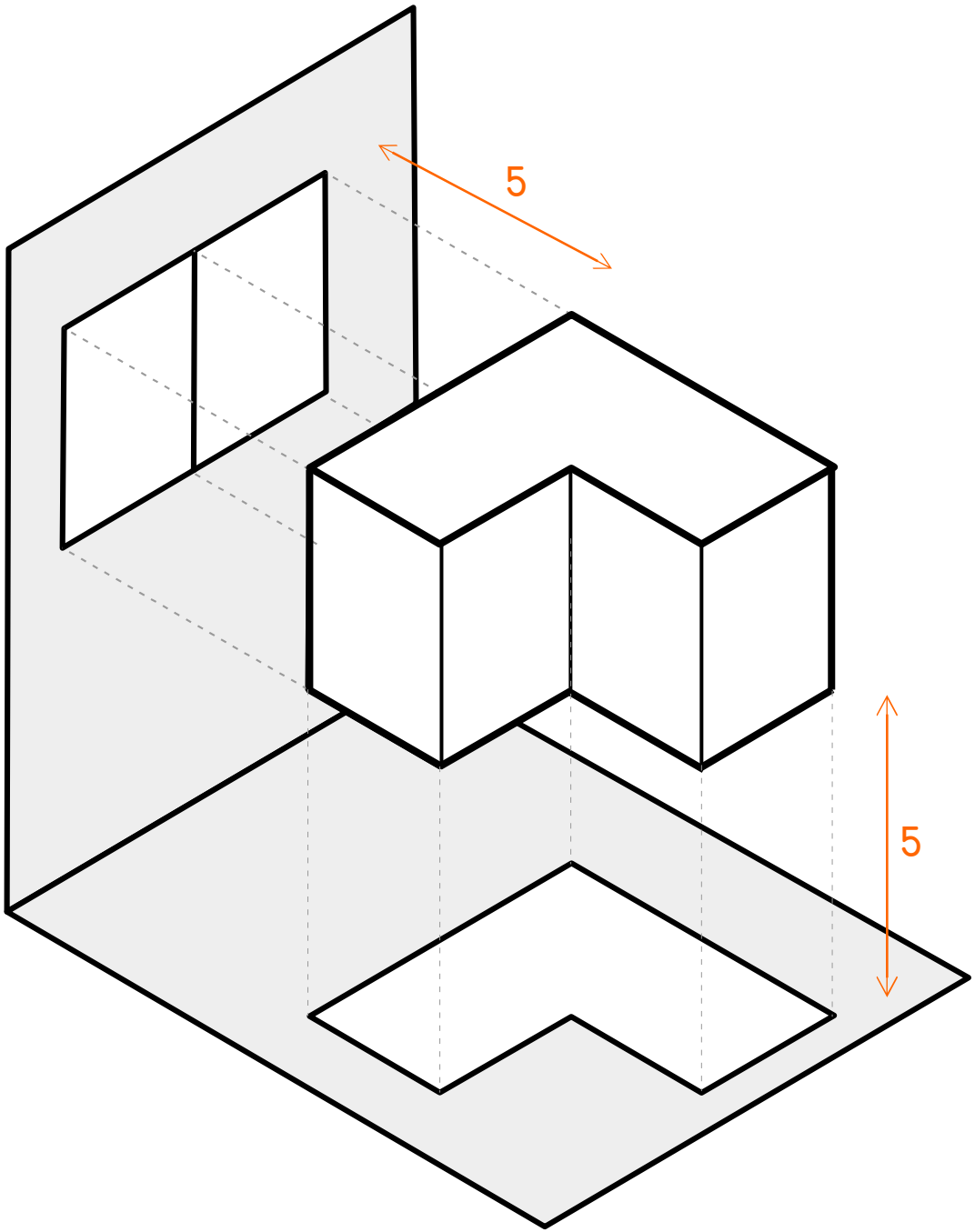
Sistema de Projeção Ortogonal:

Objeto é projetado ortogonalmente sobre planos.

plano vertical
Plano horizontal

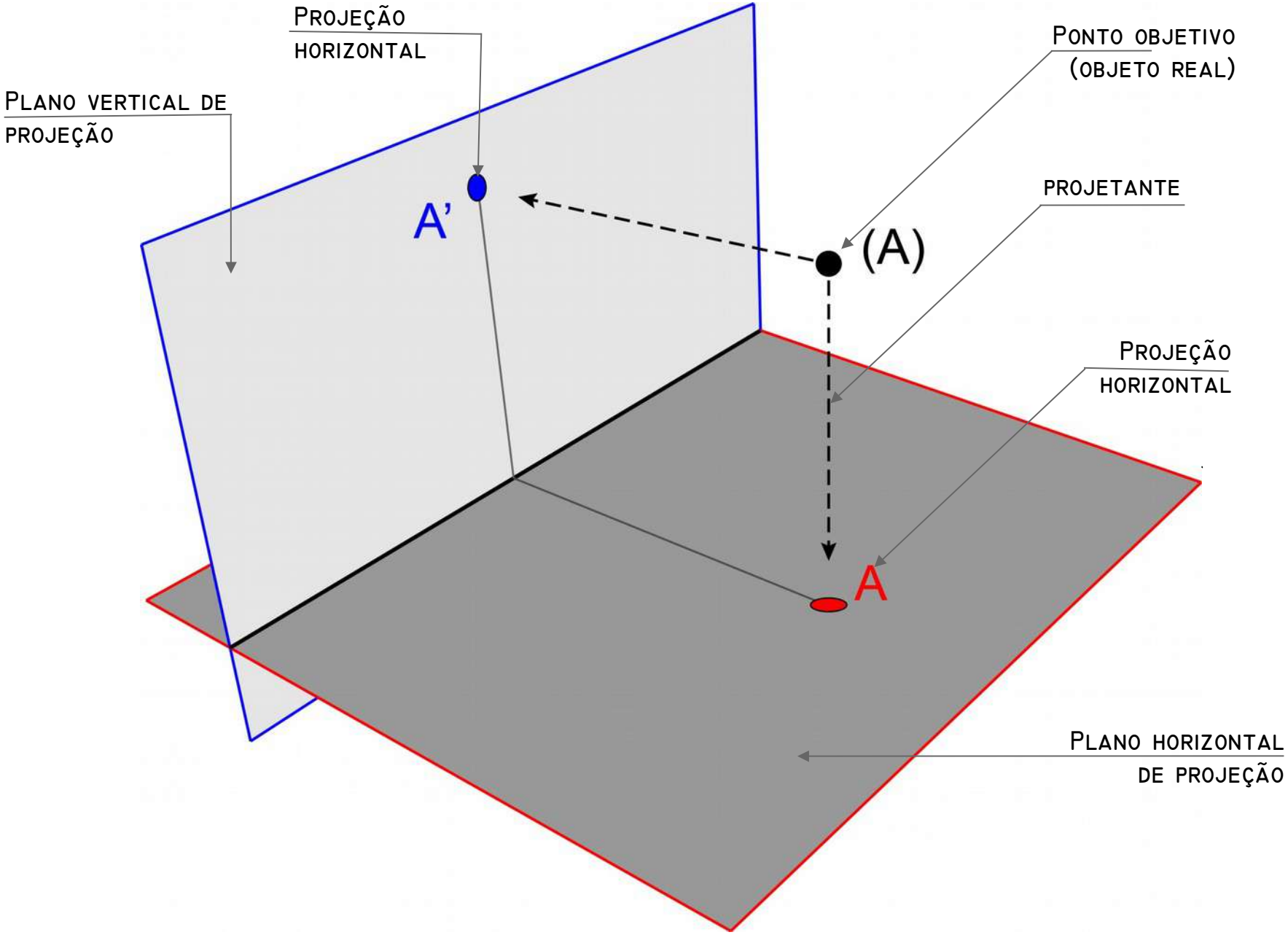
Neste caso os dois planos são perpendiculares entre si.

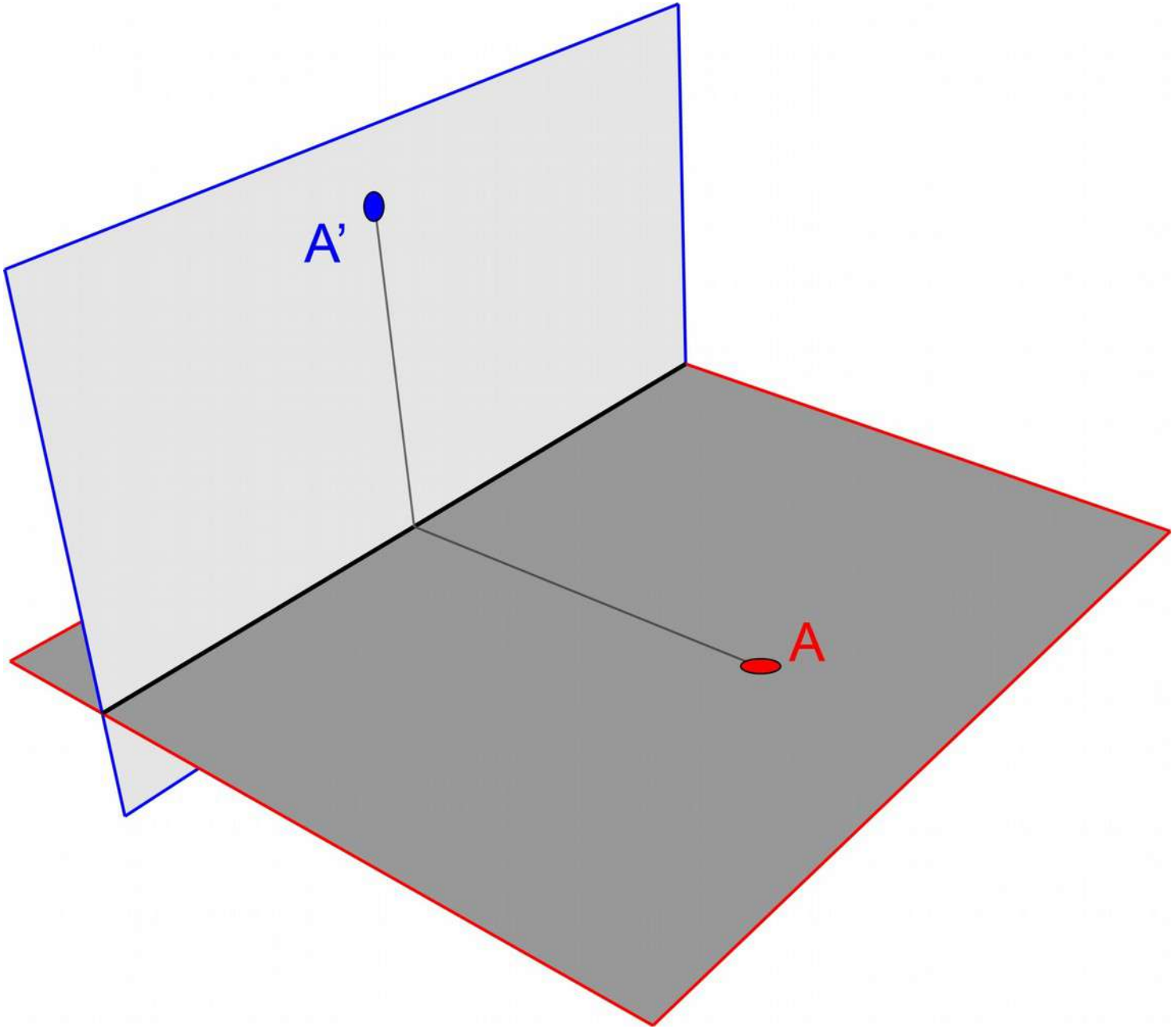
Diferente da projeção oblíqua, não acontece deformação da imagem projetada (projeção). Ela tem as mesmas medidas do elemento real

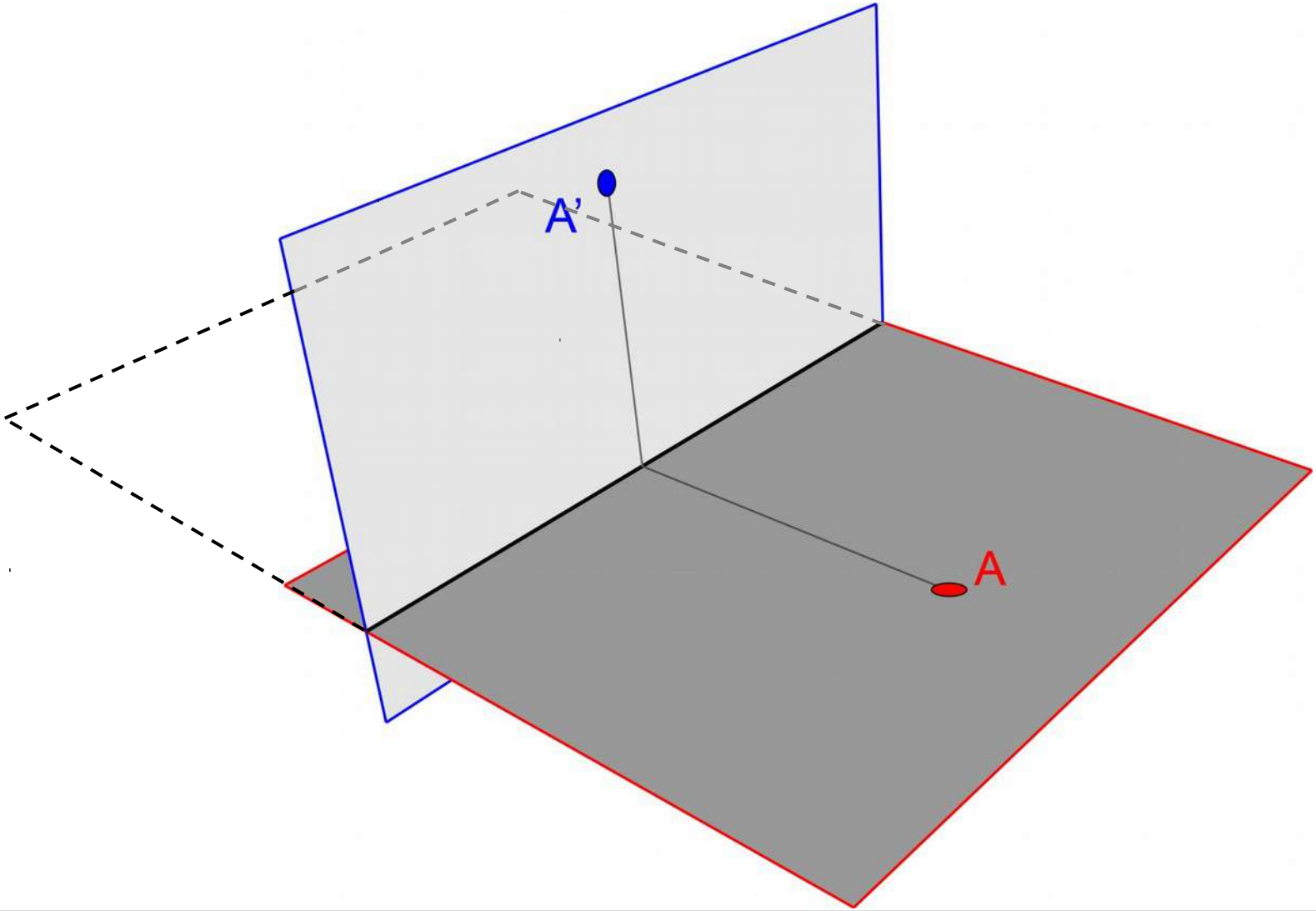


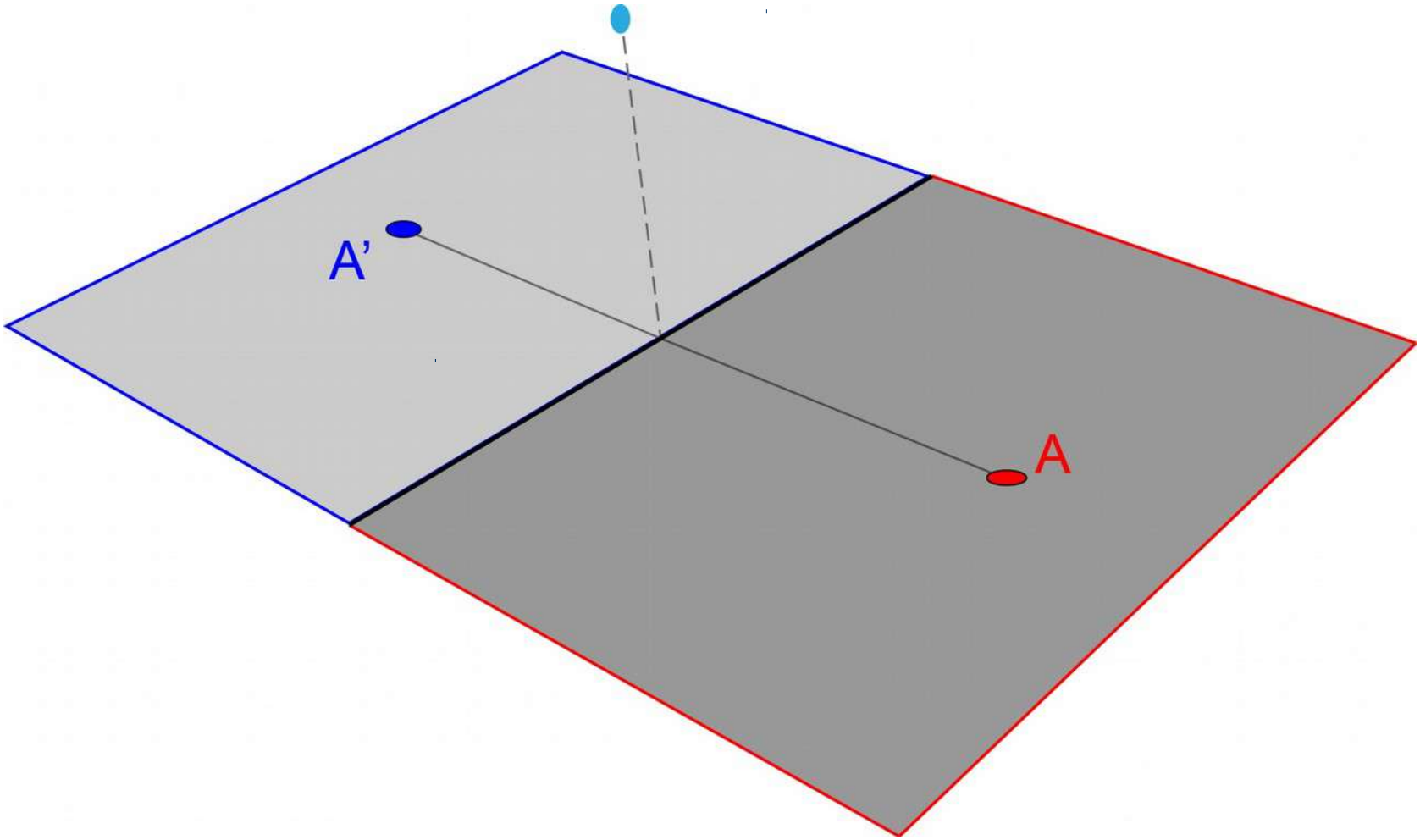
CONCEITOS INICIAIS

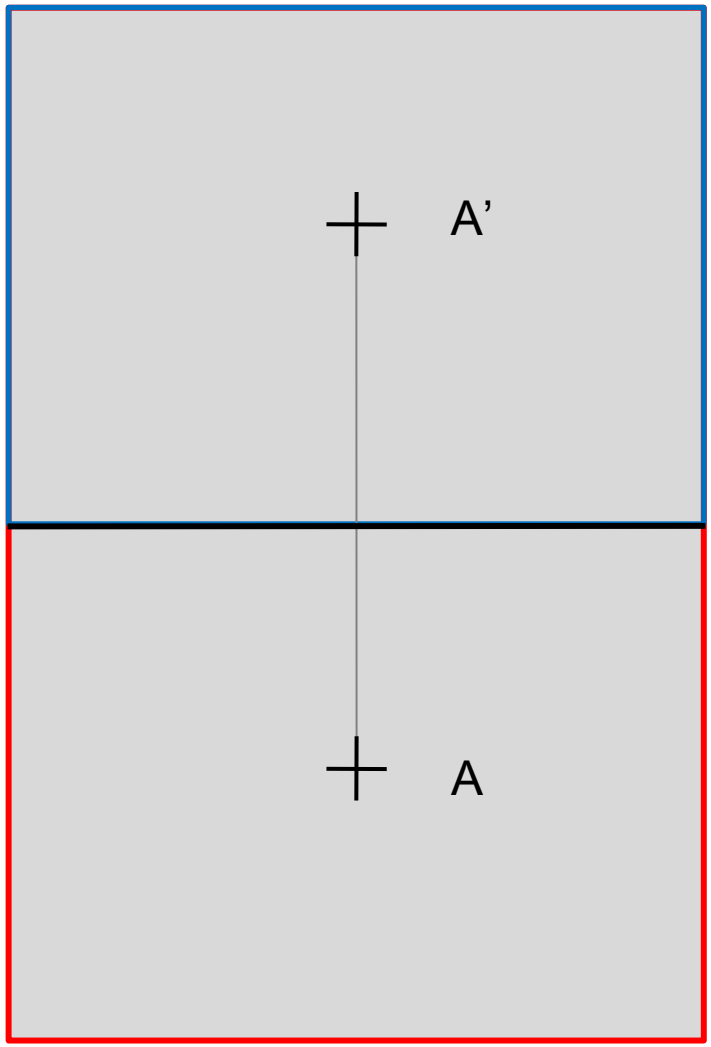
SISTEMA DE PROJEÇÕES MONGEANO

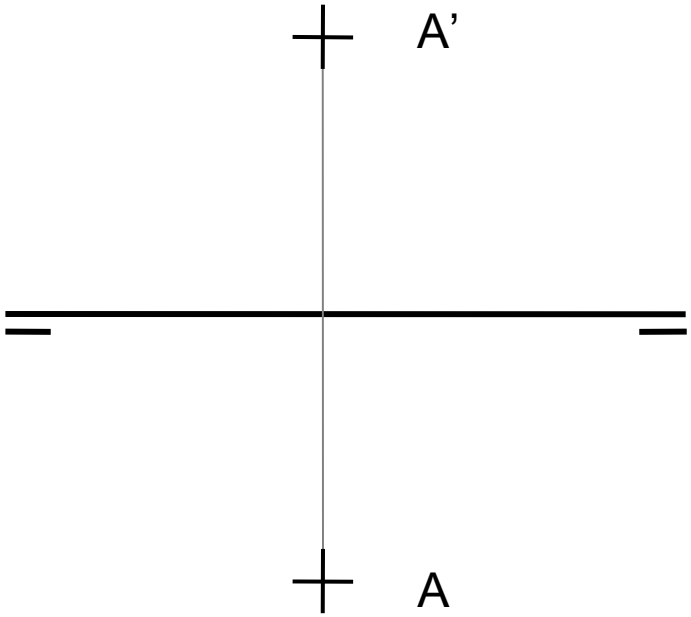

PROJEÇÃO
HORIZONTAL
PONTO OBJETIVO
(OBJETO REAL)
PLANO VERTICAL DE
PROJEÇÃO
PROJETANTE
A'
(A)
PROJEÇÃO
HORIZONTAL
A
PLANO HORIZONTAL
DE PROJEÇÃO

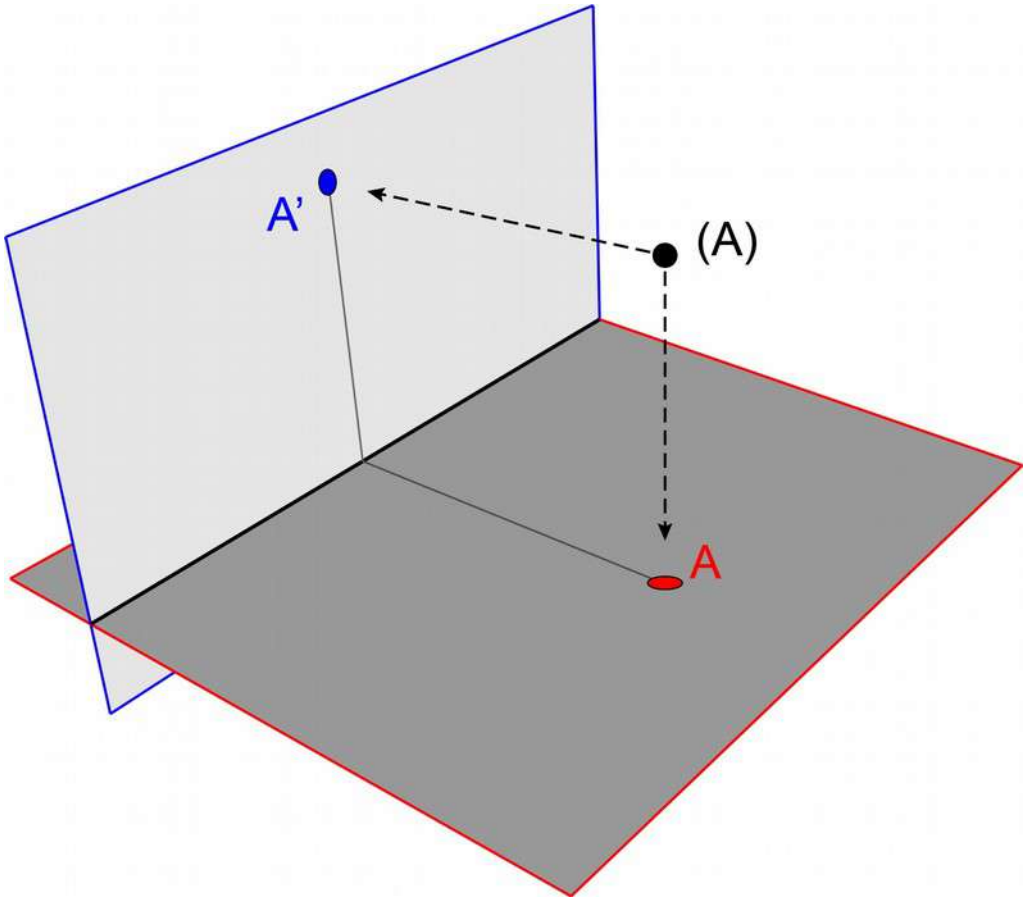

A’
A

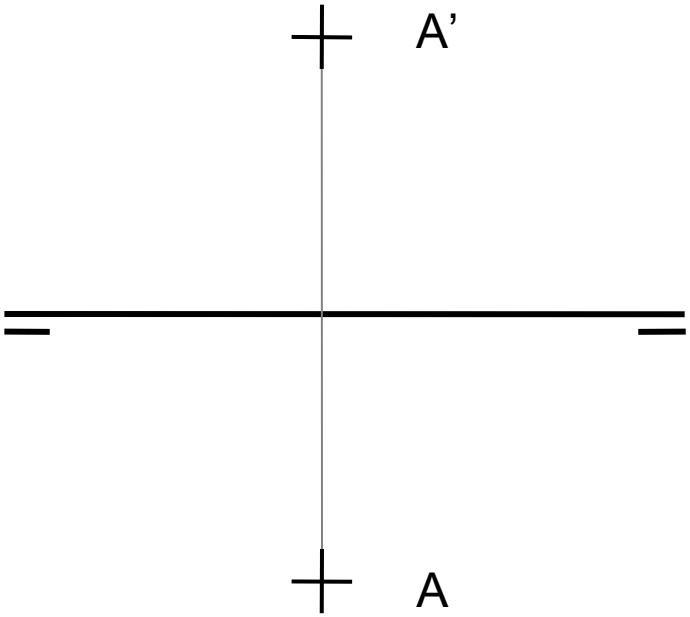

A'
A

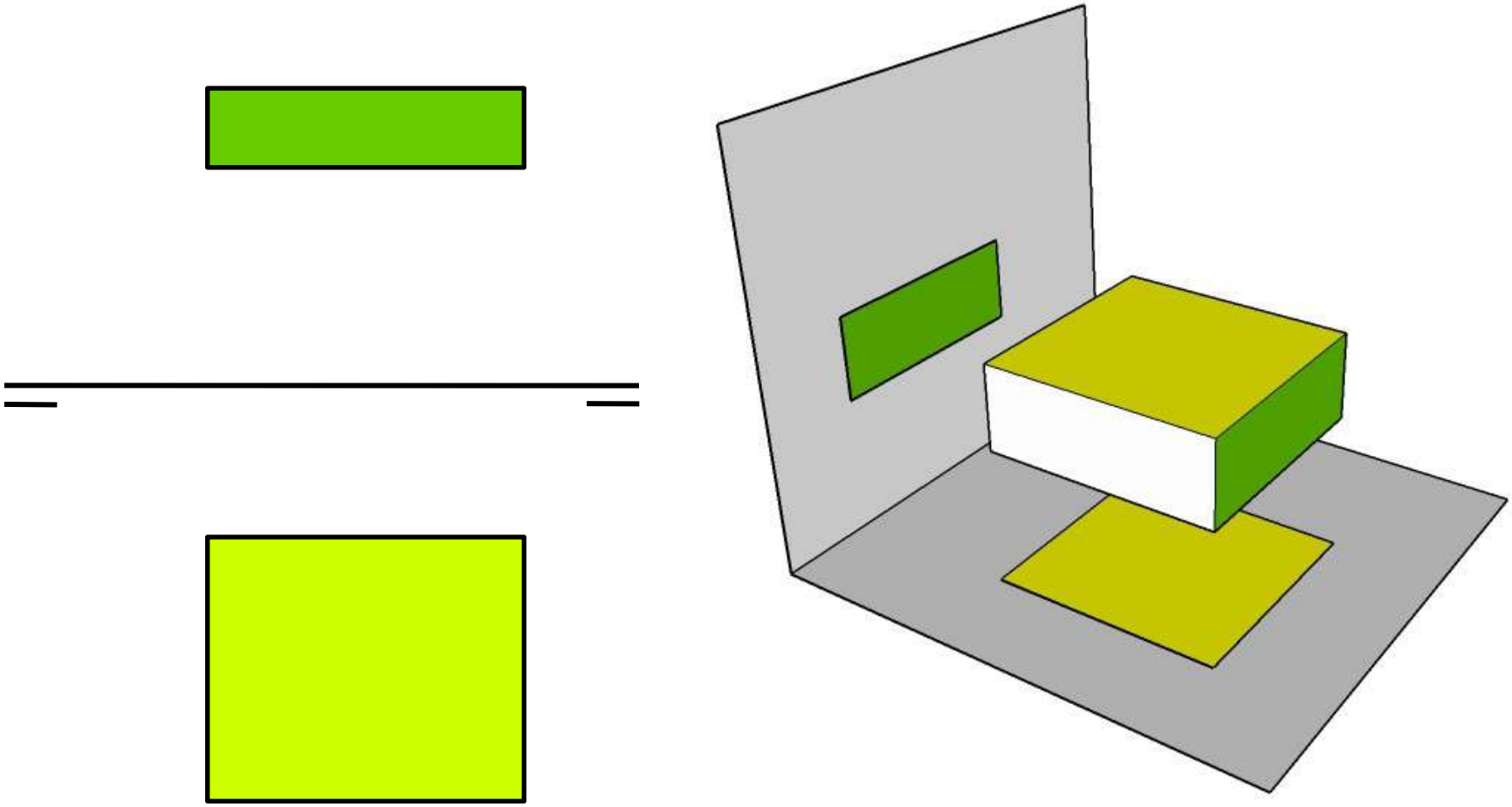

A’
A


A’
A

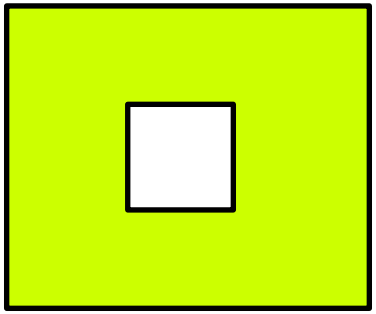

A’
A

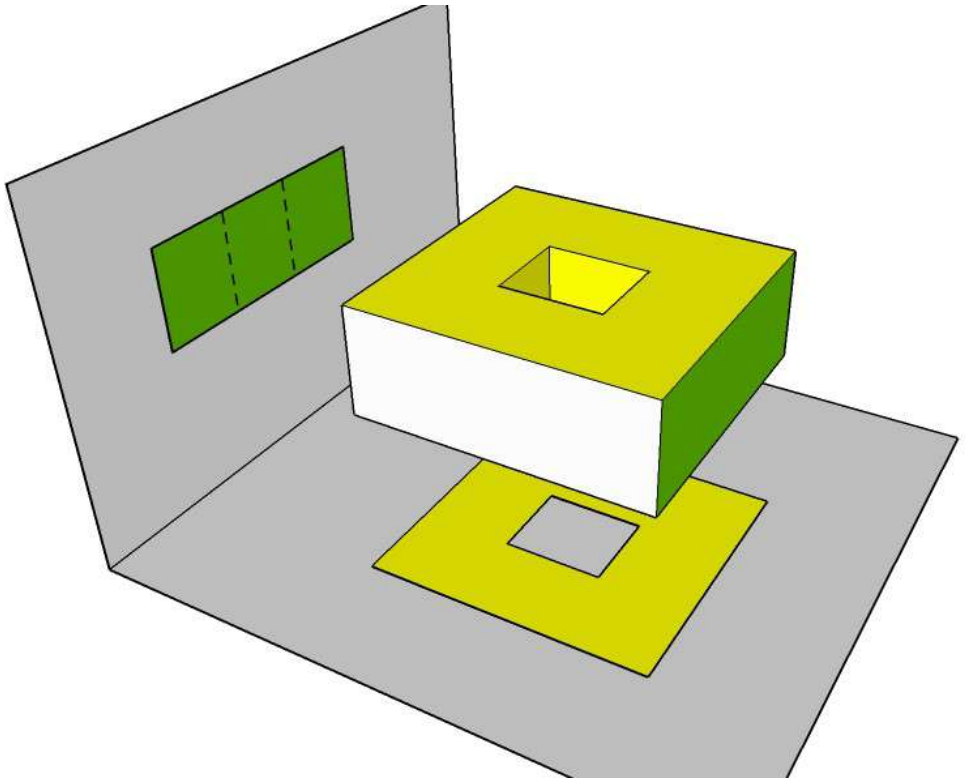

A’
(A)
A


A’
A





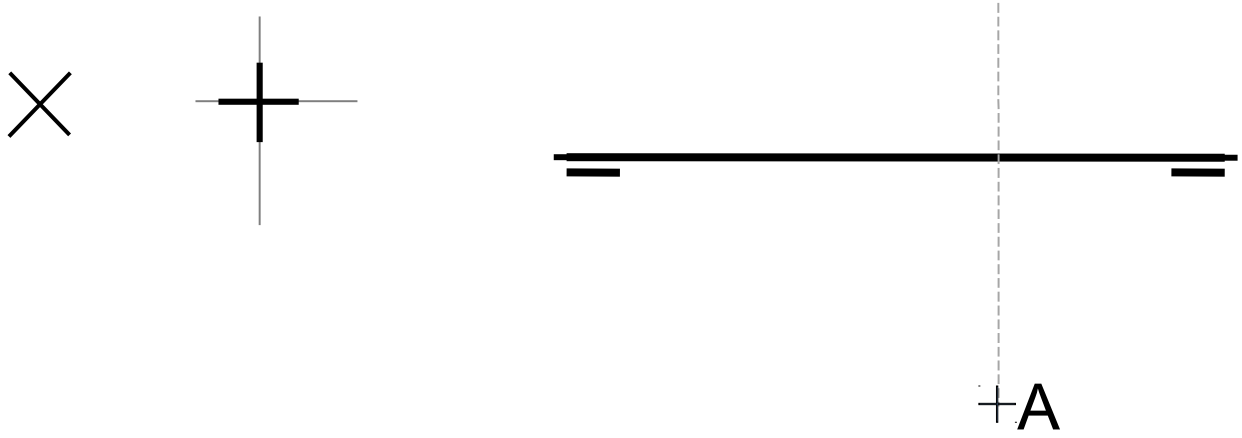




REPRESENTAÇÃO DO PONTO

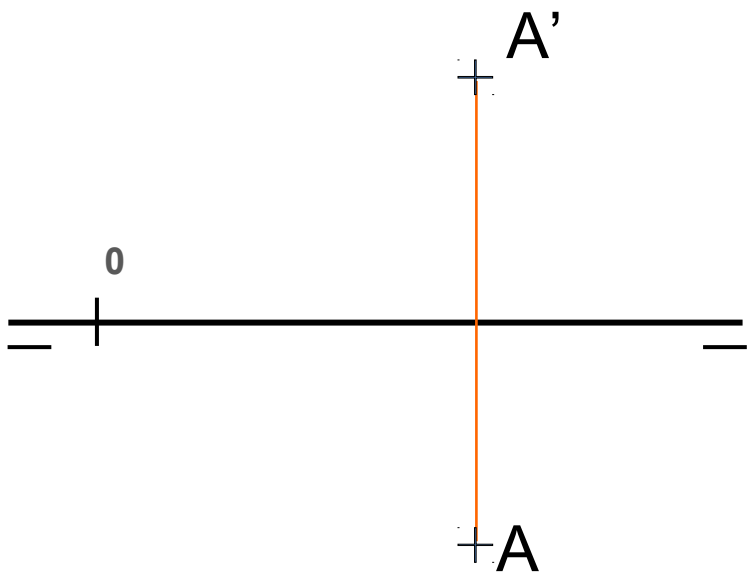
DETERMINAÇÃO DA ÉPURA DO PONTO

UM PONTO, POR DEFINIÇÃO É O LU-GAR GEOMÉTRICO DA INTERSEÇÃO ENTRE DUAS RETAS.
SEGUINDO ESTA DEFINIÇÃO, A REPRESENTAÇÃO SERÁ DOIS SEGUIMENTOS DE RETAS:



POSIÇÕES DO PONTO

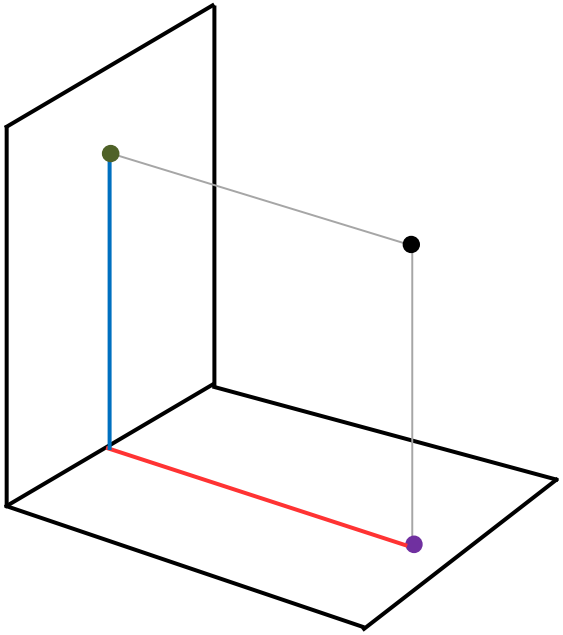
- Os segmentos de retas que unem as projeções vertical **A'** e horizontal **A** à Linha de Terra são denominadas por **Linhas de Chamada de (A)**. Na Geometria Descritiva onde se utiliza o Sistema de Projeções Ortogonais, as Linhas de Chamada serão sempre perpendiculares à Linha de Terra.

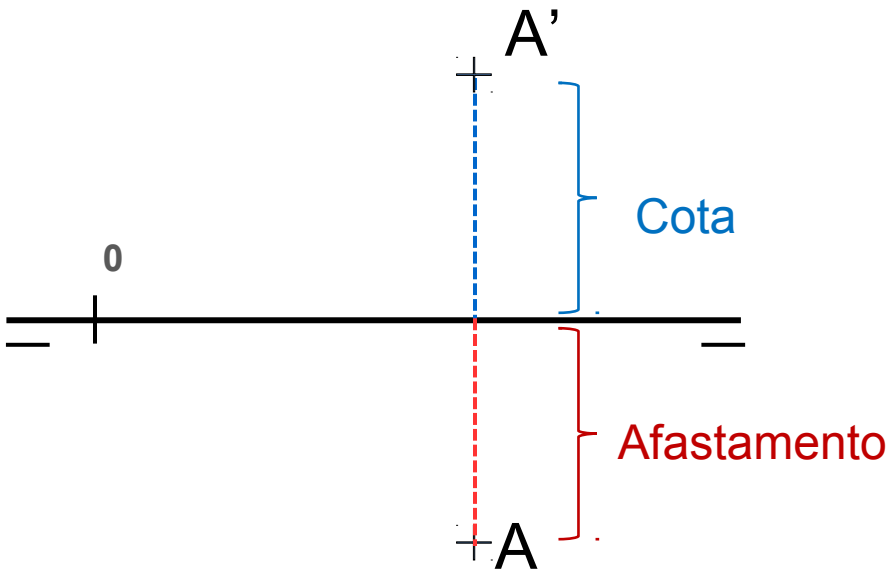


POSIÇÕES DO PONTO

- A distância da projeção vertical A’ até (ππ’) representa **a cota** do ponto (A).

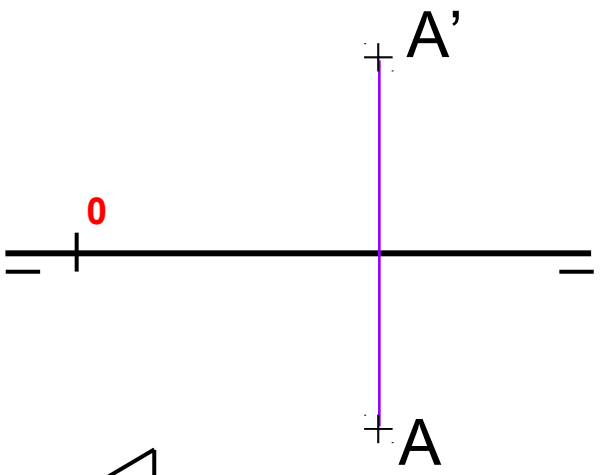
- A distância da projeção horizontal A até a (ππ’) representa **o afastamento** deste ponto.

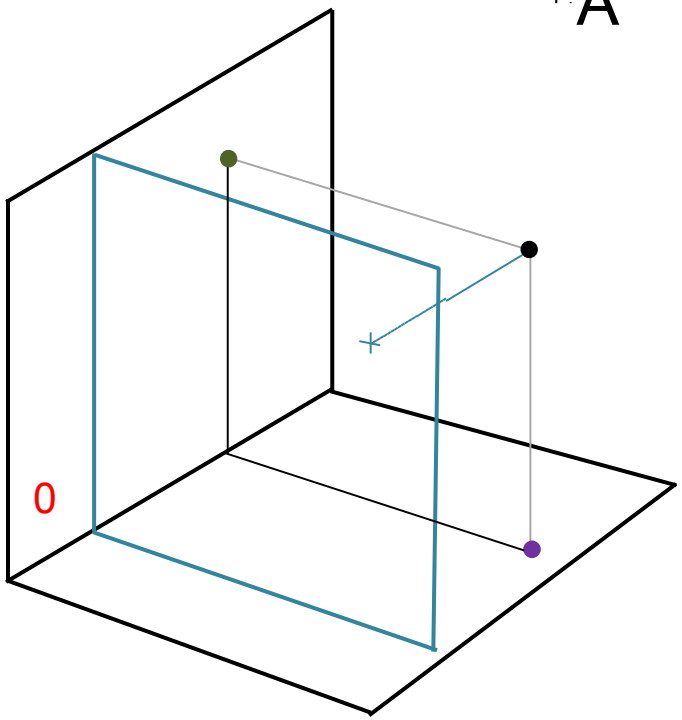




POSIÇÕES DO PONTO

- A abscissa do ponto **(A)**, que corresponde no espaço à distância do ponto objetivo até um plano lateral de projeções será, em épura, representada pela distância horizontal da linha de chamada de **(A)** até a interseção do plano lateral com a (**ππ'**), ponto este marcado arbitrariamente sobre (**ππ'**).
- Adotamos uma origem das abscissas na linha de terra do sistema quando esta não for indicada.



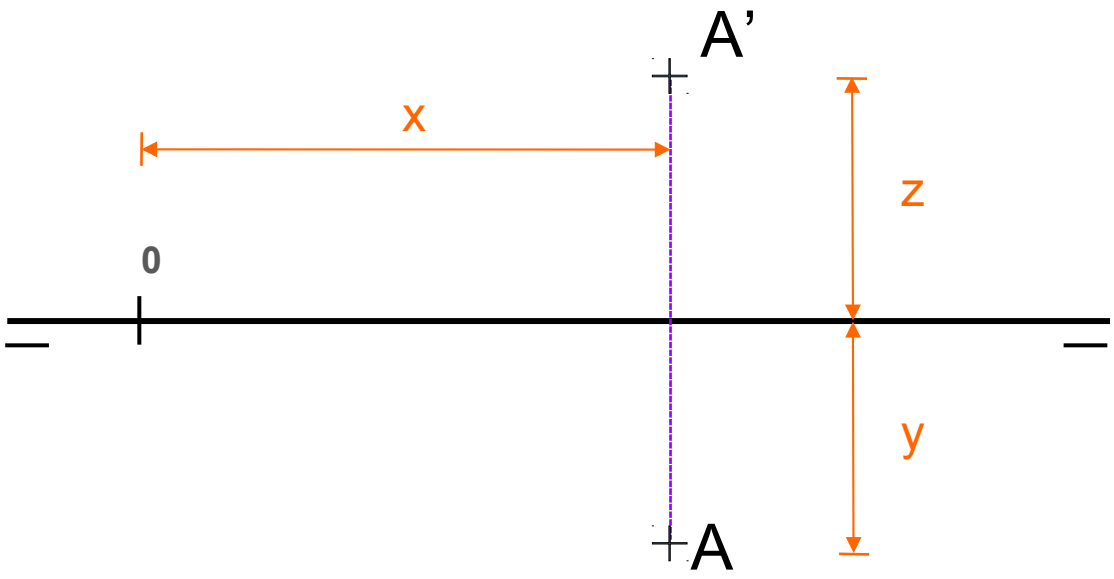


Coordenadas descritivas

- As coordenadas descritivas de um ponto objetivo serão sempre apresentadas conforme a ordem abaixo:

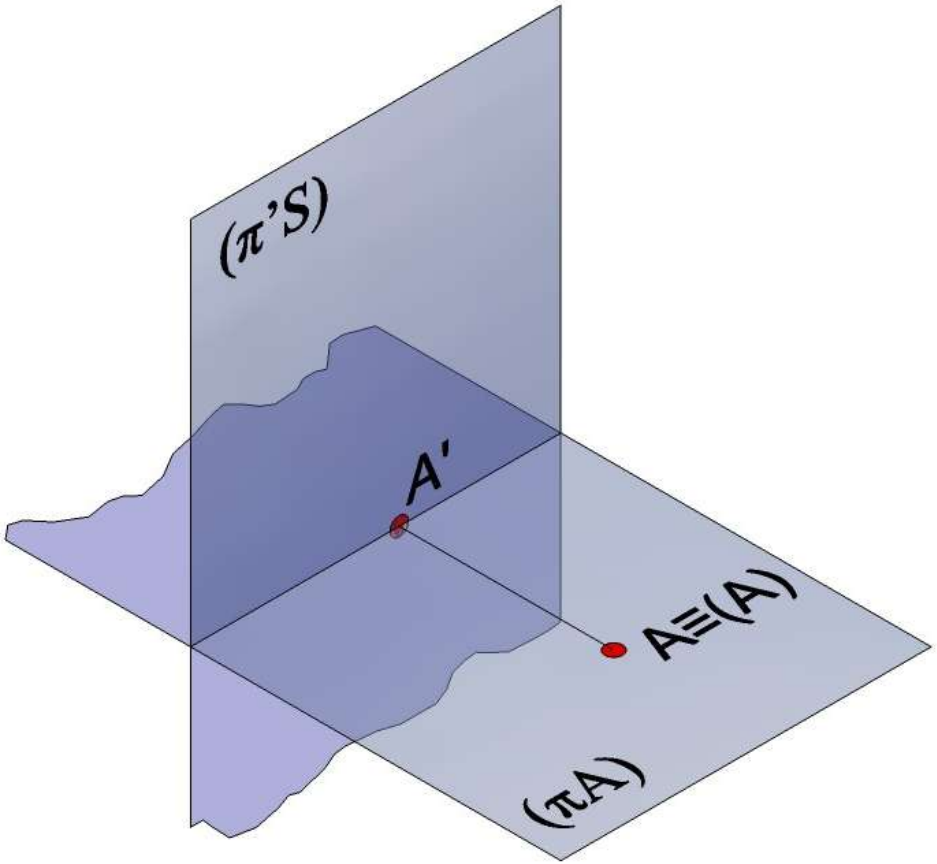
Abscissa = x; **Afastamento = y;** **Cota = z.**

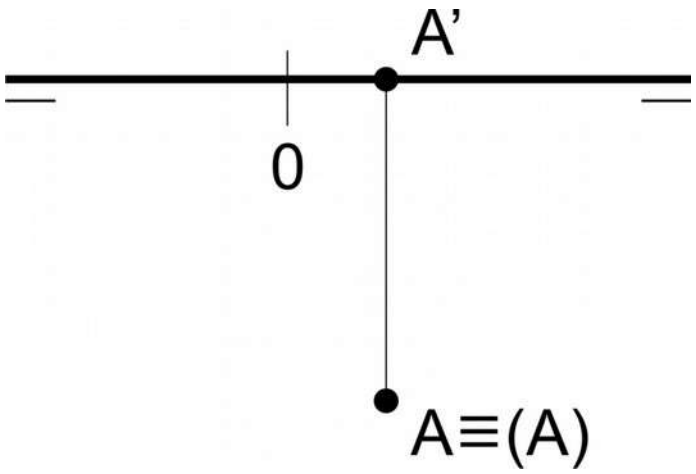
- Assim, para o ponto (A) tem-se a seguinte notação: **(A) [x ; y ; z]**



Ponto localizado no semi-plano horizontal anterior (πA)

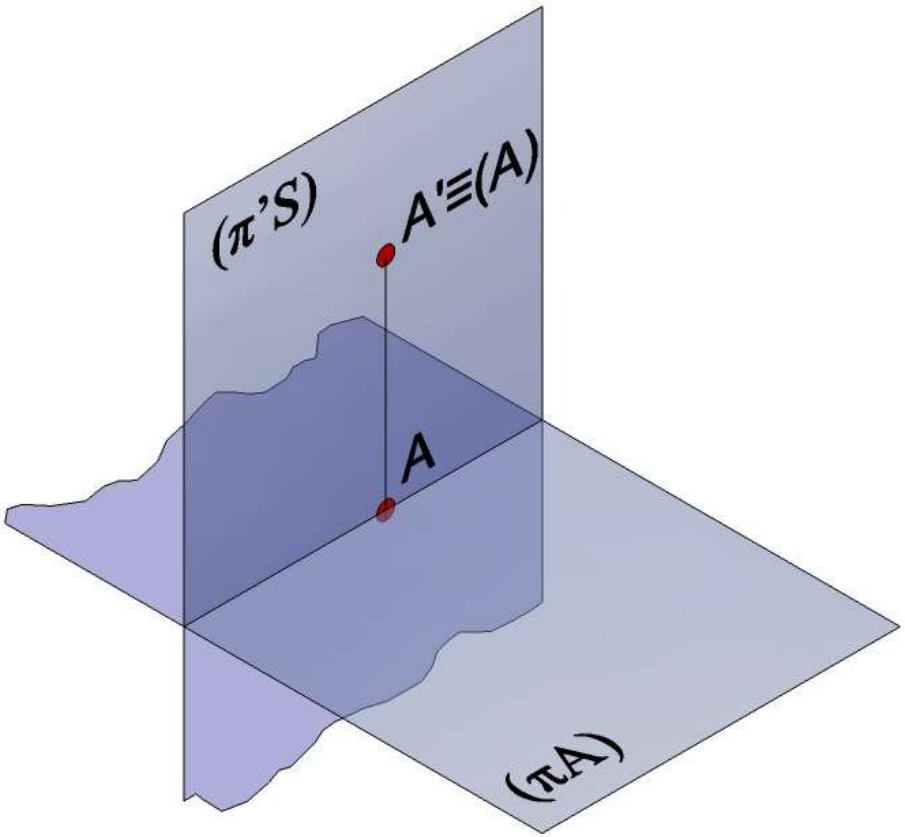
- A cota é nula, o afastamento é positivo. Em épura, a projeção vertical apresenta-se sobre (**ππ'**) e a projeção horizontal, abaixo de (**ππ'**).

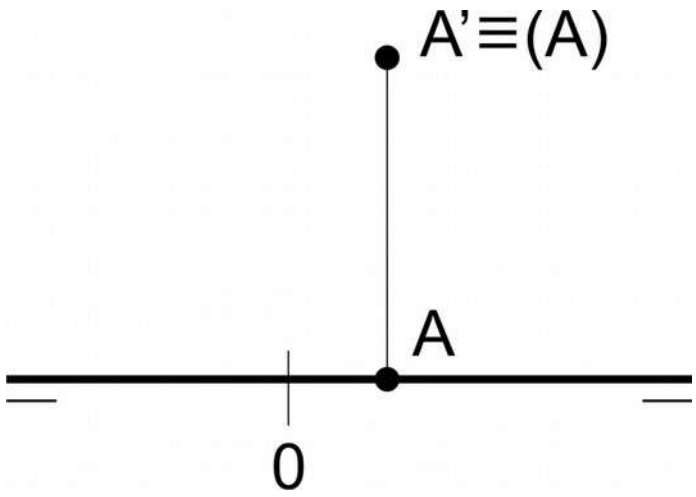




Ponto localizado no semi-plano vertical superior (π'S)

- O afastamento é nulo, a cota é positiva. Em épura, a projeção horizontal apresenta-se sobre (**ππ'**) e a projeção vertical, acima de (**ππ'**).





Dar a épura dos pontos:

a) (A) [3 ; 2 ; 0]

b) (B) [0 ; 1; 1]

c) (C) [0 ; 1,5 ; 3]

d) (D) [5; 0; 0]